

Mathematical research report

Ginibre and Poisson Boolean percolation

Proofs, recovered arguments, and obstructions

Investigation of 15 September 2026

Amended 21 September 2026

Status of the original target. The strict inequality

$$r_c(G) < r_c(\Pi_1)$$

for equal unit intensity and deterministic disk grains remains unproved and undisproved in this investigation.

The critical-radius limit under area-preserving stretching is established by the proof included here. The recovered small- β expansion is preserved in detail, but its central uniform geometric estimate has not received a complete fresh verification. Neither result is silently identified with the original target.

A theoretical investigation; no numerical experiments.

Abstract

This report records the investigation of the critical-radius comparison between planar Ginibre and Poisson Boolean models. It contains complete proofs of an optimal homogeneous Poisson domination for thinned and rescaled Ginibre, exact Palm interpolation and Fock variational identities, buffered conditional Bergman domination, an obstruction to uniform exterior Palm comparison, and finite-ensemble deletion calculations. It restores the ellipse route: fixed-eccentricity large-area comparison, quantitative local Poisson convergence under area-preserving stretching, and a negative-association block argument proving convergence of the actual critical radii. It also preserves the small- β asymptotic argument, with its principal geometric estimate explicitly separated from its verified algebraic and limiting consequences. The remaining routes are organized by their mathematical outcomes and exact obstructions. The amended account also restores the quantitative stretching rate, shape-variation and switching calculations, coupling barriers, the positive-bath mixed-model comparison, conditional clearing laws, and further certificate, transport, and finite-rank refinements. The full original notes accompany the report; historical claims of audit success in those notes are not used as substitutes for the proofs here. No claim of novelty or resolution of the original strict inequality is made.

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Scope, conventions, and verification status

The mathematical target

Identify $\mathbb{R}^2$ with $\mathbb{C}$, and write dA for Lebesgue area. For a locally finite configuration η , define

$$\mathcal{O}_r(\eta) = \bigcup_{x \in \eta} \overline{B}(x, r).$$

Its critical radius is defined by

$$r_c(\mathcal{L}(\eta)) = \inf\{r > 0 : \mathbb{P}(\mathcal{O}_r(\eta) \text{ has an unbounded component}) > 0\}.$$

Every grain is a full closed disk. Thus the center connection range is $2r$, not r . A restricted crossing means that its occupied path is restricted to the observation rectangle; centers outside that rectangle are still allowed when their disks enter it. Such an event is determined by the radius- r expansion of the rectangle.

Let Π_λ be homogeneous Poisson of intensity λ , write $\Pi = \Pi_1$, and put $r_P = r_c(\Pi_1)$. The unit-intensity Ginibre process G has kernel

$$H(z, w) = \exp\left(\pi z \bar{w} - \frac{\pi}{2}(|z|^2 + |w|^2)\right), \quad |H(z, w)|^2 = e^{-\pi|z-w|^2}.$$

Kernels are relative to dA , unless a change of gauge is stated. The n -point factorial density is $\det[H(z_i, z_j)]$. The kernel is an orthogonal projection in the physical Gaussian gauge.

For a finite distinct tuple A , $G^{\dagger A}$ denotes its canonical reduced Palm law. The prescribed points are omitted. This is defined through factorial densities, not ordinary conditioning on a probability-zero event. Whenever those points play the role of occupied parents or arms, their restoration is explicitly part of the configuration. The corresponding reduced kernel is $Q_A = H - P_A$, where P_A projects onto the span of the coherent evaluation vectors at A .

For any functional F , write

$$D_q F(\eta) = F(\eta \cup \{q\}) - F(\eta).$$

For increasing Boolean F , the mixed difference $D_q D_z F$ can have either sign. It equals $+1$ when two insertions cooperate and -1 when they are individually successful alternatives. This sign distinction is central to the investigation.

Two deformations must be kept separate:

- $G_\beta = \text{DPP}(\beta H_\beta)$ is unit intensity after thinning by β and spatial contraction by $\sqrt{\beta}$. The small- β endpoint is Poisson; the full Ginibre endpoint is $\beta = 1$.
- $\Phi_t = S_t^{-1} G$, with $S_t(x, y) = (e^t x, e^{-t} y)$, has unit intensity because $\det S_t = 1$. The parameter t changes anisotropy without thinning.

The notation $G_s = \text{DPP}(sH)$ used in the signed interpolation section instead has intensity s ; a Poisson bath restores total intensity one on the path $\nu = 1 - s$.

How to read the status statements

1. **Verified in this report.** The displayed argument has been reconstructed and checked, using the cited published inputs. This wording is a statement about the supplied proof, not external certification or novelty.

2. **Recovered argument.** A substantive derivation survives in the original notes, sometimes with older positive audit records. Its complete dependency chain has not been freshly verified. It is preserved with its exact unresolved verification dependency; it is not declared false.
3. **Disproved auxiliary assertion.** A counterexample or exact calculation contradicts the assertion. The scope of that counterexample is stated; failure of a stronger auxiliary assertion does not disprove the original critical-radius comparison.

Result or route	Present status	Consequence and limitation
Optimal upper Poisson domination of G_β	Verified	Gives a lower bound on $r_c(G_\beta)$, for $\beta < 1$.
Palm interpolation and Fock variational formulas	Verified	Exact identities; their pivotal average has no established favorable sign.
Buffered Bergman bound and exterior Palm likelihood obstruction	Verified	Control conditioned remainders and rule out uniform arbitrary-exterior comparisons.
Fixed-shape large-area ellipse comparison	Verified	Strict comparison of large-grain percolation probabilities, away from an identified critical comparison.
$r_c(S_t^{-1}G) \rightarrow r_P$	Verified	Actual infinite-volume thresholds converge; no ordering of the value at $t = 0$ follows.
Small- β expansion with coefficient $-r_P/4$	Recovered argument	Depends on a uniform geometric pivotal estimate, including the zero-bath regime.
Further certificate, motion, and multiple-mark constructions	Status specified individually	Their source proofs are preserved; none supplies the missing full-Ginibre pivotal average.
Quantitative shape rate, mixed-model gap, and additional local refinements	Recovered arguments	Restored with source hypotheses, quantitative constants, and remaining dependencies.
$r_c(G) < r_P$	Unresolved	No positive radius gap at $\beta = 1$ is established.

Completeness and provenance

The supplied research-history PDF and ellipse note are retained unchanged in the companion source archive. The archive also includes every surviving individual mathematical Markdown note and the earlier consolidated checkpoint. The report provides a coherent mathematical account rather than treating successive checkpoint descriptions as independent proofs. Its source index maps back to the original files. Earlier labels such as “audited” and “proved” are historical assertions when quoted in the supplement; the status statements in this report govern its own conclusions.

The amendment of 21 September 2026 incorporates the twelve main coverage findings and six additional quantitative refinements identified in the companion coverage audit. Their result-level locations are listed in Section 9.26. New summaries retain the recovered-argument status and their explicit conditional or finite-volume scope; restoration does not certify every historical proof afresh. The pre-amendment report and PDF are preserved in `history/report_2026-09-15/`.

In particular, the two arguments shown in the user's screenshots were not lost. The small- β note contains both the original $O(\beta^{6/5})$ remainder and its later optimized remainder. The stretching-limit proof survives a fresh check and is restored here to the verified results. The previous selective write-up's omission of that limit was an omission, not a mathematical retraction.

1 Sharp homogeneous Poisson upper domination

For $0 < \beta < 1$, let

$$H_\beta(z, w) = \frac{1}{\beta} \exp\left(\frac{\pi z \bar{w}}{\beta} - \frac{\pi(|z|^2 + |w|^2)}{2\beta}\right), \quad G_\beta = \text{DPP}(\beta H_\beta).$$

Here H_β is an orthogonal projection of diagonal $1/\beta$, and G_β has intensity one. Set

$$L_\beta = -\frac{\log(1 - \beta)}{\beta}.$$

Theorem 1.

$$G_\beta \preceq \Pi_{L_\beta}. \quad (1.1)$$

Moreover, if $G_\beta \preceq \Pi_\lambda$ for a homogeneous Poisson process, then $\lambda \geq L_\beta$. Consequently,

$$\boxed{r_c(G_\beta) \geq \frac{r_c(\Pi_1)}{\sqrt{L_\beta}}.} \quad (1.2)$$

The stochastic order in (1.1) is inclusion order on locally finite configurations. The critical-radius bound is a lower bound.

1.1 Proof of domination

Write $\mu_s = \mathcal{L}(\text{DPP}(sH_\beta))$, for $0 < s < 1$, and $\mu_s^{!q}$ for its reduced Palm law.

Goldman's one-point Ginibre–Palm coupling, after spatial scaling, couples the full projection process with its reduced Palm process by adding exactly one point. Independently retain every point with probability s . The added point is not retained with probability $1 - s$, independently of the thinned Palm remainder. Therefore, as measures,

$$\mu_s \geq (1 - s)\mu_s^{!q}. \quad (1.3)$$

This use of Goldman's theorem does not assert independence of the added point's location and the Palm remainder.

Let F be a bounded increasing function determined by a bounded set W . For an independent Poisson process Π_ν , put

$$C(s, \nu) = \mathbb{E}F(\text{DPP}(sH_\beta) \cup \Pi_\nu), \quad D_q F(\eta) = F(\eta \cup \{q\}) - F(\eta).$$

Independent thinning, followed by the Campbell formula of the full projection process, gives

$$\partial_s C(s, \nu) = \frac{1}{\beta} \int_W \mathbb{E}_{\mu_s^{!q}, \Pi_\nu} D_q F dA(q). \quad (1.4)$$

Poisson insertion gives

$$\partial_\nu C(s, \nu) = \int_W \mathbb{E}_{\mu_s, \Pi_\nu} D_q F dA(q). \quad (1.5)$$

These differentiations are justified directly: conditional on the full projection process in W , thinning involves finitely many Bernoulli variables. The derivative is bounded by $\text{osc}(F)N(W)$, whose expectation is finite. The same couplings give continuity of both derivatives, with one-sided derivatives at the endpoints.

Since $D_q F \geq 0$, (1.3)–(1.5) imply

$$\partial_s C(s, \nu) \leq \frac{1}{\beta(1-s)} \partial_\nu C(s, \nu). \quad (1.6)$$

Integrate this inequality along

$$\nu(s) = \frac{1}{\beta} \log \frac{1-s}{1-\beta}, \quad 0 \leq s \leq \beta.$$

The chain rule shows that $C(s, \nu(s))$ is nonincreasing. Its initial law is Π_{L_β} , and its final law is G_β . Thus (1.1) holds for every bounded increasing local function.

For completeness, local stochastic order extends to whole-plane inclusion order. On each bounded open disk, the order on locally finite counting measures is closed, being determined by integration against nonnegative continuous compactly supported functions. Strassen's closed-order theorem gives a coupling of the restrictions. Extend each to a coupling of the full laws using regular conditional exterior laws. These couplings form a tight family because their marginals are fixed. Any subsequential weak limit preserves all compactly supported order inequalities, and hence is a whole-plane inclusion coupling.

The external input here is [Goldman's Theorem 1](#), whose printed normalization has intensity $1/\pi$. Spatial scaling gives the projection normalization used above.

1.2 Proof that L_β is optimal

Let $D_R = B(0, R)$ and $Q_R = 1_{D_R} H_\beta 1_{D_R}$. The Fredholm hole formula gives

$$-\log \mathbb{P}(G_\beta(D_R) = 0) = \sum_{j \geq 1} \frac{\beta^j}{j} \text{Tr}(Q_R^j). \quad (1.7)$$

Also

$$\text{Tr } Q_R = \frac{|D_R|}{\beta}, \quad \text{Tr}(Q_R - Q_R^2) = \int_{D_R} \int_{D_R^c} |H_\beta(x, y)|^2 dA(y) dA(x) = O_\beta(R). \quad (1.8)$$

To verify the last estimate, use $|H_\beta(x, y)|^2 = \beta^{-2} e^{-\pi|x-y|^2/\beta}$. For displacement h , the area of $\{x \in D_R : x+h \notin D_R\}$ is at most $C(R|h| + |h|^2)$. Integrating this bound against the Gaussian proves (1.8).

The eigenvalues of Q_R lie in $[0, 1]$. Hence, for fixed j ,

$$0 \leq \text{Tr } Q_R - \text{Tr}(Q_R^j) \leq (j-1) \text{Tr}(Q_R - Q_R^2).$$

Therefore $\text{Tr}(Q_R^j)/|D_R| \rightarrow 1/\beta$. The bound $\text{Tr}(Q_R^j) \leq |D_R|/\beta$, together with $\sum_{j \geq 1} \beta^j/j < \infty$, justifies dominated convergence in (1.7):

$$\lim_{R \rightarrow \infty} \frac{-\log \mathbb{P}(G_\beta(D_R) = 0)}{|D_R|} = L_\beta. \quad (1.9)$$

If $G_\beta \preceq \Pi_\lambda$, its hole probability is at least $e^{-\lambda|D_R|}$. Equation (1.9) forces $L_\beta \leq \lambda$.

1.3 Infinite-volume implication

In an inclusion coupling, every occupied component of G_β is contained in an occupied component of Π_{L_β} , using the same disk radius. Poisson scaling gives $r_c(\Pi_\lambda) = r_c(\Pi_1)/\sqrt{\lambda}$, proving (1.2). Alternatively, apply the local comparison to connection from a fixed bounded seed to distance L , then let $L \uparrow \infty$. Only finitely many grains touch the seed, so the decreasing intersection is exactly connection of that seed to an unbounded component. No claim about percolation at the Poisson critical radius is needed.

2 Exact signed interpolation formula

Let $H = H_1$, $G_s = \text{DPP}(sH)$, and

$$C_F(s, \nu) = \mathbb{P}(F(G_s \cup \Pi_\nu) = 1)$$

for a local increasing Boolean event F . For two absent points q, z , define

$$\begin{aligned}\mathcal{R}_{q,z}(\eta) &= \mathbf{1}\{F(\eta) = 0, F(\eta + q) = F(\eta + z) = 1\}, \\ \mathcal{C}_{q,z}(\eta) &= \mathbf{1}\{F(\eta) = F(\eta + q) = F(\eta + z) = 0, F(\eta + q + z) = 1\}.\end{aligned}$$

Thus $\mathcal{R}$ records two individually successful alternatives, whereas $\mathcal{C}$ records success requiring both insertions.

Translate one fixed Goldman coupling at the origin to obtain a jointly measurable family of couplings at q :

$$X^q \sim G^{\dagger q}, \quad X^q + \{Z_q\} \sim G.$$

Its added point has density $e^{-\pi|z-q|^2}$, but is generally dependent on X^q . Let X_s^q be an independent s -thinning of X^q , and set $Y_s^q = X_s^q \cup \Pi_\nu$.

Theorem 2.

$$(\partial_s - \partial_\nu)C_F(s, \nu) = s \int \mathbb{E}[\mathcal{R}_{q,Z_q}(Y_s^q) - \mathcal{C}_{q,Z_q}(Y_s^q)] dA(q). \quad (2.1)$$

The integrals are over a bounded active region of F . Along the equal-intensity path $\nu = 1 - s$, (2.1) is its ordinary derivative.

2.1 Proof

For every increasing Boolean F , checking its four possible input configurations gives

$$D_q F(\eta) - D_q F(\eta + z) = \mathcal{R}_{q,z}(\eta) - \mathcal{C}_{q,z}(\eta). \quad (2.2)$$

Thin the full coupled process with common independent marks. The extra point survives with an independent Bernoulli variable B_s of parameter s . The ordinary background has law $Y_s^q + B_s\{Z_q\}$, while Y_s^q is its DPP reduced-Palm background with the same independent Poisson bath. Consequently, writing

$$J_q = \mathbb{E}D_q F(Y_s^q), \quad I_q = \mathbb{E}D_q F(G_s \cup \Pi_\nu),$$

equation (2.2) gives

$$J_q - I_q = s \mathbb{E}[\mathcal{R}_{q,Z_q}(Y_s^q) - \mathcal{C}_{q,Z_q}(Y_s^q)].$$

The derivative formulas are $\partial_s C_F = \int J_q dA(q)$ and $\partial_\nu C_F = \int I_q dA(q)$: the full parent has intensity one. This proves (2.1).

For each fixed bounded active window, the common thinning marks eventually retain all its points as $s \uparrow 1$. Bounded convergence therefore justifies this endpoint, also after integration over active roots. This does not give convergence uniform in growing windows.

There is also an exact formulation under the ordinary current law. Disintegrate the coupling given G_s , and let $w_s(q, G_s, z)$ be the conditional probability that z is its surviving deleted point. For $0 < s < 1$, its no-deletion probability is

$$(1 - s) \frac{d\mu_s^{\dagger q}}{d\mu_s}(G_s) = 1 - T_q^s(G_s), \quad \sum_{z \in G_s} w_s(q, G_s, z) = T_q^s(G_s).$$

Then the right side of (2.1) is equivalently

$$\int \mathbb{E} \sum_{z \in G_s} w_s(q, G_s, z) [\mathcal{R}_{q,z} - \mathcal{C}_{q,z}] ((G_s \setminus \{z\}) \cup \Pi_\nu) dA(q). \quad (2.3)$$

There is no additional s in (2.3); its weights already include the survival probability. Neither formula supplies the sign.

3 Exact Fock variational formula, and a negative local example

Let

$$\mathcal{F} = \left\{ f \text{ entire} : \|f\|_{\mathcal{F}}^2 = \int_{\mathbb{C}} |f(z)|^2 e^{-\pi|z|^2} dA(z) < \infty \right\}.$$

Fix a finite distinct set A , let $Q = H - P_A$ be the projection onto the corresponding Gaussian-gauged functions vanishing on A , and fix $q \notin A$. For bounded Borel B , let $Q_B = 1_B Q 1_B$. For $0 \leq s \leq 1$, define

$$M_{A,B,s}(q) = Q(q, q) + s \left\langle 1_B Q(\cdot, q), (I - sQ_B)^{-1} 1_B Q(\cdot, q) \right\rangle.$$

Theorem 3.

$$\boxed{\frac{1}{M_{A,B,s}(q)} = \min_{\substack{f \in \mathcal{F}, f|_A=0 \\ f(q)=e^{\pi|q|^2/2}}} \int |f(z)|^2 e^{-\pi|z|^2} (1 - s1_B(z)) dA(z).} \quad (3.1)$$

In particular $s = 1$ gives the norm outside B .

3.1 Proof

In the physical L^2 gauge, set $T = I - sQ 1_B Q$ on $\text{ran } Q$. The operator $Q 1_B Q$ is compact: its trace is $\int_B Q(z, z) dA(z) \leq |B|$. Its norm is strictly less than one. Otherwise compactness gives a nonzero eigenvector supported in bounded B , whereas its analytic representative vanishes on the open complement of a sufficiently large disk. Analyticity makes that impossible.

Thus T is positive and invertible, including at $s = 1$. Evaluation at q in the T -inner product has representer $T^{-1}Q(\cdot, q)$. Its squared norm is $\langle Q(\cdot, q), T^{-1}Q(\cdot, q) \rangle = M_{A,B,s}(q)$; the last equality is the resolvent identity. Cauchy–Schwarz gives (3.1), with equality at the normalized representer.

For every $q \notin A$, whether or not $q \in B$, the determinant lemma identifies $M_{A,B,s}(q)$ as

$$Q(q, q) \frac{\mathbb{P}(\text{DPP}(sQ^{lq})(B) = 0)}{\mathbb{P}(\text{DPP}(sQ)(B) = 0)}. \quad (3.2)$$

For the process $\text{DPP}(sQ)$, the actual root intensity times this void ratio is $sM_{A,B,s}(q)$.

3.2 A local multiplier can be below one

Fix a connection range $R > 0$. Let $A = \{-a, a\}$ on the real axis, where $R/2 < a < R$, and let

$$B_a = B(-a, R) \cap B(a, R).$$

The parents are disconnected, and inserting the origin joins them. Direct inversion of their 2×2 coherent Gram matrix gives

$$P_A(0, 0) = \frac{2e^{-\pi a^2}}{1 + e^{-2\pi a^2}} = \operatorname{sech}(\pi a^2).$$

Also $|B_a| \rightarrow 0$ as $a \uparrow R$. Since $Q(z, z) \leq 1$,

$$M_{A, B_a, s}(0) \leq 1 - \operatorname{sech}(\pi a^2) + \frac{|B_a|}{1 - |B_a|}, \quad |B_a| < 1.$$

For a sufficiently close to R , this is at most

$$1 - \frac{1}{2} \operatorname{sech}(\pi R^2) < 1$$

uniformly for $0 \leq s \leq 1$. Thus an empty connecting lens does not give a multiplier at least one for every two-parent geometry. This counterexample concerns a proposed pointwise estimate, not the critical radius.

4 Buffered conditional domination with prescribed zeros

Let $D = B(p, R)$, let $P \subset D$ be a fixed finite distinct set, and let C have compact closure in D . Write $Q_P = H - P_P$. Let B_D^P be the physical reproducing kernel of the Gaussian weighted analytic Bergman space on D whose functions vanish on P .

Theorem 4. For any DPP with kernel $0 \leq K \leq Q_P$, conditional on its entire configuration outside D , its restriction to C is stochastically dominated by $\text{DPP}(B_D^P|_C)$. In particular, almost surely in the observed exterior,

$$\mathbb{P}(N(C) > 0 \mid X_{D^c}) \leq \int_C B_D^P(z, z) dA(z). \quad (4.1)$$

The buffer $D \setminus C$ is unobserved. This is a statement about the reduced remainder process. To infer a hole in C after restoring the prescribed Palm marks, those marks must also lie outside C .

4.1 Proof

Use the usual unitary Ginibre translation to center the Gaussian weight at p . The accompanying conjugate phase factors preserve operator order and determinantal laws; all kernels below use this common gauge.

First observe the exact configuration in the finite annulus $A_L = B(p, L) \setminus D$, $L > R$. On the bounded disk $W = B(p, L)$, all compressed kernels have norm strictly below one. Their Janossy L -operators satisfy

$$L_K = K_W(I - K_W)^{-1} \leq Q_{P, W}(I - Q_{P, W})^{-1} = L_Q.$$

After conditioning on the finite annular configuration, the conditional interior L -operator is a Schur complement, so is no larger than the D -block of L_Q . Applying $L \mapsto L(I + L)^{-1}$ shows that the conditional correlation kernel is no larger than the kernel of Q_P conditioned to have no points in A_L . Singular tuples of zero Janossy density may be omitted.

The latter kernel is the reproducing kernel of Fock functions vanishing on P , with norm

$$\int_D |f|^2 e^{-\pi|z-p|^2} dA + \int_{B(p, L)^c} |f|^2 e^{-\pi|z-p|^2} dA.$$

Restriction to D is contractive into the weighted Bergman space with the same zeros. Taking dual norms of finite linear combinations of evaluations therefore bounds this reproducing kernel by B_D^P .

Its restriction to C is a positive trace-class contraction. Indeed choose $M < R$ with $C \subset B(p, M)$. The unconstrained weighted disk Bergman compression has eigenvalues $p_n(M)/p_n(R)$, where

$$p_n(t) = \frac{1}{n!} \int_0^{\pi t^2} e^{-u} u^n du.$$

These ratios decrease in n , since increasing n tilts the radial density toward larger radii. Its norm is $(1 - e^{-\pi M^2})/(1 - e^{-\pi R^2}) < 1$. The constrained kernel is smaller. Kernel-order stochastic domination follows from [Lyons, Theorem 3.8](#): the kernels are locally trace-class positive contractions on the same area-measure space. Extend them by zero to the plane if the Borel set C is not itself locally compact.

For each bounded increasing local test, let $L \uparrow \infty$. The conditional expectations converge by the bounded martingale theorem. A countable determining class gives the asserted conditional-law domination. Truncating the count and then using monotone convergence proves (4.1). No inverse on the full exterior is taken.

4.2 Explicit clustered-zero bound

For a k -fold zero at p , interpreted as deletion of the first k derivative-evaluation modes, let $C \subset B(p, M)$ and $\rho = M/R < 1$. Orthogonality of the centered monomials gives

$$\int_D |z - p|^{2n} e^{-\pi|z-p|^2} dA \geq e^{-\pi R^2} \frac{\pi R^{2n+2}}{n+1}.$$

Summing the remaining modes $n \geq k$ yields

$$\boxed{\int_C B_D^{(k,p)}(z, z) dA \leq \frac{|C| e^{\pi R^2} (k+1) \rho^{2k}}{\pi R^2 (1 - \rho^2)^2}.} \quad (4.2)$$

For fixed k , distinct zeros tending to p converge to this derivative constraint: use divided differences of the evaluation vectors and convergence of their nonsingular Gram matrices. Thus sufficiently close distinct marks give (4.2) with any prescribed positive additive tolerance. No coincident actual points are asserted. These marks must still be supplied, with their probabilities, in a percolation argument.

5 Unbounded exterior Palm likelihood ratios

Theorem 5. Let $D = B(0, T)$, let $A \subset D$ be a fixed finite distinct set, and take distinct $q, z \in D \setminus A$ with $q^2 \neq z^2$. Let ν_q, ν_z be the restrictions to D^c of the canonical reduced Ginibre Palm laws at $A \cup \{q\}$ and $A \cup \{z\}$. Then

$$\nu_z \sim \nu_q, \quad \operatorname{ess\,sup}_{\nu_q} \frac{d\nu_z}{d\nu_q} = \infty, \quad \operatorname{ess\,inf}_{\nu_q} \frac{d\nu_z}{d\nu_q} = 0. \quad (5.1)$$

Hence no finite constant bounds either exterior law by the other. The restriction $q^2 \neq z^2$ suffices for this theorem; no assertion about the omitted symmetric case is needed here.

The published inputs are the radial conditional-density and same-order Palm-equivalence results of [Bufetov–Qiu, Proposition 1.1 and Corollary 1.6](#), and Ginibre number rigidity and fixed-count conditional tolerance from [Ghosh–Peres](#). The proof below explains the support construction and fixed-tuple passage.

5.1 5.1 Control of the full distant product

For $L > 2T$, take centered radial cutoffs in

$$g_{>L}(v) = \prod_{|x|>L} (1 - v/x), \quad |v| \leq T.$$

The analytic logarithm of each factor is unambiguous, and

$$\log g_{>L}(v) = -vS_1(L) - \frac{1}{2}v^2S_2(L) - \sum_{|x|>L} \sum_{j \geq 3} \frac{v^j}{jx^j}.$$

Here $S_1(L) = \lim_{U \rightarrow \infty} \sum_{L < |x| \leq U} x^{-1}$ requires care. It is not justified by an exterior $L^2(dA)$ bound on $1/x$, which diverges.

For a smooth compactly supported complex function f , the projection variance identity and the fundamental theorem of calculus give

$$\text{Var} \sum_{x \in G} f(x) = \frac{1}{2} \iint |f(x) - f(y)|^2 e^{-\pi|x-y|^2} dA(x) dA(y) \leq \frac{1}{2\pi} \int |\nabla f|^2 dA. \quad (5.2)$$

The displayed constant is a sufficient bound; no optimal constant is needed. Approximate the radial cutoff of $1/x$ by cutoffs changing over strips of width one at radii L, U . Their Dirichlet integrals are $O(L^{-1} + U^{-1})$. The difference from the sharp cutoff has $L^2(dA)$ norm squared of the same order; the elementary DPP variance bound applies to that difference. All means vanish by angular integration. Thus, for an absolute C ,

$$\mathbb{E}|S_1(L, U)|^2 \leq C/L, \quad U > L \geq 2.$$

Applying this to differences of outer cutoffs proves the radial L^2 limit. The elementary variance estimate also gives

$$\mathbb{E}|S_2(L)|^2 \leq \pi/L^2, \quad \mathbb{E} \sum_{|x|>L} |x|^{-3} = 2\pi/L.$$

As $\sum_{j \geq 3} |v/x|^j/j \leq (2/3)|v/x|^3$, it follows that

$$\mathbb{E} \sup_{|v| \leq T} |\log g_{>L}(v)| \leq \frac{C^{1/2}T}{\sqrt{L}} + \frac{\sqrt{\pi}T^2}{2L} + \frac{4\pi T^3}{3L} \rightarrow 0. \quad (5.3)$$

The normalized radial product in the BQ conditional-density formula has the same limit, by uniqueness of convergence in probability. There is no free unaccounted-for linear analytic factor.

5.2 5.2 Realizing a large quadratic exterior field

Fix an inside count n , an error $\varepsilon > 0$, and $M \geq 16\pi$. Choose L large. With positive probability both the product in (5.3) is within $\varepsilon/3$ of zero in logarithm and

$$\pi L^2/2 \leq N(B_L) \leq 2\pi L^2. \quad (5.4)$$

The product condition tends to probability one. The count condition does too, by its mean πL^2 and variance at most πL^2 . Both are measurable from the exterior of B_L ; this uses number rigidity for the count.

Condition on that exterior first. Its inside count is a fixed N . Set $J = \lfloor (N - n)/2 \rfloor$ and $d_J = \sqrt{J/M}$. Place n points in any positive-volume open set inside D , J points in a small

patch near $+id_J$, and J in a patch near $-id_J$. An odd remaining point goes near $3iL/4$. For large L , these patches lie outside D and inside B_L . Their template product is

$$(1 + Mv^2/J)^J \longrightarrow e^{Mv^2}$$

uniformly on $\overline{D}$, because the logarithmic error is at most M^2T^4/J . Taking patches small gives the same approximation for distinct actual points. The possible odd factor tends uniformly to one.

Fixed-count conditional tolerance assigns positive probability to each such open configuration set. There are only finitely many counts in (5.4), so the templates may depend measurably on N . Integrating proves the existence of a positive-probability set E_M of actual D -exteriors with rigid count n and

$$g_O(v) = \exp(M\omega v^2 + \eta_O(v)), \quad \sup_{\overline{D}} |\eta_O| < \varepsilon, \quad (5.5)$$

where any unit ω is obtained by rotating the patches. This ordering of conditioning is essential: the tail estimate has not been conditioned on a previously selected rare patch event.

5.3 5.3 Fixed Palm tuples, not just almost every tuple

For ordinary exterior O , BQ identifies the conditional interior process as a polynomial ensemble with kernel K_O , rank $N(O)$, and weight $e^{-\pi|v|^2}|g_O(v)|^2$. Its analytic range is contained in the weighted disk Bergman space, so $K_O \leq B_D$. Campbell disintegration initially gives, for almost every distinct m -tuple A ,

$$\frac{d\mu_{\text{out}}^A}{d\mu_{\text{out}}}(O) = \frac{\det K_O(A, A)}{\det H(A, A)}. \quad (5.6)$$

This identity holds for every fixed distinct tuple. Indeed, its right side is continuous in the tuple and dominated locally by products of disk Bergman diagonals. The canonical reduced Palm kernels depend locally uniformly on their distinct roots. Their correlation densities are bounded by one, so the absolutely convergent local Janossy series gives total-variation continuity on every bounded observation window. Equality for almost every tuple therefore extends to every fixed tuple on a countable determining class of exterior cylinder tests, and hence for the full exterior laws.

The same argument with a joint inside/exterior test identifies the conditional reduced interior law. The determinant-weighted finite polynomial-ensemble Palm integral is continuous, including the case of an empty remainder, and has the same deterministic domination. This supplies the needed disintegration without an uncountable intersection of unrelated null sets.

Apply the support construction with $n = m+1$. Polynomial interpolation gives $\det K_O(A, A) > 0$ on E_M for every fixed distinct A . Thus $\mu_{\text{out}}^A(E_M) > 0$. Under that law, the residual conditional interior process has one point with density

$$t_O(v) = \frac{1}{Z_O} |p_A(v)|^2 e^{-\pi|v|^2} |g_O(v)|^2, \quad p_A(v) = \prod_{a \in A} (v - a). \quad (5.7)$$

5.4 5.4 The likelihood ratio

Choose ω so $\Delta = \Re[\omega(z^2 - q^2)] > 0$. On E_M , (5.5)–(5.7) imply

$$\frac{t_O(z)}{t_O(q)} \geq \left| \frac{p_A(z)}{p_A(q)} \right|^2 e^{-\pi(|z|^2 - |q|^2)} e^{2M\Delta - 4\varepsilon}.$$

Divide the $(m + 1)$ -root version of (5.6) by the m -root version. The Schur determinant formula gives

$$\frac{d\nu_z}{d\nu_q}(O) = \frac{Q_A(q, q) t_O(z)}{Q_A(z, z) t_O(q)} \quad \text{on } E_M.$$

All fixed factors are positive. The event E_M has positive probability under both further Palm laws, since both conditional intensities are positive there. Increasing M proves unbounded essential supremum. Reversing ω proves essential infimum zero. Same-order Palm equivalence supplies the asserted mutual absolute continuity.

This theorem rules out a uniform bound over arbitrary exteriors. It provides no lower bound on the probability of the adverse exterior sets and no conclusion about their weight in a pivotal event.

6 Exact finite-ensemble deletion calculation and counterexample

Use the rank- n Ginibre projection generated by the first n Gaussian-weighted monomials, and the reduced Palm root $q = 0$. If X has $n - 1$ points, a deletion rule $w(G, z)$ reconstructs this Palm source from the rank- n ensemble precisely when its pointwise weights sum to one and, almost everywhere in X ,

$$\int e^{-\pi|z|^2} \prod_{x \in X} \frac{|z - x|^2}{|x|^2} w(X + z, z) dA(z) = 1. \quad (6.1)$$

This is the Campbell formula with the Vandermonde density ratio. The root kernel diagonal is one.

Proposition 6. In rank two the rule

$$w(\{x, z\}, z) = \frac{|x|^2}{|x|^2 + |z|^2}$$

is valid. In rank three, the normalized Lagrange-square rule is not valid.

For rank two, its conditional insertion density is

$$k(x, z) = e^{-\pi|z|^2} \frac{|z - x|^2}{|x|^2 + |z|^2}.$$

Angular averaging replaces $|z - x|^2$ by $|z|^2 + |x|^2$, so its integral is one. The two deletion weights also sum to one. Thus both marginals, not just an averaged normalization, are correct.

For the failed rule, put

$$L_i(0) = \prod_{j \neq i} \frac{-z_j}{z_i - z_j}, \quad S(G) = \sum_i |L_i(0)|^2, \quad w_L(G, z_i) = \frac{|L_i(0)|^2}{S(G)}.$$

The Vandermonde terms in (6.1) cancel, leaving the necessary condition

$$\int e^{-\pi|z|^2} \frac{1}{S(X + z)} dA(z) = 1. \quad (6.2)$$

In rank three, take $X = \{a, -a\}$, $a > 0$. Direct interpolation gives

$$S(\{a, -a, z\}) = \frac{a^4 + \frac{|z|^2}{2}(a^2 + |z|^2)}{|a^2 - z^2|^2}.$$

The angular mean of its reciprocal is

$$\frac{a^4 + t^4}{a^4 + \frac{1}{2}a^2t^2 + \frac{1}{2}t^4}, \quad t = |z|.$$

It tends to two for every $t > 0$ as $a \downarrow 0$. The interpolation identity $\sum_i L_i(0) = 1$ gives $1/S \leq 3$. Dominated convergence therefore makes the left side of (6.2) tend to two. Continuity in the distinct remainder points, with the same Gaussian majorant, makes the failure persist on an open set of positive Palm probability. This is not merely a coincidence example.

7 The ellipse route: large grains, the Poisson limit, and its scope

This section records the results obtained from the ellipse route, with their proofs. In particular, the critical-radius convergence theorem below is an established conclusion of the argument. It was omitted from one of the shorter summaries of this investigation; that omission was not a mathematical refutation. The theorem does not, however, order the undeformed Ginibre critical radius and the limiting Poisson critical radius.

Throughout this section, G has intensity one with respect to planar Lebesgue measure dA . Its kernel is

$$H(z, w) = \exp\left(\pi z\bar{w} - \frac{\pi}{2}|z|^2 - \frac{\pi}{2}|w|^2\right). \quad (7.1)$$

For a locally finite configuration ξ , write

$$\mathcal{O}_r(\xi) = \bigcup_{x \in \xi} \bar{B}(x, r), \quad \theta_\xi(r) = \mathbb{P}(0 \text{ belongs to an unbounded component of } \mathcal{O}_r(\xi))$$

when ξ denotes a random configuration, and set

$$r_c(\xi) = \inf\{r > 0 : \theta_\xi(r) > 0\}.$$

Thus r is the grain radius and two grains intersect when their centers are at distance at most $2r$. All probabilities in this section are ordinary probabilities at a fixed spatial point, not Palm probabilities. Open disks will be used for hole events and closed disks for grains. The intensity measures of the processes to which we apply the results are absolutely continuous, so a fixed circle contains no point almost surely.

Define the area-preserving deformation and its transformed process by

$$S_t(x, y) = (e^t x, e^{-t} y), \quad \Phi_t = S_t^{-1} G. \quad (7.2)$$

For area $A > 0$, put $r = \sqrt{A/\pi}$ and

$$E_{A,t} = S_t \bar{B}(0, r) = \{(x, y) : e^{-2t} x^2 + e^{2t} y^2 \leq A/\pi\}. \quad (7.3)$$

If $\theta_G(E)$ is the probability that the origin belongs to an unbounded component of $\bigcup_{x \in G} (x + E)$, then the deterministic homeomorphism S_t^{-1} gives

$$\theta_G(E_{A,t}) = \theta_{\Phi_t}(\sqrt{A/\pi}). \quad (7.4)$$

It preserves the origin and boundedness of components. Since $\det S_t = 1$, every Φ_t has intensity one.

7.1 A contour bound for bounded occupied components

The contour argument needs only a particular consequence of negative association. Say that a stationary point process Φ has the disjoint-hole inequality if, for pairwise disjoint bounded Borel sets $D_1, \dots, D_k$,

$$\mathbb{P} \left(\Phi \left(\bigcup_{j=1}^k D_j \right) = 0 \right) \leq \prod_{j=1}^k \mathbb{P}(\Phi(D_j) = 0). \quad (7.5)$$

For such a process, define

$$q_\Phi(r) = \mathbb{P}(\Phi(B(0, r)) = 0), \quad f_\Phi(r) = \mathbb{P}(0 \in \mathcal{O}_r(\Phi) \text{ and its component is bounded}).$$

Lemma 7 (Contour bound). *For every $\varepsilon \in (0, 1)$ there are constants $C_\varepsilon < \infty$ and $\eta_\varepsilon > 0$, independent of Φ and $r > 0$, such that a stationary locally finite process satisfying (7.5) obeys*

$$q_\Phi((1 - \varepsilon)r) \leq \eta_\varepsilon \implies f_\Phi(r) \leq C_\varepsilon q_\Phi((1 - \varepsilon)r)^2. \quad (7.6)$$

Proof. Fix $r > 0$ and use a translated square grid of mesh $h = \varepsilon r / \sqrt{2}$, with the origin at a square center. Suppose the origin is covered and its occupied component C is bounded. Every center of a grain belonging to C lies in C ; local finiteness therefore makes C a finite union of closed disks. In particular, C is compact.

Let Q be the union of all closed grid squares meeting C . There are finitely many. At every point of C , all incident grid squares are included, so $C \subset \text{int } Q$. The squares of Q are connected through shared edges. One way to see this is to follow a continuous path in the finite disk union C : transitions through square edges preserve edge connectivity, and transitions through a grid vertex include all four incident squares. The external boundary of Q can consequently be traced by a closed grid-edge walk surrounding the origin, with distinct edges, although vertices can be revisited. Let n be its length.

For every $p \in \partial C$,

$$\text{dist}(p, \Phi) \geq r. \quad (7.7)$$

Indeed, a grain whose interior contains p meets C and belongs to the same component; it would make p an interior point of C . Every vertex v of the external grid boundary is outside C and belongs to a square meeting C . Hence it is within distance $\sqrt{2}h$ of ∂C . The distance function to Φ is 1-Lipschitz, and (7.7) implies

$$\Phi(B(v, (1 - \varepsilon)r)) = 0. \quad (7.8)$$

The component contains a complete grain, so its enclosing contour has horizontal span at least $2r$. Choose the leftmost and rightmost contour vertices first. Their distance exceeds $2(1 - \varepsilon)r$. Extend this pair to a maximal collection of contour vertices separated by at least $2(1 - \varepsilon)r$. For a fixed contour this selection is made deterministically. There is an integer K_ε bounding the number of grid vertices in any disk of radius $2(1 - \varepsilon)r$, because the ratio of this radius to h depends only on ε . The contour has at least $n/2$ distinct vertices: its n distinct edges contribute $2n$ incidences and each grid vertex has degree at most four. Maximality of the selection thus gives at least

$$\max\{2, n/M_\varepsilon\}, \quad M_\varepsilon = 2K_\varepsilon,$$

selected vertices. Their open disks in (7.8) are pairwise disjoint. If $u = q_\Phi((1 - \varepsilon)r)$, stationarity and (7.5) bound the probability of all the empty disks required by this fixed contour by $u^{\max\{2, n/M_\varepsilon\}}$.

There are at most $C(n+1)^2 4^n$ length- n grid walks enclosing the origin, for an absolute constant C . In fact, their convex hull contains the origin, and their diameter is at most nh . Every possible starting vertex is therefore within distance nh of the origin; there are $O((n+1)^2)$ such vertices and at most 4^n step sequences from each. A union bound yields

$$f_\Phi(r) \leq C \sum_{n \geq 4} (n+1)^2 4^n u^{\max\{2, n/M_\varepsilon\}}. \quad (7.9)$$

For $n \leq 2M_\varepsilon$ this is bounded by a constant depending only on ε times u^2 . Writing $n = 2M_\varepsilon + k$ in the remaining terms and factoring out u^2 leaves a constant times

$$\sum_{k \geq 1} (2M_\varepsilon + k + 1)^2 (4u^{1/M_\varepsilon})^k.$$

This is uniformly bounded when $4u^{1/M_\varepsilon} \leq 1/2$. The choice $\eta_\varepsilon = 8^{-M_\varepsilon}$ is sufficient; the case $u = 0$ follows directly from the contour union bound. $\square$

Corollary 8 (Failure is asymptotically a hole). *Assume the hypotheses of Lemma 7 and*

$$\log q_\Phi(r) = -cr^4 + o(r^4), \quad c > 0. \quad (7.10)$$

Then, with $\log 0 = -\infty$,

$$\limsup_{r \rightarrow \infty} \frac{\log f_\Phi(r)}{r^4} \leq -2c, \quad f_\Phi(r) = o(q_\Phi(r)).$$

If Φ has no point on a fixed circle almost surely, then

$$1 - \theta_\Phi(r) \sim q_\Phi(r). \quad (7.11)$$

Proof. For each fixed $\varepsilon \in (0, 1)$, Lemma 7 and (7.10) give

$$\limsup_{r \rightarrow \infty} \frac{\log f_\Phi(r)}{r^4} \leq -2c(1 - \varepsilon)^4.$$

Taking the infimum of these valid upper bounds as $\varepsilon \downarrow 0$ proves the first assertion. More directly, fix any ε with $2(1 - \varepsilon)^4 > 1$. Then

$$\log \frac{f_\Phi(r)}{q_\Phi(r)} \leq -c(2(1 - \varepsilon)^4 - 1)r^4 + o(r^4) \longrightarrow -\infty.$$

No estimate uniform in ε is needed. The failure event is the disjoint union of the uncovered event and the event defining $f_\Phi(r)$. The circle-null condition makes the first probability equal to $q_\Phi(r)$ with the stated boundary conventions. $\square$

7.2 The fixed-eccentricity large-area comparison

The external determinantal input used here and below is [Lyons, Theorem 3.7](#): a locally trace-class positive contraction on a Radon measure space defines a negatively associated point process. The theorem applies to arbitrary bounded increasing functions supported on disjoint Borel sets. It also applies to decreasing functions by replacing both functions by their negatives. Iteration gives (7.5). The Ginibre projection and its images under invertible area-preserving linear transformations satisfy the operator hypotheses. Such transformations also preserve disjointness of the supporting sets.

The hole asymptotic is a published input, rather than a new result of this investigation. In the normalization of intensity $1/\pi$, [Adhikari–Reddy, Theorem 6 and Example 17](#) give, for fixed semiaxes $a, b > 0$,

$$\lim_{R \rightarrow \infty} \frac{1}{R^4} \log \mathbb{P} \left(G_{1/\pi} \left(\left\{ (x, y) : \frac{x^2}{R^2 a^2} + \frac{y^2}{R^2 b^2} < 1 \right\} \right) = 0 \right) = -\frac{(ab)^3}{2(a^2 + b^2)}. \quad (7.12)$$

Their domain theorem assumes an open domain in the unit disk and, in the case used here, an exterior-ball condition. An ellipse is convex and satisfies that condition. To remove the unit-disk restriction, choose $d > 0$ with $d \max(a, b) < 1$, use semiaxes da, db and scale R/d . The coefficient is homogeneous of degree four, so the factors d^4 cancel. The ellipse coefficient in (7.12) has been checked against their displayed Example 17; the normalization is not inferred from a disk formula alone.

Theorem 9 (Equal-area disks versus fixed noncircular ellipses). *For every fixed $t \in \mathbb{R}$, at unit intensity,*

$$1 - \theta_G(E_{A,t}) \sim \mathbb{P}(G(E_{A,t}) = 0), \quad (7.13)$$

$$\lim_{A \rightarrow \infty} \frac{-\log(1 - \theta_G(E_{A,t}))}{A^2} = \frac{1}{4 \cosh(2t)}. \quad (7.14)$$

For every fixed $t \neq 0$, there is $A_0(t) < \infty$ such that

$$A > A_0(t) \implies \theta_G(E_{A,0}) > \theta_G(E_{A,t}). \quad (7.15)$$

More precisely,

$$\lim_{A \rightarrow \infty} \frac{1}{A^2} \log \frac{1 - \theta_G(E_{A,t})}{1 - \theta_G(E_{A,0})} = \frac{1}{4} \left(1 - \frac{1}{\cosh(2t)} \right). \quad (7.16)$$

Proof. The dilation $x \mapsto \sqrt{\pi}x$ maps unit-intensity Ginibre to the intensity- $1/\pi$ normalization. Apply (7.12) with $a = e^t$, $b = e^{-t}$, and $R = \sqrt{\pi}r$. Since $ab = 1$ and $a^2 + b^2 = 2 \cosh(2t)$, this gives

$$\log q_{\Phi_t}(r) = -\frac{\pi^2 r^4}{4 \cosh(2t)} + o_t(r^4). \quad (7.17)$$

The process Φ_t is stationary and satisfies the disjoint-hole inequality, and its intensity measure charges no fixed circle. Corollary 8 applies with $c = \pi^2/(4 \cosh(2t))$. The affine equivalence (7.4), and $A = \pi r^2$, prove (7.13) and (7.14). Subtracting the two logarithmic asymptotics gives (7.16). For fixed $t \neq 0$, its right side is strictly positive, so the failure-probability ratio eventually exceeds one. This proves (7.15) and identifies exactly where strictness enters. $\square$

Remark 10 (Intensity scaling and quantifiers). For Ginibre intensity $\lambda > 0$, replace A^2 in the denominators of (7.14) and (7.16) by $\lambda^2 A^2$: dilation by $\sqrt{\pi\lambda}$ reduces the claim to (7.12). The eccentricity is fixed before $A \rightarrow \infty$. Neither the error term nor $A_0(t)$ has been made uniform as $t \rightarrow 0$ or $|t| \rightarrow \infty$. A comparison at sufficiently large area does not compare the areas where percolation first begins.

7.3 Exact kernel and local convergence in total variation

We now let $t \rightarrow +\infty$. Write $z = x + iy$ and $w = x' + iy'$. Because (7.2) preserves area, the kernel of Φ_t with respect to dA is $K_t(z, w) = H(S_t z, S_t w)$. Expanding the exponent gives

$$K_t(z, w) = \exp \left\{ -\frac{\pi}{2} \left[e^{2t}(x - x')^2 + e^{-2t}(y - y')^2 \right] + i\pi(yx' - xy') \right\}. \quad (7.18)$$

In particular, the phase is unchanged. Let W be a bounded Borel window of area m , contained in a horizontal strip of height H_W . On $L^2(W, dA)$ let A_t be the integral operator with kernel $|K_t(z, w)|$, and set

$$a_t = \sqrt{2}H_W e^{-t}, \quad s_t = \int_{W^2} |K_t(z, w)|^2 dA(z)dA(w). \quad (7.19)$$

The absolute kernel is a positive definite Gaussian convolution kernel on the plane, compressed to W . Thus A_t is positive and trace class, with trace m . Direct integration and Schur's bound give

$$\|A_t\| \leq a_t, \quad \text{Tr}(A_t^2) = s_t \leq mH_W e^{-t}. \quad (7.20)$$

For example, the first estimate follows by integrating the horizontal Gaussian over the whole line and bounding the vertical integral by H_W . Positivity and the spectral theorem then yield

$$\text{Tr}(A_t^k) \leq a_t^{k-2} s_t \quad (k \geq 2). \quad (7.21)$$

Proposition 11 (Quantitative local Poisson approximation). *For $2a_t < 1$,*

$$d_{\text{TV}}(\mathcal{L}(\Phi_t|_W), \mathcal{L}(\Pi_1|_W)) \leq \frac{e^{2m}}{2} \left[\exp\left(\frac{2s_t}{1-2a_t}\right) - 1 \right]. \quad (7.22)$$

Consequently the restrictions converge in total variation for every fixed bounded W .

Proof. Write $\rho_n(z_1, \dots, z_n) = \det[K_t(z_i, z_j)]_{i,j=1}^n$, and put $\rho_0 = 1$. The diagonal is one, so Hadamard's inequality gives $0 \leq \rho_n \leq 1$. In the determinant expansion the identity permutation contributes one. Bound all other terms in absolute value, integrate over W^n , and group permutations by cycle type. A fixed point contributes m and a cycle of length $k \geq 2$ contributes $\text{Tr}(A_t^k)$. The exponential generating function for permutations with these cycle weights gives, for $ua_t < 1$,

$$\begin{aligned} & \sum_{n \geq 0} \frac{u^n}{n!} \int_{W^n} |\rho_n - 1| dA^n \\ & \leq e^{um} \left[\exp\left(\sum_{k \geq 2} \frac{u^k}{k} \text{Tr}(A_t^k)\right) - 1 \right]. \end{aligned} \quad (7.23)$$

For clarity, the exponential arises because j_k cycles of length k have combinatorial weight $1/(k^{j_k} j_k!)$. Every term in this majorant is nonnegative, so all regroupings are justified by Tonelli's theorem. By (7.21),

$$\sum_{k \geq 2} \frac{2^k}{k} \text{Tr}(A_t^k) \leq 2s_t \sum_{j \geq 0} (2a_t)^j = \frac{2s_t}{1-2a_t}.$$

The Janossy density for exactly n points in W is

$$j_n(z_1, \dots, z_n) = \sum_{k \geq 0} \frac{(-1)^k}{k!} \int_{W^k} \rho_{n+k}(z_1, \dots, z_n, w_1, \dots, w_k) dA^k(w). \quad (7.24)$$

The bound $0 \leq \rho_j \leq 1$ makes this inclusion–exclusion series absolutely convergent, also after integration over the first n coordinates. The Poisson Janossy density is e^{-m} , obtained by

replacing every ρ_j by one. Hence

$$\begin{aligned} 2d_{\text{TV}}(\Phi_t|_W, \Pi_1|_W) &= \sum_{n \geq 0} \frac{1}{n!} \int_{W^n} |j_n - e^{-m}| dA^n \\ &\leq \sum_{n,k \geq 0} \frac{1}{n!k!} \int_{W^{n+k}} |\rho_{n+k} - 1| dA^{n+k} \\ &= \sum_{j \geq 0} \frac{2^j}{j!} \int_{W^j} |\rho_j - 1| dA^j. \end{aligned}$$

Apply (7.23) at $u = 2$. Finally $a_t, s_t \rightarrow 0$ for fixed W , proving convergence. $\square$

Remark 12. This approximation holds for every measurable event in a fixed window, including closed-grain crossing events. No continuity-set argument or tangency exception is needed for the transfer of such event probabilities. The factor e^{2m} is kept explicitly: the proposition is not a uniform approximation on windows growing arbitrarily with t .

7.4 Two finite-size criteria using negative association

The next arguments supply the passage from local convergence to infinite-volume percolation. Neither a bound on a mixing length nor isotropy of the approximating processes is used.

Fix $r > 0$ and $L > 2r$, and tile the plane by closed squares

$$S_v = L(v + [-1/2, 1/2]^2), \quad v \in \mathbb{Z}^2.$$

Overlapping square boundaries cause no ambiguity in the event definitions. Supports of events within each color class below are separated by a positive distance.

Lemma 13 (Upper finite-size criterion). *There is an absolute $b_0 > 0$ with the following property. Let Φ be stationary, locally finite and negatively associated. For each nearest-neighbor edge $e = \{v, w\}$ of $\mathbb{Z}^2$, let $R_e = S_v \cup S_w$ and declare e good if the occupied set has a long-direction crossing of R_e and a transverse crossing of each of S_v, S_w . Crossings are paths restricted to the indicated rectangles. If both orientations of edges have bad probability at most $b < b_0$, then $\mathcal{O}_r(\Phi)$ has an unbounded component with positive probability.*

Proof. All three crossing events are increasing. Since the grain radius is the deterministic number r , the good-edge event depends only on centers in $R_e + \overline{B}(0, r)$; centers outside this set cannot contribute even to a path on the rectangle boundary.

First check deterministic gluing. A long crossing of R_e contains a crossing of each endpoint square in the same direction: take the relevant subpath between the exterior side and the middle side. For two collinear good edges, the transverse crossing of their common square intersects the square crossings supplied by both long witnesses. For two good edges making a turn, the long witnesses themselves supply horizontal and vertical crossings of the common square, and these intersect. The fact that horizontal and vertical paths in a closed rectangle intersect follows from planar separation; it does not require independence or uniqueness of crossings. Thus witnesses for any connected family of good edges belong to one occupied component. If the edge family is infinite, these witnesses are spatially unbounded.

Color the lattice edges by their orientation and the coordinates modulo 5 of their canonical initial vertex (the endpoint with smaller coordinate in the edge direction). There are 50 colors. Two different edges of a color are separated by at least $5L$ in one coordinate before the sizes of their rectangles are taken into account. A rectangle has side lengths $2L, L$; after expansion

by r , both side lengths remain below $3L$, since $2r < L$. Thus expanded supports are disjoint within each color.

A prescribed set of n bad edges contains at least $n/50$ edges of one color. Their events are decreasing and supported on disjoint sets. Negative association, iterated over these sets, gives

$$\mathbb{P}(\text{all prescribed edges are bad}) \leq b^{n/50}. \quad (7.25)$$

If the good-edge cluster of the lattice origin is finite, its outer edge boundary contains a simple dual circuit of bad edges surrounding the origin. The number of length- n possible circuits is at most $C(n+1)^2 4^n$: choose a starting vertex within distance $O(n)$ of the origin, then a sequence of nearest-neighbor steps. Consequently

$$\mathbb{P}(\text{the origin's good-edge cluster is finite}) \leq C \sum_{n \geq 4} (n+1)^2 (4b^{1/50})^n. \quad (7.26)$$

Choose an absolute $b_0 > 0$ for which the series is less than one. The probability of an infinite good-edge cluster is then positive; the gluing argument proves the stated occupied percolation. $\square$

Lemma 14 (Lower finite-size criterion). *Under the same stationarity, local finiteness and negative association assumptions, let H_v be the event that $\mathcal{O}_r(\Phi)$ has a path from S_v to the boundary of the concentric square*

$$T_v = L(v + [-3/2, 3/2]^2),$$

with its path contained in T_v . If

$$p = \mathbb{P}(H_v) < 8^{-25}, \quad (7.27)$$

then almost surely $\mathcal{O}_r(\Phi)$ has no unbounded component.

Proof. The event H_v is increasing and depends only on centers in $T_v + \overline{B}(0, r)$. Color vertices v by their two coordinates modulo 5. These supports have side lengths $3L + 2r < 4L$, and hence are disjoint within each of the 25 colors. For any prescribed n distinct vertices, negative association implies that the probability that all their H_v events hold is at most $p^{n/25}$.

If an unbounded occupied component exists, the tiles it meets form an infinite connected set for star adjacency (squares may meet at an edge or a vertex). The finite-degree lattice graph then contains an infinite self-avoiding star path in that set. Every tile on that path has H_v : follow an occupied path from the component's intersection with S_v until its first meeting with ∂T_v . Such finite occupied paths exist because a locally finite union of fixed-radius disks has path-connected components. There are at most 8^n candidate n -vertex star paths from a specified tile. Thus the probability of an infinite such path from that tile is bounded, for every n , by

$$8^n p^{n/25} = (8p^{1/25})^n \longrightarrow 0.$$

Take a countable union over starting tiles. $\square$

Remark 15 (Existence and origin membership). For a stationary process with fixed radius $r > 0$, positive probability of some unbounded occupied component is equivalent to positive probability of origin membership in such a component. Indeed, an unbounded disk component contains the interior of a disk and hence a rational point. The existence event is contained in the countable union of the events that a specified rational point belongs to an unbounded component. If the union has positive probability, one event does; stationarity makes its probability equal to the origin probability. The converse is immediate. No ergodicity assertion is needed for this equivalence.

7.5 Convergence of the actual critical radii

The Poisson input is [Ahlberg–Tassion–Teixeira, Theorem 1.1\(i\),\(iii\)](#). For a radius distribution with finite second moment, fixed-aspect-ratio occupied rectangle crossings tend to one above the Poisson critical intensity and to zero below it. The deterministic unit-radius distribution meets that hypothesis. Dilation $x \mapsto x/r$ changes intensity-one radius- r disks into intensity- r^2 unit disks. Thus their theorem gives precisely the off-critical crossing limits needed at a fixed radius r on either side of $r_c(\Pi_1)$.

Theorem 16 (A local-convergence criterion for critical radii). *Let (Ξ_n) be stationary, locally finite, negatively associated planar point processes. Suppose that for every bounded Borel set W ,*

$$d_{\text{TV}}(\mathcal{L}(\Xi_n|_W), \mathcal{L}(\Pi_1|_W)) \longrightarrow 0.$$

For deterministic disk grains,

$$r_c(\Xi_n) \longrightarrow r_c(\Pi_1). \quad (7.28)$$

Proof. Write $r_P = r_c(\Pi_1)$. First fix $r > r_P$. The cited Poisson crossing limit, and a union bound for the three crossings defining a good edge, allow one to choose a finite $L > 2r$ for which the Poisson bad-edge probability is below $b_0/2$. Rotation invariance gives this for both orientations. Now keep r and L fixed. Each good-edge event depends on one fixed bounded expanded rectangle. Local total-variation convergence makes both bad-edge probabilities below b_0 for all sufficiently large n . Lemma 13 and Remark 15 give percolation at radius r . Thus $r_c(\Xi_n) \leq r$ eventually. Since this holds for every $r > r_P$,

$$\limsup_{n \rightarrow \infty} r_c(\Xi_n) \leq r_P.$$

Next fix $0 < r < r_P$. An H_v path implies a short-direction crossing of one of four strips of dimensions $3L$ by L . For example, translate v to zero and suppose the path first reaches the top side of T_0 at height $3L/2$. Its subpath after its last visit to height $L/2$ stays in $[-3L/2, 3L/2] \times [L/2, 3L/2]$ and crosses that strip vertically. The other first-exit sides give the other three strips. Consequently the subcritical Poisson crossing limit and a union bound give $\mathbb{P}_{\Pi_1}(H_0) \rightarrow 0$ as $L \rightarrow \infty$. Choose one finite $L > 2r$ with

$$\mathbb{P}_{\Pi_1}(H_0) < \frac{1}{2}8^{-25}.$$

For this fixed choice, local total-variation convergence gives $\mathbb{P}_{\Xi_n}(H_0) < 8^{-25}$ for sufficiently large n . Lemma 14 proves nonpercolation at r . Monotonicity in the grain radius then gives $r_c(\Xi_n) \geq r$. Letting $r \uparrow r_P$ yields

$$\liminf_{n \rightarrow \infty} r_c(\Xi_n) \geq r_P.$$

This proves (7.28). □

Corollary 17 (The anisotropic Ginibre threshold limit). *For (7.2),*

$$\boxed{\lim_{t \rightarrow +\infty} r_c(S_t^{-1}G) = r_c(\Pi_1)}. \quad (7.29)$$

Equivalently, the critical areas of equally oriented Ginibre ellipse grains of shape t converge to the Poisson disk critical area $\pi r_c(\Pi_1)^2$.

Proof. Each Φ_t is stationary, locally finite and negatively associated by the determinantal input stated above. Proposition 11 supplies local total-variation convergence. Apply Theorem 16 along every sequence $t_n \rightarrow +\infty$; the sequential characterization of a real limit proves (7.29). Finally use (7.4) and $A = \pi r^2$. □

Remark 18 (The order of choices). The proof fixes a radius strictly away from the Poisson threshold, then one finite block size, and only then sends the deformation parameter to infinity. Infinite-volume percolation is established by the finite-size criteria, not by exchanging an infinite-volume limit with local convergence. This qualitative proof alone supplies neither a rate for (7.29) nor a sign for $r_c(\Phi_t) - r_c(\Pi_1)$. The separate recovered rate in Section 7.8 uses an additional near-critical input.

7.6 A joint large-area and large-deformation regime

The fixed- t quartic hole exponent is not uniform in the deformation parameter. The following direct Fredholm estimate describes a different joint regime and makes that distinction quantitative.

Proposition 19 (Thin-ellipse hole regime). *Let $q_t(r) = \mathbb{P}(\Phi_t(B(0, r)) = 0)$. If $2\sqrt{2}re^{-t} < 1$, then*

$$0 \leq -\log q_t(r) - \pi r^2 \leq \frac{\pi r^3 e^{-t}}{1 - 2\sqrt{2}re^{-t}}. \quad (7.30)$$

Along every sequence with $r \rightarrow \infty$ and $re^{-t} \rightarrow 0$,

$$\frac{-\log q_t(r)}{\pi r^2} \rightarrow 1, \quad 1 - \theta_{\Phi_t}(r) \sim q_t(r). \quad (7.31)$$

Proof. Let T be the compression of K_t to $B(0, r)$. It is a positive contraction with trace πr^2 . Use (7.20) with $m = \pi r^2$ and $H_W = 2r$. Taking the absolute kernel in the Schur estimate gives

$$\|T\| \leq 2\sqrt{2}re^{-t}, \quad \text{Tr}(T^2) \leq 2\pi r^3 e^{-t}.$$

The Fredholm determinant hole formula and the spectral theorem give, under the displayed norm restriction,

$$\begin{aligned} -\log q_t(r) - \pi r^2 &= \sum_{k \geq 2} \frac{\text{Tr}(T^k)}{k} \\ &\leq \frac{\text{Tr}(T^2)}{2(1 - \|T\|)}. \end{aligned}$$

The series is absolutely convergent; all traces are nonnegative. This proves (7.30) and the first limit.

Fix $\varepsilon \in (0, 1)$ with $2(1 - \varepsilon)^2 > 1$. The same estimate applies at $(1 - \varepsilon)r$ along the given sequence. The constants in Lemma 7 are uniform over the processes Φ_t , so for all sufficiently large indices it gives

$$\log \frac{f_{\Phi_t}(r)}{q_t(r)} \leq \log C_\varepsilon - \pi(2(1 - \varepsilon)^2 - 1)r^2 + o(r^2) \rightarrow -\infty.$$

The hole and finite-component decomposition proves the second assertion in (7.31). $\square$

Remark 20. At fixed t , (7.17) has an exponent proportional to r^4 . In Proposition 19, the exponent is asymptotic to πr^2 . The latter regime makes the minor semiaxis re^{-t} tend to zero while the area tends to infinity. Both assertions concern large grains; neither by itself is a near-critical comparison.

7.7 The first finite-window perturbation at the Poisson end

This calculation retains the geometry of the leading interaction. Unlike the isotropically concentrated interaction in a thinned Ginibre limit, the anisotropic interaction concentrates only the horizontal separation of two insertion locations.

For a bounded measurable functional F determined by the points in a fixed bounded window W , define

$$D_z F(\xi) = F(\xi \cup \{z\}) - F(\xi), \quad D_{z_1, \dots, z_n} F = D_{z_1} \cdots D_{z_n} F.$$

Insertion differences are zero outside W . Coincident insertion locations form a null set in the integrals below and cause no ambiguity. Keep m, A_t, a_t, s_t from (7.19)–(7.20), and put

$$B_t = \sum_{k \geq 2} \frac{2^k}{k} \text{Tr}(A_t^k). \quad (7.32)$$

For fixed W , $B_t = 2s_t + O_W(e^{-2t})$ and $B_t \leq 2s_t/(1 - 2a_t)$ when $2a_t < 1$.

Proposition 21 (Exact leading local correction). *For $2a_t < 1$,*

$$\mathbb{E}F(\Phi_t) = \mathbb{E}F(\Pi_1) - \frac{1}{2} \int_{W^2} |K_t(z, w)|^2 \mathbb{E}[D_z D_w F(\Pi_1)] dA(z) dA(w) + R_t, \quad (7.33)$$

$$|R_t| \leq \|F\|_\infty e^{2m} (e^{B_t} - 1 - 2s_t) = O_W(\|F\|_\infty e^{-2t}). \quad (7.34)$$

This is a fixed-window statement, with no asserted uniformity over growing percolation boxes.

Proof. For every finite configuration $\eta \subset W$, the elementary subset expansion is

$$F(\eta) = \sum_{X \subseteq \eta} D_X F(\emptyset).$$

Since $|D_X F| \leq 2^{|X|} \|F\|_\infty$ and the factorial densities are bounded by one, Campbell's formula and absolute summability give the Newton expansion

$$\mathbb{E}F(\Phi_t) = \sum_{n \geq 0} \frac{1}{n!} \int_{W^n} \rho_n(Z) D_Z F(\emptyset) dA^n(Z). \quad (7.35)$$

The sum of the absolute integrals is at most $\|F\|_\infty e^{2m}$. The same formula holds for Poisson with $\rho_n = 1$.

Expand each determinant in (7.35). The identity permutation gives $\mathbb{E}F(\Pi_1)$. A permutation with exactly one transposition (i, j) and all remaining coordinates fixed contributes $-|K_t(z_i, z_j)|^2$. By symmetry and $\binom{n}{2}/n! = 1/(2(n-2)!)$, summing these terms gives

$$-\frac{1}{2} \int_{W^2} |K_t(z, w)|^2 \sum_{k \geq 0} \frac{1}{k!} \int_{W^k} D_{z, w, u_1, \dots, u_k} F(\emptyset) dA^k(u) dA(z) dA(w).$$

The inner sum is the Poisson Newton expansion of $\mathbb{E}D_z D_w F(\Pi_1)$.

All remaining permutations contain a cycle of length at least three or at least two transpositions. The same absolute cycle majorant as in (7.23), now multiplied by $\|F\|_\infty$, bounds their total contribution by $\|F\|_\infty e^{2m} (e^{B_t} - 1 - 2s_t)$. The subtraction $2s_t$ is exactly the majorant weight of a single transposition at $u = 2$. Finally,

$$0 \leq B_t - 2s_t \leq 2s_t \frac{2a_t}{1 - 2a_t} = O_W(e^{-2t}), \quad s_t = O_W(e^{-t}),$$

so $e^{B_t} - 1 - 2s_t = O_W(e^{-2t})$. Every interchange above is covered by these absolute majorants. $\square$

Proposition 22 (The vertical-pair coefficient for a rectangle crossing). *Fix $r > 0$ and a nondegenerate closed rectangle $R = I_R \times J_R$. Let F indicate a left-to-right crossing of R by radius- r occupied disks, the crossing path being restricted to R . Choose a closed rectangle $W = I_W \times J_W$ containing $R + \overline{B}(0, r)$. Then*

$$\begin{aligned} \lim_{t \rightarrow \infty} e^t (\mathbb{E}F(\Phi_t) - \mathbb{E}F(\Pi_1)) \\ = -\frac{1}{2} \int_{I_W} \int_{J_W} \int_{J_W} \mathbb{E} \left[D_{(a,b)} D_{(a,c)} F(\Pi_1) \right] dc db da. \end{aligned} \quad (7.36)$$

The coefficient is a finite signed integral. No assertion about its sign is made.

Proof. Write $h(z, w) = \mathbb{E}[D_z D_w F(\Pi_1)]$, taking $h = 0$ if an insertion is outside the determining window. The only point requiring care beyond Proposition 21 is the existence of the horizontal trace of h at equal horizontal coordinates. We prove that for Lebesgue-almost every (a, b, c) ,

$$h((a, b), (a + \delta, c)) \longrightarrow h((a, b), (a, c)) \quad (\delta \rightarrow 0). \quad (7.37)$$

Conditional on the finite Poisson configuration in W , represent the crossing by the finite intersection graph of clipped convex grains $C_u = \overline{B}(u, r) \cap R$, with additional marks indicating intersection with the left and right sides of R . The event holds exactly when this graph connects the two marks. Its incidences are locally stable in the insertion coordinates away from the following contact equalities.

For one fixed center u with $C_u \neq \emptyset$, contact between its clipped grain and a varying disk centered at v is governed by membership in $C_u + \overline{B}(0, r)$. Its boundary is the boundary of a convex body with nonempty interior and has planar Lebesgue measure zero. The same observation applies to meeting a marked side segment, and to appearance or disappearance of a clipped grain, using respectively the segment or R in place of C_u . These facts, with Fubini's theorem, exclude all relevant contacts between inserted and Poisson grains for almost every (a, b, c) and almost every Poisson configuration. Existing grains do not move; any contact among them is in any case a null event under their joint densities.

It remains to check contact between the two inserted clipped grains when their horizontal coordinates coincide. Put $d = |b - c|/2$ and $h_0 = (b + c)/2$. Their intersection is nonempty exactly when

$$\text{dist}(a, I_R)^2 + (d + \text{dist}(h_0, J_R))^2 \leq r^2. \quad (7.38)$$

Indeed, for $(x, y) \in R$, the maximum of its squared distances to (a, b) and (a, c) is $(x - a)^2 + (d + |y - h_0|)^2$, whose minimum over the rectangle is the left side of (7.38). On each of the finitely many regions specifying the nearest points in the two intervals and the sign of $b - c$, equality in (7.38) is a nontrivial polynomial equation. Its solution set has three-dimensional Lebesgue measure zero. Coincident centers $b = c$ also form a null set.

Outside these exceptional sets every relevant graph incidence is stable under a sufficiently small horizontal displacement of the second inserted center. Therefore all four Boolean terms defining $D_z D_w F$ stabilize for almost every fixed background and almost every triple. The difference is bounded, so conditional Poisson integration and dominated convergence prove (7.37).

Now use (7.18) in the integral of (7.33) and substitute $x = a$, $x' = a + e^{-t}u$, $y = b$, $y' = c$. After multiplication by e^t , its integrand becomes

$$e^{-\pi u^2} e^{-\pi e^{-2t}(b-c)^2} h((a, b), (a + e^{-t}u, c)),$$

with the window indicator also imposed. For almost every (a, b, c, u) , (7.37) supplies the limit. A constant multiple of $e^{-\pi u^2}$, with (a, b, c) restricted to a fixed bounded box, is an integrable

dominating function. The window boundary contributes a null set. Since $\int_{\mathbb{R}} e^{-\pi u^2} du = 1$, dominated convergence gives the right side of (7.36). The remainder vanishes after multiplication by e^t by (7.34). $\square$

Remark 23 (The remaining sign is a geometric question). For an increasing Boolean event, $D_z D_w F$ equals -1 when the background fails but either insertion separately succeeds; it equals $+1$ when the background and both single insertions fail but the pair succeeds; in all other cases it is zero. Thus the minus sign in (7.36) gives a positive contribution from redundant individually sufficient insertions and a negative contribution from cooperative pairs. The two vertical coordinates b, c need not coincide. Consequently the repeated-insertion identity $D_z^2 F = -D_z F$ does not determine this coefficient. A sign estimate uniform over the boxes needed near criticality has not been proved.

7.8 A recovered quantitative rate for the stretching limit

The source `ellipse_critical_variation.md`, with `ellipse_critical_variation_audit.md`, contains a stronger conclusion than the qualitative limit above. We restore it as a recovered argument, with its additional published input explicit. Write $c(t) = r_c(S_t^{-1}G)$. The recorded estimate is

$$|c(t) - r_P| \leq Ct^{-\eta/2} \quad (t \geq t_0) \quad (7.39)$$

for some $C, t_0 < \infty$ and $\eta > 0$.

For unit-radius Poisson disks, Manolescu–Santoro, Theorem 2.4 and equation (2.16) give a transition window of order $[n^2 \pi_4(n)]^{-1}$ and a polynomial lower bound $\pi_4(n) \geq cn^{-2+\eta}$ on the critical four-arm probability. The critical specialization of the latter follows by fixing n , approaching critical intensity, and using finite-window continuity and critical RSW. The authors give a proof sketch of this near-critical input; this report uses their stated result rather than supplying a replacement proof.

For windows of diameter $O(n)$, (7.22) is at most $Cn^3 \exp(C_0 n^2 - t)$ once its norm condition holds. Choose $n = \lfloor a_0 \sqrt{t} \rfloor$, with $C_0 a_0^2 < 1/2$. The local error then tends to zero exponentially. Choose the Poisson crossing tolerance below the two fixed tolerances in the upper and lower negative-association criteria. Intensity deviations of order $n^{-\eta}$ give the required Poisson block tests; the scaling $\lambda = r^2$ converts them to radius deviations of the same order near $r_P > 0$. Transfer these tests to Φ_t in total variation and apply the criteria. This gives (7.39). The scale is selected before the local comparison; no unrestricted exchange of a growing-window and deformation limit is used. The rate has no asserted sign.

7.9 Recovered orientation and critical-radius variation identities

Use the center-terminal convention of `ellipse_critical_variation.md`: in the fixed original configuration G , connect a center in B_m to a center outside the open B_L using edges $|S_{-t}(p - q)| \leq 2r$. The terminal labels do not move with t or r . For $|t| \leq T$ and $r \leq R$, the event is local in B_{L+2ReT} , by stopping a witness at its first center outside B_L . Let $U_{m,L}(t)$ be its minimax threshold capped at R , assigning R when no path exists below that cap.

For an edge of weight $w(t) = |S_{-t}v|/2$ and transformed angle θ_t ,

$$w' = -w \cos(2\theta_t), \quad w'' = w[2 - \cos^2(2\theta_t)].$$

At each fixed t , distinct pair weights tie only on Lebesgue-null hypersurfaces. Thus the active edge below the cap is almost surely unique, with angle $\Theta_{m,L}(t)$, and

$$U' = -U \cos(2\Theta) \mathbf{1}_{\{U < R\}} \quad (7.40)$$

almost everywhere. The capped threshold is R -Lipschitz. Absolute continuity and Fubini justify integration of (7.40) before taking expectations.

If $\mu_t(dr, d\theta)$ is the joint law of the uncapped threshold and active angle, the finite-window crossing probability satisfies the distributional contact identity

$$\partial_t C(t, r) = rM(t, r)\partial_r C(t, r), \quad M(t, r) = \mathbb{E}_{\mu_t}[\cos(2\Theta) \mid U = r], \quad |M| \leq 1,$$

where the conditional mean is understood relative to the radius contact measure. This identity supplies no favorable sign.

Deterministic grain inclusion gives $e^{-|t-s|}c(s) \leq c(t) \leq e^{|t-s|}c(s)$. Choose $R > \sup_{|t| \leq T} c(t)$ and a slow diagonal of seeds and outer radii valid for rational t and noncritical rational radii. The seed exhaustion gives convergence of capped thresholds in probability and mean at rational t . The common Lipschitz bound upgrades it to $\sup_{|t| \leq T} \mathbb{E}|U_n(t) - c(t)| \rightarrow 0$. Passing only the endpoint expectations through the integrated identity therefore gives the recovered exact representation

$$c(T) - c(0) = -\lim_n \int_0^T \mathbb{E}[U_n(t) \mathbf{1}_{\{U_n(t) < R\}} \cos(2\Theta_n(t))] dt. \quad (7.41)$$

For each fixed $\delta > 0$, contributions with $|U_n(t) - c(t)| > \delta$ vanish in absolute value after integration. No pointwise limit of angular contact measures is assumed. Evenness of c does not exclude a cusp at zero.

Combining (7.41) with the recovered rate, an orientation integral with limit at most $-CT^{-\eta/2} - \delta$ would give $c(0) \leq r_P - \delta$. This states a sufficient quantitative sign estimate; that estimate remains unproved.

7.10 Switching destroys the proposed convexity argument

The finite calculation in `ellipse_minmax_switching.md` concerns $T(t) = \min_\gamma \max_{e \in \gamma} |S_t v_e|/2$ on a finite Euclidean graph with fixed terminal vertices. On intervals with one active weight, $T = w_e$ and $w_e'' > 0$. At a switch, however,

$$T'' = w_{\text{active}}''(t) dt + \sum_j [T'(t_j+) - T'(t_j-)] \delta_{t_j}.$$

Contract edges strictly below a tied threshold and discard those strictly above it. Giving each remaining critical edge slope v_e , the two directional derivatives are

$$T'(t_0+) = \min_\gamma \max_{e \in \gamma} v_e, \quad T'(t_0-) = \max_\gamma \min_{e \in \gamma} v_e.$$

These follow by expanding finitely many edge weights to first order. Series alternatives give positive switching atoms; parallel alternatives give negative ones.

Both occur with all Euclidean edges present. The points $(0, 0), (1, 0), (1, 1)$, with the first and last as terminals, give $T(t) = e^{|t|}/2$ near zero and atom $+1$. For source $(0, 0)$, target $(1, 0)$, and additional points $(0, 1/10), (3/5, 9/10)$, the two short within-cluster edges leave two competing unit-length bridges. Their slopes are $1/2$ and $-7/50$, giving atom $-16/25$. After a common uniform rotation, the right derivatives of the mean thresholds are respectively $1/\pi$ and $-4/(5\pi)$.

The negative example can be smoothed: replace $(1, 0)$ by $(e^u, 0)$, sample u from a smooth density h supported in a sufficiently small interval about zero, and then rotate uniformly. Splitting the u -integral at the moving equality of the two bridge weights yields

$$M'(0) = 0, \quad M''(0) = \frac{3}{2}M(0) - \frac{16}{25}h(0). \quad (7.42)$$

Since $M(0) \leq 1/2$, taking $h(0) > 75/64$ makes the curvature negative. The moving-boundary term in the second derivative is essential; ties at a fixed time have probability zero. The analogous smoothed series example has $M''(0) = \frac{3}{2}M(0) + h(0)$. Thus neither isotropy nor positive edge curvature implies convexity of an averaged connecting threshold. These finite-law examples do not determine the Ginibre critical-radius curvature.

7.11 Riesz coefficients and the two-point Palm angular response

Further recovered calculations in `ellipse_critical_variation.md`, Sections 6–7, distinguish two finite approximations. For a fixed rotation-invariant local event in the transformed process, isotropy rewrites the covertical Poisson coefficient as

$$\lim_{t \rightarrow \infty} e^t [\mathbb{E}F(\Phi_t) - \mathbb{E}F(\Pi_1)] = -\frac{1}{2\pi} \iint \frac{\mathbb{E}_\Pi D_x D_y F}{|x - y|} dx dy. \quad (7.43)$$

The kernel is locally integrable and insertion support is bounded. This is not repeated insertion at one point.

For two disjoint disks of radius a , separation $D > 2a$, put $u = \pi a^2$, $J_A = 16\pi a^3/3$, and $J_{AB} = \int_{A \times B} |x - y|^{-1} dx dy$. For the event that both disks contain a point, averaged over common rotations, its Riesz integral is $2e^{-2u}[J_{AB} - (e^u - 1)J_A]$. It is positive if $u \leq 1$ and $a(D + 2a) < 3/32$. The one-disk occupancy event instead has integral $-e^{-u}J_A$. These examples give opposite leading signs without a claim about the near-critical annulus.

Return now to fixed original centers and moving ellipse edges. At a contact pair $x, y = x + 2rS_t(\cos \theta, \sin \theta)$, let $p_{t,r}(x, \theta)$ be the reduced two-point Palm probability that the edge xy is pivotal when toggled, with both centers restored. Set

$$P_{t,r}(\theta) = \int p_{t,r}(x, \theta) dx, \quad a_{t,r}(\theta) = 1 - e^{-4\pi r^2(e^{2t} \cos^2 \theta + e^{-2t} \sin^2 \theta)}.$$

Two-point Campbell and the contact Jacobian, divided by two for unordered edges, give

$$\partial_r C = 2r \int_0^{2\pi} a_{t,r} P_{t,r} d\theta, \quad M = \frac{\int \cos(2\theta) a_{t,r} P_{t,r} d\theta}{\int a_{t,r} P_{t,r} d\theta}.$$

At $t = 0$, $P_{0,r} = P_0(r)$ is constant in angle. With $v = 4\pi r^2$ and $D_t = \int a_{t,r} P_{t,r} d\theta$, continuity of the Palm contact law yields, whenever $P_0(r) > 0$,

$$M(t, r) = \frac{1 - e^{-v}}{D_t} \int \cos(2\theta) [P_{t,r}(\theta) - P_0(r)] d\theta + \frac{v}{e^v - 1} t + o(t).$$

The explicit pair-density term is positive; a desired negative orientation response must overcome it through the Palm geometry. That response cannot be omitted. Its kernel is exactly $K^{\dagger x, y} = K - VG_{xy}^{-1}V^*$, with coherent columns $V = [K(\cdot, x), K(\cdot, y)]$ and their Gram matrix G_{xy} . Differentiation retains all three terms $-V'G_{xy}^{-1}V^* - VG_{xy}^{-1}(V')^* + VG_{xy}^{-1}G'_{xy}G_{xy}^{-1}V^*$. Contact pairs stay separated on compact positive radius and deformation intervals. Graph changes, including near-tie switches, remain a further contribution. The source's independent audit checks these formulas, not the missing sign.

7.12 What these results do and do not establish

Theorem 9 supplies an actual strict comparison of fixed-point percolation probabilities for equal-area disk and fixed noncircular ellipse grains at sufficiently large area. Corollary 17 supplies an actual infinite-volume critical-threshold limit for the anisotropic processes. Proposition 19

separates two large-grain regimes, and Proposition 22 identifies the finite-window first correction at the Poisson end without assigning it an unproved sign. The recovered refinements give a quantitative endpoint error, the exact integrated orientation identity, and further signed Palm calculations. The switching examples explain why positive edge curvature cannot settle their sign.

These are separate assertions. They do not establish $r_c(G) < r_c(\Pi_1)$. In particular, neither monotonicity of $t \mapsto r_c(\Phi_t)$ nor a one-sided comparison with its limit follows from (7.29). A logically sufficient additional conclusion would be a proved radius $r < r_c(\Pi_1)$ at which $\Phi_0 = G$ percolates. This section does not supply that conclusion. The large-area comparison and the Poisson limit have been connected at the level of well-defined models and valid finite-size criteria, but an additional near-critical comparison is still absent.

8 The recovered small- β argument

8.1 Status and the result recorded in the archive

The small- β work has not been lost. The principal proof note is `research/small_beta_critical_radius_asymptotic.md`; its recorded audits are `small_beta_asymptotic_audit.md`, `pure_small_beta_theorem_audit.md`, and `direct_pivotal_void_audit.md`. Those documents state that their arguments passed the earlier checks. That historical statement is not being substituted here for a complete new verification of every geometric construction used by the argument.

The precise distinction is as follows. The finite-window differentiation identities and the deterministic implication from a uniform pivotal estimate to the asserted critical-radius asymptotic are written out below. The necessary uniform geometric estimate is recorded with its actual hypotheses and a substantial reconstruction of its proposed proof. A bounded fresh review found no decisive counterexample, but did not independently reconstruct every certificate selection, boundary placement, and conditional normalization in that chain. Accordingly, the geometric estimate, and hence the asymptotic which depends on it, are designated here as *archival claims awaiting complete fresh verification*. This is neither a disproof nor a report of missing files.

Write $r_P = r_c(\Pi_1)$ and set

$$H_\beta(z, w) = \frac{1}{\beta} \exp \left\{ \frac{\pi z \bar{w}}{\beta} - \frac{\pi(|z|^2 + |w|^2)}{2\beta} \right\}, \quad G_\beta = \text{DPP}(\beta H_\beta), \quad (8.1)$$

$$L_\beta = \frac{-\log(1 - \beta)}{\beta}, \quad 0 < \beta < 1. \quad (8.2)$$

The kernel H_β is a projection of intensity $\rho = \beta^{-1}$, whereas G_β has intensity one. Thus G_β is ordinary unit-intensity Ginibre independently thinned with retention β and then contracted spatially by $\sqrt{\beta}$. In particular $G_1 = G$.

The archived conclusion is the squeeze

$$\frac{r_P}{\sqrt{L_\beta}} \leq r_c(G_\beta) \leq \frac{r_P}{\sqrt{(1 - \epsilon_\beta)L_\beta}}, \quad \epsilon_\beta = o(\beta), \quad (8.3)$$

for sufficiently small positive β . Its lower bound is the independently substantiated consequence of the exact Poisson upper domination proved in the core part of this report. Its upper bound is the part requiring the additional geometry. The screenshot's formula was

$$r_c(G_\beta) = r_P(1 - \beta/4) + O(\beta^{6/5}). \quad (8.4)$$

The later note sharpens the claimed remainder, using the same proposed estimate with a different cutoff, to

$$r_c(G_\beta) = r_P(1 - \beta/4) + O(\beta^{3/2}[\log(1/\beta)]^{3/2}). \quad (8.5)$$

Neither statement would, by itself, settle the target at $\beta = 1$.

8.2 Finite events, conditioning, and an exact transport identity

All grains in this section are full closed disks of radius r . For $m < L$, let $F_{m,L,r}$ be the indicator that an occupied component meets both $B(0, m)$ and $\{z : |z| \geq L\}$. Centers are allowed everywhere; the graph is not clipped to an annulus. A center attaches to the inner terminal exactly when $|z| \leq m+r$, and to the outer terminal exactly when $|z| \geq L-r$. A finite witnessing path can be stopped when it first attaches to the outer terminal, so the event is determined by centers in a bounded disk, which may be enlarged whenever a later local construction requires it. The closed convention also specifies the events at artificially inserted roots on a terminal boundary.

For a fixed β , let

$$X_{s,\nu} = \text{DPP}(sH_\beta) \cup \Pi_\nu, \quad 0 \leq s < 1, \quad \nu \geq 0,$$

with independent, retained colors. Let μ_s be the DPP law and $\mu_s^{!x}$ its reduced Palm law at x ; the root is omitted from this Palm configuration. For $s > 0$, the rank-one mixture identity gives the positive measure

$$\Lambda_x = \mu_s - (1-s)\mu_s^{!x}, \quad 0 \leq \Lambda_x \leq \mu_s. \quad (8.6)$$

Write $T_x = d\Lambda_x/d\mu_s$, so $0 \leq T_x \leq 1$. This is a measure definition, not an identification with the projection onto an infinite random span of coherent vectors. Restriction to an observation window gives the corresponding bounded local density and the same integrals for events observed there.

For any bounded measurable local F , define

$$C_F(s, \nu) = \mathbb{E}F(X_{s,\nu}), \quad D_x F(\eta) = F(\eta \cup \{x\}) - F(\eta).$$

The exact formulas are

$$\partial_s C_F(s, \nu) = \frac{1}{\beta} \int \mathbb{E}_{\mu_s^{!x}, \Pi_\nu} D_x F \, dx, \quad (8.7)$$

$$\partial_\nu C_F(s, \nu) = \int \mathbb{E}_{\mu_s, \Pi_\nu} D_x F \, dx. \quad (8.8)$$

Indeed, realize the DPP by thinning one process of kernel H_β . Conditional on that process, only its finitely many points in the active window W affect F . The expectation is a polynomial in s ; its derivative is the sum of the add-one differences over those points, with all other retention marks independently sampled. Its absolute value is at most $\text{osc}(F)N(W)$, whose expectation is $\text{osc}(F)|W|/\beta$. Dominated convergence and reduced Campbell give (8.7). The analogous Poisson add-one calculation gives (8.8). The same couplings give continuity of these derivatives, including the appropriate one-sided derivatives at $s = 0$ and $\nu = 0$. This is a finite-window assertion; its dominating constant is not claimed to be independent of W .

For increasing F , put

$$I_F = \int \mathbb{E}[D_x F] \, dx, \quad J_F = \int \mathbb{E}[T_x D_x F] \, dx.$$

The expectations include the independent Poisson color. Substituting (8.6) into (8.7) gives the exact identity

$$\left(\partial_s - \frac{1}{\beta(1-s)} \partial_\nu \right) C_F = -\frac{J_F}{\beta(1-s)}. \quad (8.9)$$

Both I_F and J_F are nonnegative. Without any geometric estimate, (8.9) has only the sign yielding the Poisson upper domination. A gain in the opposite direction for connection events requires a useful upper bound on J_F/I_F .

8.3 The exact additional estimate and its parameter regime

The principal input asserted in `direct_pivotal_void_split.md` is the following. Fix

$$0 < r_- \leq r \leq r_+, \quad m \geq 200000r_+, \quad L - m > 240r_+. \quad (8.10)$$

Fix also $c_* > 0$. For sufficiently small $\beta_0 < 1/2$, the proposed estimate is uniform for

$$0 < s \leq \beta \leq \beta_0, \quad 0 \leq \nu \leq 2, \quad \max\{s/\beta, \nu\} \geq c_*, \quad 0 < d < r_-/100. \quad (8.11)$$

For $F = F_{m,L,r}$ it states

$$J_F \leq \epsilon_\beta(d)I_F, \quad \epsilon_\beta(d) = Cd^3 + C \exp\left\{-\frac{\pi d^2}{8\beta}\right\}. \quad (G)$$

The constant C is asserted to depend only on the fixed radius and intensity bounds, not on m, L, d, s, ν, β in the displayed ranges. In particular, a proof with a constant growing in L , or one requiring $\nu \geq \nu_0 > 0$ throughout, does not establish (G). Independence of C from d is needed for the optimized cutoff yielding (8.5).

The next subsections reconstruct the proposed proof of (G). They retain its conditioning and repairs, but do not certify that its entire geometric chain has been freshly proved. The later transport and limit argument is a rigorous implication *if (G) holds as stated*.

8.4 The direct small-ball split

Write $f(x) = \mathbb{P}(D_x F = 1)$ and $I = \int f(x) dx$. Set

$$B_x = B(x, d), \quad S_x(d) = \{w : 2r - d < |w - x| \leq 2r\}.$$

Suppose x is insertion-pivotal and an original center z lies in B_x . Take a shortest positive terminal path after inserting x . It uses x . If both neighbors of x on that path are old centers, at least one of them, say w , is not adjacent to z : otherwise replacing x by z would produce a crossing in the old configuration. Consequently $w \in S_x(d)$. If a neighbor is a terminal, the same replacement works unless x is attached to that terminal and z is not. Such a membership change forces x into a deterministic strip of width d around the corresponding center-attachment circle.

Let $\mathcal{T}_d$ be the union of these two strips, and let N_D and N_X count the DPP color and the full union. The deterministic implication needed by the argument is therefore

$$\{D_x F = 1, N_D(B_x) > 0\} \subset \{N_D(B_x)N_X(S_x(d)) \geq 1\} \cup \{x \in \mathcal{T}_d, N_D(B_x) > 0\}. \quad (8.12)$$

The two random count regions are disjoint, with

$$|B_x| = \pi d^2, \quad |S_x(d)| = \pi(4rd - d^2) \leq 4\pi r d. \quad (8.13)$$

The current-law reduced Campbell bounds have coefficients

$$M_D = \frac{s}{\beta(1-s)}, \quad M = M_D + \nu.$$

They are bounded uniformly in (8.11). If H_x is an exterior event whose observed region excludes both B_x and $S_x(d)$, deleting the counted points does not change H_x . Applying the colored Campbell upper bounds successively gives

$$\mathbb{E}[\mathbf{1}_{H_x} N_D(B_x) N_X(S_x(d))] \leq M_D M |B_x| |S_x(d)| \mathbb{P}(H_x), \quad (8.14)$$

$$\mathbb{E}[\mathbf{1}_{H_x} N_D(B_x)] \leq M_D |B_x| \mathbb{P}(H_x). \quad (8.15)$$

These estimates retain the exterior event. Replacing it by an independent event would be unjustified for a DPP.

In the bulk, the proposed local separation lemma supplies an event H_x , observed outside a fixed multiple of r , with

$$\{D_x F = 1\} \subset H_x, \quad \mathbb{P}(H_x) \leq C f(x). \quad (8.16)$$

Then (8.14) costs $O(d^3) f(x)$. At boundary roots, the claimed estimates are an integrated replacement of (8.16) and the strip estimate

$$\int_{\mathcal{T}_d} \mathbb{P}(H_x) dx \leq C d I. \quad (8.17)$$

Multiplying (8.17) by the $O(d^2)$ factor in (8.15) recovers the third power of d . With the relocation described below, this yields the proposed conclusion

$$\int \mathbb{P}(D_x F = 1, N_D(B_x) > 0) dx \leq C d^3 I. \quad (8.18)$$

It is the geometric comparisons, not the elementary count calculation, that require the substantial uniform argument.

8.5 Canonical certificates and the empty-ball estimate

The complementary event is $E_x = \{D_x F = 1, N_D(B_x) = 0\}$. Choose a minimum-cardinality positive path through the artificial root, with ties broken by an order depending only on the candidate finite subset and its colors. Remove the artificial root from that certificate. The remaining positive requirements are paths to the two fixed terminals; the negative requirements prohibit an old crossing and a preceding certificate. Once a certificate is forced, these remainder constraints are decreasing under deletion. This would fail for an order depending on unretained random data, or for changing component labels whose terminal connections were not themselves retained.

For the DPP part A of a forced colored certificate, put

$$\begin{aligned} P_A &= H_\beta(\cdot, A) H_\beta(A, A)^{-1} H_\beta(A, \cdot), & Q_A &= H_\beta - P_A, \\ p_A(x) &= P_A(x, x) / \rho, & q_A(x) &= 1 - p_A(x). \end{aligned}$$

The reduced Campbell disintegration of (8.6) is the measure identity

$$s^k \det H_\beta(A, A) [p_A(x) \mu_A + q_A(x) \Lambda_{A,x}], \quad (8.19)$$

where $\mu_A = \text{DPP}(sQ_A)$ and $\Lambda_{A,x} = \mu_A - (1-s)\mu_A^{\downarrow x}$. It follows from the determinant ratio $\rho_k^{\downarrow x}(A) / \rho_k(A) = q_A(x)$ and commutation of finite reduced Palm operations. It is not an evaluation of an arbitrary almost everywhere version of T_x on a fixed Palm manifold. For k DPP and j Poisson certificate points, the unordered coefficient is

$$\frac{s^k \det H_\beta(A, A) \nu^j}{k! j!}.$$

Each configuration contributes its one selected colored subset, so the sum has nonnegative integrands and Tonelli applies. At $\nu = 0$ only $j = 0$ remains. Poisson certificate vertices may lie inside B_x ; only the DPP part must avoid it, and only that part enters P_A .

The void improvement can be calculated without any geometric surgery. For a projection Q , let $\mu = \text{DPP}(sQ)$, fix a root with $Q(x, x) > 0$, and write

$$u = \frac{Q(\cdot, x)}{\sqrt{Q(x, x)}}, \quad K = sQ, \quad D = I - K_B, \quad t = \|\mathbf{1}_{B^c} u\|_2^2, \quad a_0 = 1 + s\langle u_B, D^{-1} u_B \rangle.$$

The determinant lemma gives

$$\frac{\mu^{\text{!}x}(N(B) = 0)}{\mu(N(B) = 0)} = a_0.$$

The two void-conditioned exterior kernels satisfy

$$\begin{aligned} K^{\text{void}} &= K_{B^c} + K_{B^c B} D^{-1} K_{B B^c}, \\ (K^{\text{!}x})^{\text{void}} &= K^{\text{void}} - \frac{s}{a_0} v v^*, \quad v = u_{B^c} + K_{B^c B} D^{-1} u_B. \end{aligned}$$

The second formula is Sherman–Morrison applied to the first. Kernel order gives stochastic order of the conditional exterior DPPs. Since $Q_B \geq u_B u_B^*$, inverse monotonicity also gives

$$a_0 \geq \frac{1}{1 - s + st}.$$

Thus for a decreasing exterior event E_{B^c} and $E = E_{B^c} \cap \{N(B) = 0\}$,

$$\Lambda_x(E) \leq \frac{st}{1 - s + st} \mu(E) \leq \frac{st}{1 - s} \mu(E). \quad (8.20)$$

The dependence between the exterior event and the void is retained through their conditional laws.

For the remainder Q_A , its normalized row tail obeys

$$q_A(x) t_A(x) \leq 2(h_B(x) + p_A(x)), \quad h_B(x) = \rho^{-1} \int_{B^c} |H_\beta(x, y)|^2 dy. \quad (8.21)$$

This follows by writing $Q_A = H_\beta - P_A$ and using $\int |P_A(x, y)|^2 dy = P_A(x, x)$. For $B = B(x, d)$, $h_B(x) = e^{-\pi d^2/\beta}$. Combining (8.19)–(8.21) bounds each normalized certificate contribution by

$$p_A(x) + \frac{2s}{1 - s} (e^{-\pi d^2/\beta} + p_A(x)). \quad (8.22)$$

The analytic input recorded in `sparse_fock_projection.md` is

$$p_A(x) \leq C_k \exp \left\{ -\frac{\pi \text{dist}(x, A)^2}{8\beta} \right\}, \quad (8.23)$$

uniformly over finite sets with a fixed local clique or grid quota, including colliding clusters of distinct points and arbitrarily large total cardinality. The mechanism deserves emphasis. A quota forces small physical clusters to have uniformly bounded cardinality. Within each such cluster, confluent divided differences provide a collision-uniform coherent-state basis. A compactness argument is used only for this bounded number of nodes. Each cluster is then orthonormalized; Gaussian decay makes the off-diagonal blocks of the global Gram matrix have row sums below $1/2$ for small β . The inverse is consequently bounded independently of the number of clusters. Gaussian packing sums give (8.23). Applying a Gram inverse bound to the

original unnormalized near-collision matrix would be invalid; the within-cluster normalization is essential.

A shortest two-path certificate has at most four points in each radius- r clique, so its DPP subset satisfies the needed quota. Thus the claimed application of (8.23) to (8.22) gives

$$\mathbb{E}[T_x \mathbf{1}_{E_x}] \leq C e^{-\pi d^2/(8\beta)} f(x). \quad (8.24)$$

The source notes for this part are `canonical_certificate_audit.md` and `canonical_void_clearance.md`. Combining (8.18) and (8.24), using $0 \leq T_x \leq 1$, is precisely the proposed proof of (G).

8.6 How the proposed construction reaches zero bath intensity

The positive-bath argument cannot simply be continued to $\nu = 0$. The archive uses two different filling constructions to cover the parameter path.

A Poisson bath bounded away from zero. For a bounded active window, the conditional DPP L -kernel inside a cleared region B , given the exact colored exterior in that window, is a Schur complement. It satisfies

$$0 \leq L_B^\xi \leq L_{BB}, \quad L_B^\xi(z, z) \leq M_D.$$

Hence its conditional void probability is at least $e^{-M_D|B|}$, and its conditional factorial densities are bounded by M_D^n . For any event E of the interior union and the fixed exterior, including a nonmonotone arm event, the bounds imply

$$e^{-M_D|B|} \mathbb{P}_{\Pi_\nu, B}(E) \leq \mathbb{P}_X(E \mid \text{colored exterior}) \leq e^{M_D|B|} \mathbb{P}_{\Pi_{\nu+M_D}, B}(E). \quad (8.25)$$

If E has an interior certificate of at most K vertices such that arbitrary deletion of other vertices preserves it, independent thinning gives

$$\mathbb{P}_{\Pi_v, B}(E) \leq (v/\nu)^K \mathbb{P}_{\Pi_\nu, B}(E), \quad v \geq \nu > 0.$$

Shortest paths in a fixed bounded annulus give such a K by a clique cover. The recorded safe quotas are $K_0 = 4 \cdot 116^2$ in the bulk and $K_1 = 2 \cdot 200^2$ at a boundary. Clearing the DPP and using separated Poisson paths therefore transfers the ordinary Poisson separation construction to the mixed process. The constants contain $(\nu/(\nu + M_D))^{K_i}$ and Poisson filling probabilities; they degenerate at $\nu = 0$. The published bulk Poisson separation input is Franceschetti–Penrose–Rosoman, Lemma 4.1, used at graph range one on $C_{29} \setminus C_{20}$ with arbitrary fixed exterior vertex groups. Its archived application and explicit inner filling are in `cooperative_pair_bulk.md`; the mixed transfer is in `cooperative_pair_dpp_bath.md`. This report does not elevate that whole application to a new external certification.

A fixed-region determinantal filling. For s/β bounded below, the archive replaces the Poisson filling by a DPP-only filling on a favorable set of ordinary exterior configurations. The construction in `finite_window_leverage_operator.md` explicitly avoids a conditional law given an arbitrary exact infinite exterior.

For a bounded active window W , the rank-one mixture and Janossy identities construct a positive operator $0 \leq B_W(\xi) \leq I$ on the Bargmann range. Its coherent diagonal is the simultaneous continuous version

$$T_W(x, \xi) = \langle f_x, B_W(\xi) f_x \rangle, \quad f_x = H_\beta(\cdot, x) / \sqrt{\rho}.$$

Positivity is first obtained simultaneously on a countable dense set of Fock vectors, and then extended by continuity, so that one configuration null set suffices. Magnetic translation and the analytic submean inequality give

$$\sup_{x \in K} T_W(x, \xi) \leq \frac{e^\pi}{\pi\beta} \int_{K+B(0, \sqrt{\beta})} T_W(w, \xi) dw. \quad (8.26)$$

Here K is a fixed compact target region inside the cleared bounded Borel region B .

Evaluate B_W at the exterior configuration with the interior empty. For $a = s/(1-s)$, the exact conditional interior kernel is

$$L_{\text{cond}}(x, y) = a[H_\beta(x, y) - B_W[H_\beta](x, y)]. \quad (8.27)$$

In particular its diagonal is $a\rho(1 - T_W(x))$, while positivity bounds the correction off the diagonal by $a\rho\sqrt{T_W(x)T_W(y)}$. Independently, $L_{\text{cond}} \leq L_W \leq aH_{\beta, W}$ supplies the uniform combined void lower bound $e^{-(a\rho+\nu)|B|}$. This lower bound permits transfer from the distribution with an empty interior to the ordinary exterior distribution without conditioning on a null set.

Suppose H_{ext} has intrinsic sparse certificates outside B and a decreasing remainder. If $b = \text{dist}(K, B^c) > 0$, the certificate void bound and (8.26) yield the recorded estimate

$$\frac{\mathbb{P}(H_{\text{ext}}, \sup_K T_W > \delta)}{\mathbb{P}(H_{\text{ext}})} \leq e^{(2+\nu_1)|B|} \delta^{-1} \frac{e^\pi |B| C'}{\pi\beta} e^{-\pi b^2/(32\beta)}. \quad (8.28)$$

The coefficient tends to zero uniformly in the active window. Choose $\delta = 1/(8n_0)$ and then β small. If at most n_0 target sets $V_i \subset K$ have mutual distance at least $b_0 > 0$, the Gaussian off-diagonal part is also at most $1/(8n_0)$. Gershgorin applied to (8.27) gives the conditional determinant lower bound $(a\rho/2)^n$. Thus, on a good exterior, exactly one DPP point in each V_i , no other DPP point in B , and no Poisson point in B has conditional probability at least

$$e^{-(2+\nu_1)|B|} (c_*/2)^n \prod_{i=1}^n |V_i|, \quad s/\beta \geq c_*. \quad (8.29)$$

The $n!$ labelings cancel the unordered Janossy factor. Target sets may depend measurably on the exterior because the good-exterior bound is a supremum over the entire fixed compact K . A deterministic lower bound on target areas is still necessary to turn this into a fixed positive multiple of $\mathbb{P}(H_{\text{ext}})$.

The region may have finitely many deterministic holes, provided its area and the target depth are controlled. An arbitrary random stopping region is *not* authorized by this theorem. Section 8 of `finite_window_leverage_operator.md` explicitly leaves that extension unproved. The later geometry seeks to avoid needing it by a finite family of deterministic punctured regions.

8.7 The fixed-hole bulk geometry and its recorded repairs

The bulk construction of `zero_bath_bulk_separation.md` translates the root to zero and divides lengths by $2r$, so the graph range is one. Both terminals lie outside C_{29} . Let H be the event that the two terminal groups have disjoint connections to $C_{25} \setminus C_{24}$ using only centers outside C_{24} . Pivotality implies H . The desired conclusion is $\mathbb{P}(D_x F = 1) \geq \kappa \mathbb{P}(H)$, uniformly in the outer window and at zero Poisson bath.

A fixed grid of side 10^{-5} covers the compact annulus through C_{29} . A pattern retains finitely many cells with assigned terminal labels. Its cleared region is C_{29} minus those retained cells, *clipped* to C_{24}^c . Every selected outer-path cell is required to have a terminal-connected anchor outside C_{25} whose connection uses only vertices outside C_{25} . Entry cells separately retain their prescribed connected entry points. These are actual corrections recorded in `zero_bath_bulk_audit.md`:

- Without the outside- C_{25} path condition, an anchor connection could run through an inner entry, invalidating the radial gap used later.
- Without clipping at C_{24} , a retained cell could reveal extra points inside the disk excluded from the definition of H , so its exterior would no longer necessarily be a subconfiguration of the nonconnecting graph defining H .

With these corrections, each retained exterior is an ordinary fixed colored exterior. There are only finitely many patterns. Their positive requirements have shortest-path witnesses with a fixed local quota, and their remaining nonconnection requirements are decreasing. The target filling therefore seeks to apply (8.29) to a deterministic punctured region, not to a configuration-dependent Markov domain.

The retained entry parents y, z have radii between 24 and 25. Label them so that $|y| \geq |z|$, and let R_C be the supremum radius of the fixed cell containing y . The split is:

- In Case F, the opposite terminal has an outside- C_{25} connected point z_0 with $|y| \leq |z_0| \leq R_C + 0.05$. Retain its exterior path in place of the inner opposite entry.
- In Case G, no such exterior connection reaches the fixed collar $C_{R_C+0.05} \setminus C_{25}$. This is a decreasing requirement after the parent cell has been fixed. Opposite outer-hole points then have radius at least $|y| + 0.049$, while opposite inner-entry points lie within 0.001 of z .

Here are the principal strict margins recorded in the construction. In Case F the two parent radii differ by at most 0.051. Parent nonadjacency gives

$$\sin(\theta/2) > 0.998/50.2.$$

Move each branch 0.6 inward and 0.1 tangentially away from the other, then continue to radial parameter 23. Their vertical separation is greater than

$$46(0.998)/50.2 + 0.14 > 1.05.$$

For an opposite hole vertex with radius deficit at most 0.061, the radial-distance identity gives squared clearance greater than 1.261 before the tangential shift, hence clearance greater than $\sqrt{1.261} - 0.1 > 1.02$ afterward. Diverging circular arcs and opposite radii then lead to endpoints at distance 0.75 from the root.

In Case G, parents more than three apart admit simple inward radial paths, with cross-branch separation exceeding 2.6. For closer parents, rotate so that $y = (R, 0)$, and choose the exterior predecessor of z on the upper side. The radial gap and the edge to that predecessor force $z_2 > 0.5$. The first lower-branch target is constructed by intersecting the unit circle around y with the circle of radius $R + 0.049$, and then taking a point at distance 0.999 from y and 1.01 from the lower intersection. In the source notation it is

$$\begin{aligned} U &= 0.049 + \frac{0.049^2 - 1}{2R}, \quad V = \sqrt{1 - U^2}, \\ t &= (0.999^2 + 1 - 1.01^2)/2, \quad q = \sqrt{0.999^2 - t^2}, \\ b &= y + (tU - qV, -tV - qU). \end{aligned}$$

The listed bounds are $b_1 - R < -0.855$ and $-0.516 < b_2 < -0.513$. The first upper-branch target is $c = z - 0.999(1, 0)$, with $c_2 > 0.5$. A two-circle distance calculation is used to keep b more than 1.01 from the opposite outer holes; the vertical separation keeps it more than 1.013 from z . The corresponding calculation for c gives clearance exceeding 1.05 from the opposite

outer holes. Horizontal corridors to coordinate 20, followed by vertical separation and diverging arcs, preserve these margins.

The final recorded discretization uses at most 1000 targets. A maximal separated subcollection retains connectivity of each branch; disks of radius 10^{-4} around its centers have mutual separation at least 0.0048 and fixed depth at least 0.3 in the punctured region. Every connection or nonconnection survives displacement inside these disks. The positive lower target area and the finite number of patterns turn (8.29) into the claimed bulk comparison. These explicit margins explain the proposed uniformity; complete fresh checking of all cases and event reductions remains part of the unresolved verification of (G).

8.8 Boundary construction, strips, and deep-root relocation

At boundary roots, ordinary insertion pivotality can involve direct terminal attachment. The archive therefore does not extend the bulk two-arm lemma to the boundary by assertion. In normalized range-one coordinates the nearby attachment circle has radius at least 100000. The construction in `zero_bath_boundary_separation.md` uses the fixed radii

$$R_j = 49 + j/100, \quad j = 0, \dots, 99.$$

Its event H_j observes only the graph outside C_{R_j} and asks for a distant-terminal connection to the shell $[R_j, R_j + 0.01]$, together with terminal nonconnection. A witnessing path first entering C_{50} gives at least one such event. Each H_j has a shortest-path certificate outside its deterministic cleared disk.

If z is the distant-connected shell parent, $r_0 = |z|$, and $e = z/r_0$, then every nearby-terminal exterior vertex v obeys $|v| \geq r_0 - 0.01$ and $|z - v| > 1$. The exact radial identity gives

$$|z - 0.5e - v|^2 = (1 - 0.5/r_0)|z - v|^2 + 0.25 + (0.5/r_0)(|v|^2 - r_0^2) > 1.229.$$

An allowed-side displacement of 0.05 corrects the small penetration of the large attachment circle. The recorded first-target clearance is then greater than $\sqrt{1.229} - 0.05 > 1.05$. A far branch follows an allowed circular arc and a positive radius to +0.75; a separate near branch ends at -0.75 and reaches the nearby terminal. For normalized signed distance in $[-0.01, 30]$, the construction makes every insertion within 0.1 of the original root pivotal. In physical coordinates this supplies comparison to roots within $0.2r$.

The discretization keeps at most 1000 targets, with target radius 0.001 and mutual separation at least 0.008 in normalized units. They have fixed positive depth in the cleared disk. The fixed-region DPP filling applies to each of the 100 bins; summing costs only a fixed factor. On terminal strips, this robust comparison allows averaging over nearby roots. Circle coarea, rather than a stationarity assertion about a conditional exterior, gives (8.17).

There is a further correction. A pivotal root can have signed normalized distance as low as -1, whereas the unshifted robust construction was justified only down to -0.01. For those deeper roots, `direct_pivotal_void_split.md` moves the artificial root to +2, replaces the branch endpoints by +1.25 and +2.75, and extends the near branch to $-(a + 0.25)$, where $a \geq -1$ is the signed distance. The original root need not itself be made robustly pivotal. Instead the claimed comparison is

$$\mathbb{P}(H_x) \leq Cf(x + 4rn(x)), \quad (8.30)$$

where $n(x)$ points into the unattached side of the nearby terminal. The recorded correction of the radial penetration bound is from 0.002 to less than 0.025, still covered by the same 0.05 allowed-side shift. The two branches retain separation 1.5; both endpoints are 0.75 from the relocated root.

On either circular belt, the map in (8.30) changes the radial coordinate by $\pm 4r$ and leaves the angle fixed. Its Jacobian is $(|x| \pm 4r)/|x|$, with a uniformly bounded inverse under (8.10); the two belt images have bounded multiplicity. Thus integrating (8.14) against (8.30) still costs Cd^3I . On the actual width- d membership-change strips, the original unshifted comparison is in its valid range and supplies the extra factor d . The explicit recorded check is `zero_bath_bounded_audit.md`.

8.9 The complete implication from (G) to an asymptotic

The remainder of the argument has no geometric freedom once (G) is available with its asserted uniform constants. For this application fix $c_* = 1/3$ in (8.11); the characteristic below is then covered for all sufficiently small β . Choose a cutoff $d = d_\beta$ so that $\epsilon_\beta = \epsilon_\beta(d_\beta) = o(\beta)$, and reduce β_0 until $0 \leq \epsilon_\beta \leq 1/2$. Combining (G) with (8.9) gives

$$\left(\partial_s - \frac{1 - \epsilon_\beta}{\beta(1 - s)}\partial_\nu\right) C_F = \frac{\epsilon_\beta I_F - J_F}{\beta(1 - s)} \geq 0. \quad (8.31)$$

For $0 \leq s \leq \beta$, define

$$\nu_s = \frac{1 - \epsilon_\beta}{\beta} \log \frac{1 - s}{1 - \beta}, \quad \lambda_\beta = \nu_0 = (1 - \epsilon_\beta)L_\beta. \quad (8.32)$$

This path has nonnegative bath, ends at $\nu_\beta = 0$, and has $\nu'_s = -(1 - \epsilon_\beta)/[\beta(1 - s)]$. It stays in the claimed parameter regime: for $s \leq \beta/2$,

$$\nu_s \geq \frac{1 - \epsilon_\beta}{\beta} \int_{\beta/2}^\beta \frac{dt}{1 - t} \geq \frac{1 - \epsilon_\beta}{2},$$

which is at least $1/3$ for sufficiently small β ; for $s \geq \beta/2$, the DPP intensity s/β is at least $1/2$. Also $\nu_s \leq L_\beta \leq 2$ for $\beta \leq 1/2$.

At each fixed finite window, the continuously differentiable formulas above permit the ordinary chain rule on every characteristic, not merely on almost every curve. One-sided continuity supplies $s = 0$ and $\nu = 0$. Alternatively, an almost-everywhere formulation can be straightened and mollified before taking the prescribed curve; the source note `positive_bath_pde_comparison.md` explains that procedure. No derivative is taken after an infinite-volume limit. Integration of (8.31) therefore gives

$$\mathbb{P}(F_{m,L,r}(G_\beta) = 1) \geq \mathbb{P}(F_{m,L,r}(\Pi_{\lambda_\beta}) = 1) \quad (8.33)$$

for every admissible m, L, r . It is a comparison of these connection events, not lower stochastic domination for all increasing observables.

Fix the compact radius interval to contain r_P , for example $[r_P/2, 2r_P]$. For small β , the Poisson threshold $r_P/\sqrt{\lambda_\beta}$ lies in this interval. Take $r > r_P/\sqrt{\lambda_\beta}$ and one admissibly large finite seed $B(0, m)$. Poisson spatial scaling makes the process on the right of (8.33) supercritical at this radius. The seed has positive probability of meeting an unbounded occupied component: otherwise stationarity and a countable cover by translates of that seed would give zero probability of any unbounded component.

Now let $L \rightarrow \infty$ while holding m, β, r fixed. The events $F_{m,L,r}$ decrease. Local finiteness gives only finitely many grains, and hence finitely many occupied components, meeting the seed. Thus their decreasing intersection is exactly the event that the seed meets an unbounded component; arbitrarily distant connections cannot be provided by infinitely many different seed components. Continuity of probability from above passes (8.33) to this event. It follows that G_β percolates

with positive probability. Its stationarity and mixing identify this with the usual almost-sure percolation conclusion. Letting r decrease to the Poisson threshold gives the upper bound in (8.3). No assertion about percolation at the critical Poisson radius is used.

For the two error choices, first take $d_\beta = \beta^{2/5}$. For small β it is in the allowed range, and

$$\epsilon_\beta \leq C\beta^{6/5} + C \exp\{-(\pi/8)\beta^{-1/5}\} = O(\beta^{6/5}).$$

The exponential is smaller than every fixed positive power of β , as follows by taking logarithms. Next take

$$d_\beta^2 = (16/\pi)\beta \log(1/\beta).$$

Then the exponential term is exactly β^2 , and

$$\epsilon_\beta = O(\beta^{3/2}[\log(1/\beta)]^{3/2}). \quad (8.34)$$

Both choices give $\epsilon_\beta = o(\beta)$.

Finally,

$$L_\beta = 1 + \frac{\beta}{2} + \frac{\beta^2}{3} + O(\beta^3), \quad L_\beta^{-1/2} = 1 - \frac{\beta}{4} - \frac{7\beta^2}{96} + O(\beta^3).$$

For $0 \leq \epsilon \leq 1/2$, $(1 - \epsilon)^{-1/2} - 1 \leq 2\epsilon$. Since $L_\beta \geq 1$, the width of the squeeze (8.3) is at most $2r_P\epsilon_\beta$. Hence that squeeze rigorously implies

$$|r_c(G_\beta) - r_P(1 - \beta/4)| \leq Cr_P(\beta^2 + \epsilon_\beta).$$

This proves the algebraic implications (8.4) and (8.5) from the respective proposed error estimates. The asserted right derivative $-r_P/4$ would follow by this deterministic squeeze, with no differentiation of an infinite-volume percolation event.

Strictness in this implication comes from a positive intensity gain, not from an unquantified limiting comparison. Since $\epsilon_\beta = o(\beta)$ and $L_\beta = 1 + \beta/2 + O(\beta^2)$, $\lambda_\beta > 1$ for all sufficiently small positive β . For an explicit conservative version, take $\epsilon_\beta \leq \beta/8$ and $\beta \leq 1/2$. The inequalities $L_\beta \geq 1 + \beta/2$ and $L_\beta \leq 2$ give $\lambda_\beta \geq 1 + \beta/4$, and therefore

$$r_c(G_\beta) \leq \frac{r_P}{\sqrt{1 + \beta/4}} < r_P, \quad r_P - r_c(G_\beta) \geq r_P\beta/16.$$

This last display, like the upper squeeze bound, remains dependent on the complete verification of (G).

8.10 A separate strict comparison with a positive Poisson bath

The recovered result of `positive_bath_pde_comparison.md` concerns a different unit-intensity endpoint. Fix $a \in (0, 1)$, put $p = 1 - a$, $b = \beta p$, and

$$Y_{\beta,a} = \text{DPP}(bH_\beta) \cup \Pi_a.$$

Its hypothesis is the uniform distributional differential estimate

$$\left(\partial_s - \frac{1}{\beta(1-s)} \partial_\nu \right) C \geq -K_r \frac{s}{\sqrt{\beta}} \partial_r C - K_\nu s \partial_\nu C. \quad (8.35)$$

Here $C = C_{m,L}(s, \nu, r)$, the constants are independent of the connection windows and small beta, and the estimate is required for $s \leq \beta$, $\nu \in [a/2, 2]$, $r \in [r_P/2, 2r_P]$ and the seed regime (8.10). Its geometric dependencies are the positive-bath cooperative and current-law cofactor estimates; this subsection records their consequence without certifying that entire chain anew.

The characteristic ending at bath a is

$$\nu_s = a + \int_s^b \left[\frac{1}{\beta(1-t)} - K_\nu t \right] dt, \quad r_s = r_0 + \frac{K_r s^2}{2\sqrt{\beta}}.$$

Thus its initial intensity and radius cost are

$$\lambda_{\text{eff}} = a - \beta^{-1} \log(1 - \beta p) - \frac{1}{2} K_\nu \beta^2 p^2, \quad \Delta_r = \frac{1}{2} K_r p^2 \beta^{3/2}.$$

Finite-window probabilities are locally Lipschitz in all parameters. Straightening the characteristic and mollifying the transverse variables justifies integration of the distributional inequality on every prescribed admissible curve. For small beta the entire path stays in the allowed domain, and

$$C_{m,L}(b, a, r_0 + \Delta_r) \geq \mathbb{P}(F_{m,L,r_0}(\Pi_{\lambda_{\text{eff}}}) = 1).$$

Keep one admissible seed fixed and send $L \rightarrow \infty$. The decreasing-event argument already used above gives $r_c(Y_{\beta,a}) \leq r_P / \sqrt{\lambda_{\text{eff}}} + \Delta_r$. If $K_\nu \beta \leq 1/2$, then $\lambda_{\text{eff}} \geq 1 + p^2 \beta / 4$; if also $\sqrt{\beta} \leq r_P / (16K_r)$, with zero-constant constraints omitted, the recovered quantitative conclusion is

$$r_c(Y_{\beta,a}) \leq r_P - \frac{r_P(1-a)^2 \beta}{32} < r_P. \quad (8.36)$$

This conditional implication is a positive-bath theorem at each fixed a , not a uniform removal of the bath and not a comparison at $\beta = 1$. The recorded checks are `weak_hybrid_threshold_audit.md` and `mixed_cooperative_audit.md`.

8.11 What this recovered argument does and does not settle

The archive contains the proof chain, its repairs, and its claimed audits. It is inaccurate to describe the small- β argument as lost or disproved. It is also inaccurate to treat its audit labels as a substitute for checking the uniform geometric statement (G) in full. The exact unresolved verification is whether all fixed-hole pattern events, boundary targets, and conditional filling comparisons provide the constants in (G) uniformly in the growing connection windows and through the zero-bath endpoint. The transport calculation and its infinite-volume consequence have been separated so that this question is not hidden in a derivative of r_c .

Even a completely verified small- β result is a separate statement from the original full-Ginibre target. Ordinary thinning gives only

$$r_c(G) \leq \frac{r_c(G_\beta)}{\sqrt{\beta}},$$

which does not turn a small improvement of $r_c(G_\beta)$ over r_P into an improvement for G . No monotonicity of $r_c(G_\beta)$ in β has been proved in these notes. Thus continuing from small β to $\beta = 1$ remains an additional mathematical task, even if the archival geometric claim is confirmed.

8.12 Sharp obstructions to coupling the small-beta endpoint to Ginibre

The recovered calculations in `research/ginibre_coupling_bridge_obstructions.md` rule out several stronger ways to transfer the small-beta gain. Write $G_{\beta,\lambda}$ for intensity- λ beta-Ginibre, with kernel

$$K_{\beta,\lambda}(z, w) = \lambda \exp \left\{ \frac{\pi \lambda}{\beta} z \bar{w} - \frac{\pi \lambda}{2\beta} (|z|^2 + |w|^2) \right\}.$$

If a coupling gives $G_{\beta,\lambda} \subset G$, products of counts in disjoint sets order the continuous factorial densities. At distinct fixed $a_1, \dots, a_n$, the least-degree Vandermonde term gives

$$\lim_{\epsilon \downarrow 0} \frac{\rho_n^{\beta,\lambda}(\epsilon a_1, \dots, \epsilon a_n)}{\rho_n^{1,1}(\epsilon a_1, \dots, \epsilon a_n)} = \lambda^{n(n+1)/2} \beta^{-n(n-1)/2}.$$

This must be at most one for every n . First taking the coalescence limit at fixed n , then $n \rightarrow \infty$, proves the necessary condition

$$\lambda \leq \beta. \quad (8.37)$$

Equality is realized by ordinary independent beta-thinning of G . Dependent thinning with an exact beta-Ginibre output cannot improve this intensity loss.

The same restriction holds for an injection with any fixed finite displacement bound D . The count in a disk for a kernel pH_γ is a sum of independent Bernoulli variables with parameters $p\mu_j(u)$, where $u = \pi R^2/\gamma$ and $\mu_j(u) = \mathbb{P}(\text{Poisson}(u) \geq j+1)$. This is independence of count summands, not spatial points. The bounds $e^{-u}u^{j+1}/(j+1)! \leq \mu_j(u) \leq u^{j+1}/(j+1)!$, applied to the first n successes and to $\mathbb{E}(\binom{N}{n})$, give the fixed-parameter expansion

$$\log \mathbb{P}(N \geq n) = -\frac{1}{2}n^2 \log n + \left(\frac{3}{4} + \frac{1}{2} \log u\right)n^2 + O_{u,p}(n \log n). \quad (8.38)$$

For the upper bound, writing indices as $j-1+\ell_j$ bounds the remaining sum by a product $\prod_{j=1}^n \sum_{\ell \geq 0} u^\ell j!/(j+\ell)! \leq C_u n^{2u}$; summing Stirling's formula supplies the expansion. An injection would order the source count in B_R by the Ginibre count in B_{R+D} . If $\lambda > \beta$, choose a fixed R such that $\lambda R^2/\beta > (R+D)^2$. Equation (8.38) then contradicts this order as $n \rightarrow \infty$.

Allowing collisions does not permit an arbitrarily small-displacement covering at nearly unit intensity. Assume only every source point is within D of a Ginibre point, with $0 < \lambda \leq 1$. Translation averaging makes an admissible coupling stationary. Assign each source point to a nearest Ginibre point and retain one assignment per used target. The discarded source density is at most $2\pi\lambda^2 D^2$, by the source pair-density bound $\rho_2 \leq \lambda^2$; mass transport bounds unused target density by $1 - \lambda + 2\pi\lambda^2 D^2$. Comparing holes in a disk of radius $1/\sqrt{\pi}$, and paying for its boundary strip, gives, for $\beta \leq 1/10$,

$$\delta_0 := e^{-10/9} - 2e^{-2} \leq 1 - \lambda + 2\pi D^2 + 2\sqrt{\pi}D. \quad (8.39)$$

The source hole is at least $e^{-\lambda/(1-\beta)}$; the first two Ginibre disk eigenvalues bound its hole above by $2e^{-2}$. Thus $\delta_0 > 0$, and at $\lambda = 1$, $D \geq (\sqrt{1+2\delta_0} - 1)/(2\sqrt{\pi})$. When a small-beta radius improvement is of order beta, a useful transfer with $\lambda \leq 1$ would require $\lambda \rightarrow 1$ and $D = O(\beta)$, contradicting (8.39).

Two related decomposition shortcuts also fail. An independent union of positive-intensity components has a nonzero cross-color pair density at coincidence, unlike Ginibre's vanishing pair density. Orthogonal spectral subspaces therefore do not in general yield independent spatial subprocesses. The known independent-block identity after a finite Ginibre power map concerns the powered process; its unbounded spatial derivative does not transfer bounded disk grains. None of these obstructions disproves the critical-radius conjecture; they close the specified exact-output coupling routes.

9 Attempted routes, exact obstructions, and recovered calculations

This section records how the investigation developed beyond the principal results proved elsewhere in this report. It includes the earlier research history supplied by the user, not just

the later working notes. Its purpose is to preserve the mathematics and the reasons for changing direction. It is not a second list of established theorems.

Three distinctions govern the presentation. An elementary identity or counterexample whose argument is reproduced below can be checked from that argument. A *recovered calculation* is an assertion with a substantial derivation in the preserved notes; its historical “audited” label is not being adopted as a fresh certification of every premise. An explicitly *unproved comparison* remains a missing estimate, even when several inputs to it have been established. None of the negative examples below disproves the original Ginibre–Poisson critical-radius conjecture.

Throughout, r is the grain radius and $R = 2r$ is the center connection range. Write $H(z, w) = \exp\{\pi z \bar{w} - \pi(|z|^2 + |w|^2)/2\}$ for the unit-intensity Ginibre projection. For a finite tuple A of distinct points,

$$P_A = H(\cdot, A)H(A, A)^{-1}H(A, \cdot), \quad Q_A = H - P_A.$$

For an increasing Boolean event F , $D_x F(\eta) = F(\eta \cup \{x\}) - F(\eta)$. The words “root”, “parent”, and “mark” always refer to point locations; a reduced Palm root is restored only when this is expressly stated.

9.1 What kind of advance would reach the target?

An exact local improvement is useful only if its normalization and its scale range permit an infinite-volume conclusion. For example, consider

$$C_{m,L}(s, \nu, r) = \mathbb{P}(\text{DPP}(sH) \cup \Pi_\nu \text{ connects } B_m \text{ to } \partial B_L \text{ with radius } r).$$

If one could prove, uniformly in appropriate m, L , a differential comparison

$$\partial_s C_{m,L} - a(s) \partial_\nu C_{m,L} \geq 0 \tag{9.1}$$

along the actual path

$$\nu(s) = \int_s^1 a(t) dt \geq 0, \quad \lambda_* := \int_0^1 a(t) dt > 1,$$

then the endpoint comparison would be $C_{m,L}(1, 0, r) \geq C_{m,L}(0, \lambda_*, r)$. After valid endpoint and infinite-volume passages, Poisson scaling would give

$$r_c(G) \leq \frac{r_c(\Pi_1)}{\sqrt{\lambda_*}} < r_c(\Pi_1).$$

Here the strictness is the definite excess $\lambda_* - 1$, not merely a strict finite-window probability inequality. A version allowing a radius increase must pay that increase from the same excess.

The exact local derivative can be written using the reduced Palm likelihood ratio $R_x^{(s)}$:

$$\partial_s C = \int \mathbb{E}[R_x^{(s)} D_x F] dx, \quad \partial_\nu C = \int \mathbb{E}[D_x F] dx.$$

For $s < 1$, let $T_x = 1 - (1 - s)R_x^{(s)}$. If the pivotal integral is positive and

$$\ell(s) = \frac{\int \mathbb{E}[T_x D_x F] dx}{\int \mathbb{E}[D_x F] dx},$$

the relevant normalized coefficient is $(1 - \ell(s))/(1 - s)$. The desired gain is therefore related to the improper integral

$$\int_0^1 \frac{s - \ell(s)}{1 - s} ds > 0. \tag{9.2}$$

This display does not assert convergence or positivity. The dependence of ℓ on the current bath, radius, and window is part of the problem. Several attempted proofs failed precisely by suppressing one of these dependencies.

9.2 Voids, negative association, and the unknown vacant path

The most immediate determinantal advantage is the smaller probability of a prescribed hole. For a bounded Borel set A , the Fredholm expansion gives

$$-\log \mathbb{P}(G(A) = 0) \geq |A| + \frac{1}{2} \iint_{A \times A} e^{-\pi|x-y|^2}, dx, dy. \quad (9.3)$$

The strict positive second term suggested comparing vacant crossings. A vacant curve γ requires an empty radius- r neighborhood $\gamma + B_r$. Thus (9.3) does improve the cost of every *fixed* candidate curve.

The missing operation is taking the union over all possible curves. Comparisons of each event, or of their finite intersections, do not compare their union because inclusion–exclusion has alternating signs. Nor is the length of a curve a lower bound for the area of its tube: repeated traversals and closely spaced turns invalidate that substitution. The earlier Voronoi route made this problem concrete. A Voronoi vertex has degree three in general position, but very short edges allow many successive vertices in a small region. A path count in a realized Voronoi graph does not bound the integral over its random nuclei.

For forced nuclei $X = \{x_1, \dots, x_k\}$, the correct weight is

$$\det H(X, X) \mathbb{P}^{!X}(G(A) = 0).$$

Rank- k eigenvalue interlacing bounds the second factor after discarding the first k eigenvalues of H_A . It does not permit discarding the factor $\det H(X, X)$ when integrating nucleus positions. The archive’s straight-strip calculation supplies the transverse spectral values

$$\lambda_w(a) = \sqrt{2} \int_{-w/2}^{w/2} e^{-2\pi(x-a)^2}, dx$$

and a negative log-hole rate per longitudinal unit length $\int_{\mathbb{R}} -\log(1 - \lambda_w(a)) da$. Its use for arbitrary curved, overlapping tubes would require geometric and spectral estimates not supplied by that calculation.

Conditioning only on an empty region A is favorable for further emptiness estimates: the exterior conditional kernel is

$$H^A = H_{A^c, A^c} + H_{A^c, A}(I - H_A)^{-1}H_{A, A^c} \succeq H_{A^c, A^c}.$$

An exploration also finds occupied points. Palm conditioning subtracts a coherent direction and can reverse the desired hole comparison. The supplied history records a boundary-root disk example where $\mathbb{P}^{!x}(G(B_s) = 0) > e^{-\pi s^2}$ for $1 \leq \pi s^2 \leq 11/10$. This is a recovered counterexample from that history; its original detailed note is not being replaced by an assertion of uniform exploratory domination here.

Negative association has a valid but narrower role. It bounds products of increasing functions, or products of decreasing functions, on *disjoint* spatial supports. It can supply a block criterion once a single good-block probability is sufficiently large. It does not order Ginibre and Poisson crossing probabilities. Even the two-site projection DPP which chooses exactly one of two sites has a larger probability of “at least one” and a smaller probability of “both” than independent Bernoulli-1/2 occupancy. This also explains why directionally convex count orders, factorial-moment bounds, Laplace-functional orders, and negative association cannot be used as crossing orders without an additional argument.

The history contains successive finite combinatorial refinements of a block criterion, culminating in the recorded sufficient condition

$$g_{L, 2L}^{\text{in}}(r) + 2g_{L, L}^{\text{in}}(r) > 23/8.$$

Its use at $r = r_c(\Pi_1)$ would establish strictness by continuity of the finite-window probabilities. No required Ginibre crossing probability was proved to satisfy it. The improved finite criterion and its original finite checks are historical inputs, not fresh certifications in this section.

9.3 Affine deformations: the correct limit and the wrong sign claim

The ellipse calculations and the justified Poisson-limit argument are treated separately in this report. Two additional obstructions belong to the general investigation.

First, isotropy does not imply that disks minimize a critical scale over equal-area ellipses. The archive gives a direct counterexample using a Poisson process of parents, each carrying an independently rotated finite δ -net of the unit disk. Let

$$\lambda = (100\pi)^{-1}, \quad a = 6400\pi, \quad b = a^{-1}, \quad \delta = b/10, \quad E = \{x_1^2/a^2 + x_2^2/b^2 \leq 1\}.$$

Every radius-two offspring disk is contained in its parent's radius-three disk. The parent connection path bound is $36\pi\lambda = 9/25 < 1$, proving nonpercolation of these radius-two offspring disks. Conversely, the ellipses centered at a cloud's net points cover the rectangle $[-a/2, a/2] \times [-1/2, 1/2]$ about its parent. Indeed, choose a cloud point within δ of $(0, x_2)$; the ellipse quadratic form is at most $(3/4)^2 + (1/4)^2 < 1$.

An area-preserving linear map turns this parent rectangle into a square of side $\sqrt{a}$. Squares of side $\sqrt{a}/2$ contain independent parent points with vacancy probability e^{-16} , and nearest-neighbor occupied squares have overlapping grains. An elementary closed-circuit sum is less than one, hence these ellipse grains percolate. After deterministic rescaling to intensity one the strict shape comparison persists. The process is stationary, isotropic and mixing because the clouds have bounded radius. It is clustered; this is not a Ginibre counterexample.

Second, an anisotropic approach to Poisson does not improve all crossings in a common direction. Put $\varepsilon = e^{-t}$ and $\Phi_t = S_t^{-1}G$ with $S_t = \text{diag}(e^t, e^{-t})$. For a bounded observable F supported in a fixed window W , the recovered determinantal difference expansion is

$$\mathbb{E}F(\Phi_t) = \mathbb{E}F(\Pi_1) - \frac{1}{2} \iint |K_t(x, y)|^2 \mathbb{E}D_x D_y F(\Pi_1) dx dy + O_W(\varepsilon^2). \quad (9.4)$$

Only one coordinate collapses. For disk crossings its leading term is

$$-\frac{\varepsilon}{2} \int_{\mathbb{R}^3} \mathbb{E}D_{(u,v)} D_{(u,w)} F(\Pi_1) du, dv dw + o(\varepsilon).$$

Replacing the two insertions by a repeated insertion at one point is incorrect: v and w remain distinct integration variables.

The sign mechanism is explicit. For complete coverage of the horizontal segment $[0, 3r] \times \{0\}$, disks centered on a common vertical line produce nested intervals on the segment. The mixed difference is nonpositive for every background and strictly negative on a positive measure set of backgrounds and insertions. For the vertical segment of length $3r$, an empty background can instead require both insertions. The notes bound its collapsed integral below by

$$r^3 \left[\frac{1}{100} e^{-(6+\pi)r^2} - 50(1 - e^{-(6+\pi)r^2}) \right] > 0$$

when $r^2 < \log(1 + 1/5000)/(6 + \pi)$. By decreasing thin-rectangle limits the signs persist for nondegenerate rectangles. These are actual Ginibre examples. They disprove the orientation-independent crossing claim, while leaving the scalar critical-radius comparison open. The remainder in (9.4) is fixed-window; its constants do not justify an unrestricted near-critical window limit.

9.4 Structure factors, clusters, and exposed contacts

The structure-factor route used

$$S_G(k) = 1 - e^{-\pi|k|^2}, \quad S_{\Pi_1}(k) = 1, \quad \text{Var } G(A) = \iint_{A \times A^c} e^{-\pi|x-y|^2} dx dy.$$

The supplied history records the perimeter bound $\text{Var } G(A) \leq \text{Per}(A)/(2\pi)$ and stronger spectral hole estimates involving both $|A|$ and this variance. These improve prescribed-hole costs but do not solve the unknown-curve union. Second-order information alone does not determine connectivity.

The cluster route tried to compare the Palm susceptibility, namely the expected size of the component containing a restored root. The proposed global inequality in the favorable direction is already false at small range. With $V = \pi R^2 \leq 1$, the recorded path estimate is

$$\chi_G(R) \leq \frac{1}{1 - (V - 1 + e^{-V})} < 1 + V \leq \chi_{\Pi_1}(R).$$

The strict middle inequality follows from $e^{-V} < 1/(1 + V)$ for $V > 0$. This is consistent with Ginibre's repulsion suppressing immediate Palm neighbors. It does not rule out a different comparison near criticality. The connectedness Ornstein–Zernike identity, when its integrability assumptions hold, contains the full pair connectedness density, not just the ordinary structure factor; replacing one by the other was invalid.

The contact route compared the rate of exposed contacts as the radius increases. The history records a favorable exact local ratio over a specified radius interval, but an exposed contact may either merge two components or close a cycle within one component. Euler characteristic counts components minus holes and therefore does not resolve that distinction. A macroscopic comparison needs the contact weighted by the exterior connections that make it pivotal.

For finite Ginibre, angularly averaging the conditional law of a pair $m \pm se^{i\theta}$ gives a half-separation density proportional to

$$s^3 e^{-2\pi s^2} \sum_k |c_k|^2 s^{4k}.$$

On a sufficiently short truncated interval this is more spread out than the corresponding Poisson density proportional to s . Its angular law also changes, and spreading a pair can break its exterior attachments. No connection-preserving transport to the Ginibre law was obtained.

9.5 Coulomb dynamics and finite spherical systems

These calculations occur in Sections 7–8 of the user's research history. They are retained here as recovered finite-system work; this report does not certify a Poisson-start infinite Ginibre diffusion.

An unrenormalized Poisson force cannot be used for such a construction. The annular increment

$$Z_R = \sum_{z \in \Pi_1, R < |z| < 2R} \frac{z}{|z|^2}$$

has mean zero and covariance $\pi \log 2$ times the identity. Its summands are bounded by $1/R$, and the Poisson characteristic-function expansion therefore yields a nondegenerate Gaussian limit. In particular the radial cutoff force is not Cauchy in probability. Force differences at two fixed roots have square-integrable far tails, so this rules out the unrenormalized construction,

not all relative-coordinate constructions. Equilibrium stochastic differential equations do not by themselves construct the intended process from Poisson initial data.

Two proposed instantaneous conclusions were also too strong. If a coupling gives a Poisson displacement process Y_t and $\sup_i |X_i(t) - Y_i(t)| \leq a(t)$, then

$$\mathcal{O}_{r-a(t)}(Y_t) \subseteq \mathcal{O}_r(X_t) \subseteq \mathcal{O}_{r+a(t)}(Y_t), \quad |r_c(X_t) - r_c(\Pi_1)| \leq a(t).$$

If $a(t) \rightarrow 0$, there is no instantaneous jump of fixed size. Likewise, local convergence to Poisson plus sufficiently uniform spatial decorrelation can imply $\liminf_{t \downarrow 0} r_c(X_t) \geq r_c(\Pi_1)$. Neither hypothesis was proved for the proposed unscreened dynamics.

On a sphere of area A , let μ_n be n independent uniform points and let $p_n(r) = \mathbb{E}_{\mu_n} F_r$. For the repulsive diffusion with generator

$$L_n^+ = \sum_i \Delta_i + 2 \sum_i \sum_{j \neq i} \nabla_i \log |x_i - x_j| \cdot \nabla_i,$$

the archived initial-response calculation gives

$$\left. \frac{d}{dt} \mathbb{E}_{\mu_n} F_r(X_t) \right|_{t=0+} = \frac{4\pi n(n-1)}{A} [p_n(r) - p_{n-1}(r)] \geq 0. \quad (9.5)$$

Distances in the logarithm are chord distances; gradients and Laplacians are spherical. The invariant density is proportional to $\prod_{i < j} |x_i - x_j|^2$. The local equilibrium limit at $A = n \rightarrow \infty$ is the intended Ginibre law. The limit of the coefficients in (9.5), for a fixed physical observation window, is $4\pi \partial_\lambda \mathbb{E} F_r(\Pi_\lambda)|_{\lambda=1}$. This is a limit of initial derivatives, not a derivative of a constructed infinite-volume evolution. No common positive-time interval uniform in particle number and crossing scale was obtained.

The attractive process killed at collisions supplies an exact finite duality in the archive. With $c_n = 4\pi n(n-1)/A$, the adjoint repulsive semigroup is $e^{c_n t}$ times the attractive killed semigroup. Thus

$$\mathbb{E}_{\mu_n}^+ F(X_t) = \mathbb{E}_{\mu_n}^- [F(Y_0) \mid \tau > t].$$

A useful proof would compare crossing with attractive survival. The identity alone gives no such correlation sign.

Merging particles at collisions yields a finite coalescent whose positions, conditional on its count, remain independent uniform at deterministic times. Its count decreases from n at rate c_n , and the recorded density limit is $(1 + 4\pi t)^{-1}$. Reversing it creates births at existing points. A renewal identity yields the positive linear term in (9.5) and, under specified contact-surface regularity hypotheses, a negative correction of order $t^{3/2}$. Those remainders were fixed-system. Removing a newborn changes later forces on all particles, so a birth is not an independent local enhancement. Movement, feedback, and rare-pivotal normalization were not controlled.

Finally, the conjectured explicit identification of this evolution with the β -Ginibre family fails. The pair equation agrees with $\beta' = 8\pi(1 - \beta)$, but at three collinear points of adjacent squared distance $\beta \log 2/\pi$ the recorded three-point discrepancy is

$$\frac{\pi(1 - \beta)}{2\beta} (5 - 6 \log 2) > 0, \quad 0 < \beta < 1.$$

Pair-correlation agreement therefore cannot identify the law. This countercalculation must be retained if the dynamics route is resumed.

9.6 Static interaction-strength interpolation

On a sphere of area N and radius R_N , the separate static family in the history is

$$d\mu_{N,\theta} = Z_{N,\theta}^{-1} e^{-\theta H_N} d\mu_N, \quad H_N = - \sum_{i < j} \log \frac{|x_i - x_j|}{2R_N}, \quad 0 \leq \theta \leq 2.$$

For a crossing-failure event B of positive probability, differentiation of the normalized finite integral gives exactly

$$\log \frac{\mu_{N,2}(B)}{\mu_{N,0}(B)} = - \int_0^2 (\mathbb{E}_{N,\theta}[H_N | B] - \mathbb{E}_{N,\theta} H_N) d\theta. \quad (9.6)$$

Thus a uniform positive integrated energy excess for macroscopic failure would be useful. That is the unproved principal quantity, not a consequence of logarithmic repulsion alone.

The recorded Poisson-endpoint calculation is singular in ambient volume: for a fixed local observable F ,

$$\partial_\theta \mathbb{E}_{N,\theta} F|_{\theta=0} = - \frac{\log N}{4} \partial_\lambda^2 \mathbb{E} F(\Pi_\lambda) \Big|_{\lambda=1} + O_F(1).$$

Some small rectangular crossings have positive intensity curvature, which rules out an unrestricted positive crossing derivative in this interpolation. Conditioning on exactly n points in a fixed planar region W removes that divergent count contribution. The limiting derivative then involves

$$- \text{Cov}_{\mu_{n,W}}(F, J_W), \quad J_W = - \sum_{i < j} \log |x_i - x_j| + \sum_i \int_W \log |x_i - y|, dy.$$

A favorable conditional example in the history does not control the unconditioned count distribution or the entire integral in (9.6).

9.7 Common-realization square replacement and its genuine endpoint gain

Partition the plane into half-open squares of side ℓ . Independently mark each square with probability q ; use the restriction of one common Ginibre realization in marked squares and an independent Poisson realization in the others. Write $\Sigma_{q,\ell}$ for the result. Its endpoints are exactly Poisson and Ginibre, but the intermediate law is generally not determinantal. For a local event,

$$\partial_q \mathbb{E} F(\Sigma_{q,\ell}) = \sum_Q \mathbb{E} [F(\Sigma_{q,\ell}^{Q,1}) - F(\Sigma_{q,\ell}^{Q,0})].$$

At $q = 0$, the exterior of the replaced square is independent Poisson. When the square is a clique, write $A, B \subset Q$ for locations attaching an inserted disk to the two relevant exterior components. Put $h(S) = e^{-|S|} - \mathbb{P}(G(S) = 0)$. The exact replacement gain, conditional on this exterior, is

$$h(A) + h(B) - h(A \cup B). \quad (9.7)$$

Even though $h \geq 0$, (9.7) need not be nonnegative. The positive endpoint argument instead compares robust favorable geometries, such as $A = B = Q$, with adverse thin-shell geometries.

The supplied history records, using its Poisson arm inputs and local modifications, a strictly positive derivative

$$\partial_q C_{L,\ell}(0, r_c(\Pi_1)) \geq c\ell^2 I_L, \quad I_L = \int \mathbb{E} D_x F_{L,r_c(\Pi_1)}(\Pi_1) dx.$$

This is a substantial historical endpoint result. The detailed earlier arm proofs are not reproduced and freshly certified here. It was not lost when the later report excluded an unjustified positive-density continuation.

The exact obstruction to a pointwise continuation is the dependence of the common Ginibre realization across squares. The relevant void law becomes conditional on observed Ginibre exterior points. The history records essential conditional hole-probability extrema zero and one for every $q > 0$, using rigidity and finite fully marked annuli. Hence uniform improvement for every exterior cannot replace an averaged argument. Moreover, when the Goldman extra point lands in a square, its mark forces the entire square's color; it is not an independent mark on one point.

For fixed L the recorded mesh expansion, with $a = \ell^2$, is

$$\partial_q C_{L,\ell}(0, r_c(\Pi_1)) = \frac{a}{2} I_L + o_L(a), \quad \partial_q^2 C_{L,\ell}(0, r_c(\Pi_1)) \longrightarrow -J_L,$$

where

$$J_L = \iint e^{-\pi|x-y|^2} \mathbb{E} D_x D_y F_{L,r_c(\Pi_1)}(\Pi_1) dx dy.$$

The history's controlled positive-density regime is $L \leq c\ell^{-1/3}$ and $q \leq c\ell^2$. It is not a macroscopic critical-radius comparison. Indeed, in the natural coupling,

$$d_{\text{TV}}(\Sigma_{q,\ell}|_W, \Pi_1|_W) \leq 2q|W|.$$

For its growing windows this is $O(L^{-4})$, so the total crossing gain tends to zero. Also a lower derivative bound by a fixed positive multiple of $I_L \rightarrow \infty$ cannot hold throughout a fixed interval in q : integrating would make a probability increase by more than one. The necessary repair is a comparison normalized by influences of the *current* law.

The subsequent fine-mesh calculation identifies the local limit

$$\Sigma_{q,\ell} \longrightarrow \text{thin}_q(G) \cup \Pi_{1-q}$$

in total variation, uniformly in q on a fixed window. Couple square marks to point marks unless two source points share a square; the union of the two source processes has second factorial density at most four, giving an error at most $2\ell^2|W|$. In this limit the first derivative at zero vanishes and the second derivative is $-J_L$. A real order- a endpoint effect therefore disappears with the mesh; it cannot be silently retained as a positive limiting derivative.

9.8 Fixed-kernel Poisson expansion and a positive long-arm family

The recovered source `research/fixed_kernel_poisson_second_variation.md`, with `fixed_kernel_poisson_second_variation_audit.md`, goes beyond the second derivative in the preceding subsection. For $X_s = \text{DPP}(sH_1) \cup \Pi_{1-s}$ and a bounded functional F supported in W , put $m_W = |W|$ and let B_n have zero diagonal and off-diagonal entries $H_1(x_i, x_j)$. The exact centered factorial density is $s^n \det B_n$, so

$$\mathbb{E} F(X_s) = \sum_{n \geq 0} \frac{s^n}{n!} \int_{W^n} \det B_n \mathbb{E}_\Pi [D_{x_1} \cdots D_{x_n} F] dA^n. \quad (9.8)$$

Hadamard's bound $|\det B_n| \leq (n-1)^{n/2}$ and $|D^n F| \leq 2^n \|F\|_\infty$ make this an entire series in s . Writing $C(s) = \mathbb{E} F(X_s)$, a cycle majorant gives, for $|s| \leq 1/8$,

$$|C(s) - C(0) - \frac{1}{2}s^2 C''(0)| \leq \|F\|_\infty \left[\frac{64}{3} m_W |s|^3 + \frac{25}{2} m_W^2 s^4 e^{5m_W s^2} \right].$$

This is explicitly a fixed-window error.

For a deterministic background with two terminal-attached parents at distance d , where $R < d < 2R$ and $R = 2r$, let V be their radius- R lens and A, B the two exclusive crescents. With $q(h) = e^{-\pi|h|^2}$, the joint pair contribution is

$$\text{Red} = \int_{V^2} q(y-x) dx dy, \quad \text{Coop} = 2 \int_{A \times B} q(y-x) \mathbf{1}_{\{|x-y| \leq R\}} dx dy.$$

Gaussian translation and bounded variation imply $\text{Red} \geq |V| - \text{Per}(V)/(2\pi)$. The two exclusive sets lie on opposite sides of the perpendicular bisector; integrating the Gaussian over these halfplanes and their vertical span gives $\text{Coop} \leq 2R/\pi$. Consequently

$$\liminf_{d \downarrow R} (\text{Red} - \text{Coop}) \geq R \left[(2\pi/3 - \sqrt{3}/2)R - \frac{2}{3} - \frac{2}{\pi} \right].$$

It is strictly positive if

$$R > \frac{2/3 + 2/\pi}{2\pi/3 - \sqrt{3}/2}, \quad (9.9)$$

including the source's rationally checked value $\pi R^2 = 9/2$. The limit is one-sided: at $d = R$ the old parent edge opens. For comparison, at $d = 5R/2$ there is no singleton bridge but an open set of cooperative pairs, hence a strictly negative contribution. This proves neither a universal conditional sign nor a Poisson average.

The positive estimate extends to arbitrarily long arms and their neutral Poisson background. Put the tips at 0 and d on the horizontal axis. Keep all other left-arm centers at first coordinate at most $-\ell_0$, and all other right-arm centers at least $d + \ell_0$, with $2R - d < \ell_0 < R$; all centers lie in a strip $|y| \leq b_0$. Let U_S, U_T be the radius- R neighborhoods of the two connected arms. Their intersection is exactly the tip lens. For every neutral configuration outside $U_S \cup U_T$,

$$\text{Red} - \text{Coop} \geq |V| - \frac{\text{Per}(V)}{2\pi} - \frac{2(R + b_0)}{\pi}. \quad (9.10)$$

The bound permits cooperative paths through any number of neutral vertices. It is positive for d sufficiently close to R and b_0 small under (9.9). Conditional on the arms and no additional points in $U_S \cup U_T$, the entire remaining Poisson process can therefore be integrated. The actual void factor $e^{-|U_S \cup U_T|}$ is retained.

For the true seed-to-outer event, take collinear arms from the deterministic terminal attachment sets to tips $u_0, v_0 = u_0 + d$. With positive margins $D_{\text{seed}} = v_0 - R - (m + r)$ and $D_{\text{outer}} = L - r - (u_0 + R)$, restrict the other centers to the free part of the active window, outside both attachment sets and arm neighborhoods. The bound becomes

$$\text{Red} - \text{Coop} \geq |V| - \frac{\text{Per}(V)}{2\pi} - \frac{2R}{\pi} - 2\pi(m+r)^2 e^{-\pi D_{\text{seed}}^2} - 2|U_S| e^{-\pi D_{\text{outer}}^2}.$$

At $u_0 = (L + m)/2$ with fixed m, d , the errors vanish as $L \rightarrow \infty$, since $|U_S| = O(L)$. Strict edge and terminal slack allow open perturbations of the arm positions. This is a favorable positive-probability family for the actual event, with all free neutral points integrated. Its probability may be very small, and no uniform fraction of pivotal weight or verification of (9.9) at r_P is asserted.

9.9 Bottleneck thresholds, cloning, and contact surgery

For a finite terminal graph the first crossing radius is a minimax:

$$\mathcal{R}(\eta) = \min_{\gamma} \max_{e \in \gamma} w_e, \quad w_{xy} = |x - y|/2.$$

Terminal edge weights are distances to the relevant terminal. With $T = \max(a, \min(b, \mathcal{R}))$, the deterministic layer-cake identity is

$$T = b - \int_a^b F_r, dr.$$

This avoids differentiating a nonsmooth threshold at every radius.

A stronger deterministic comparison of a two-point gain with integrated one-point gains is false. In $[0, 6] \times [0, 1]$, take centers

$$(1/2, 1/2), (1, 1/2), (3, 1/2), (5, 1/2), (11/2, 1/2).$$

The left–right threshold is one and remains one after any single insertion. Insert instead $(3 - h, 1/2)$ and $(3 + h, 1/2)$, where $0 < h < 2/3$. The threshold becomes $1 - h/2$. The insertions can be arbitrarily close. Perturbing the central old point removes the tied bottleneck while making the integral of one-point gains arbitrarily small and retaining a positive two-point gain. Probability weighting, rather than a deterministic inequality, is essential.

For the whole-plane seed-to-outer event, adding a clone at displacement h decreases T by at most $|h|$. The parent integration must include the whole plane: a parent outside a nominal center window may have a clone entering the active region. Dropping this term was one of the corrected cofactor-boundary errors.

For Poisson intensity λ , define the Gaussian clone gain I_ε and the mean absolute move gain M_ε using displacement εZ , with density $e^{-\pi|z|^2}$ for Z . The exact decomposition is

$$I_\varepsilon = \frac{M_\varepsilon}{2\lambda} + J_\varepsilon, \quad J_\varepsilon \geq 0, \quad (9.11)$$

where J_ε integrates

$$\min\{T(\eta + x), T(\eta + y)\} - T(\eta + x + y).$$

It counts connections requiring both insertions. The movement part satisfies

$$M_\varepsilon \leq \frac{\varepsilon}{\pi} \mathbb{P}(a < \mathcal{R} < b).$$

The coefficient follows because a generic unique center–center bottleneck has two endpoint gradients of norm $1/2$, a terminal bottleneck has one gradient of norm one, and $\int e^{-\pi|z|^2} |z \cdot v| dz = |v|/\pi$. Mecke’s formula, followed by translation of the moved point, makes this bound uniform in the window. In contrast, the dominated convergence giving $I_\varepsilon/\varepsilon \rightarrow \mathbb{P}(a < \mathcal{R} < b)/(2\pi\lambda)$ is fixed-window.

The cooperative-pair notes develop a more ambitious bound of order $|h|^2$ times the ordinary pivotal integral. Their construction retains two occupied arms, excludes bypasses, and treats circular terminal strips separately. The positive-bath version fills a resampling region with Poisson points and its constants deteriorate as the bath vanishes. The zero-bath version uses sparse Fock projection estimates and conditional Janossy densities. These lengthy geometric chains are *recovered calculations*; their exact relationship to the claimed small- β theorem and their outstanding verification are set out in the separate small- β section.

Contact surgery isolated a further precise issue. At a regular artificial-grain contact $y - w$, $|y - w| = 2r$, the open lens $B(y, 2r) \cap B(w, 2r)$ contains no background point. A predecessor z may nevertheless approach y from the rear. Its normalized coherent contribution is $e^{-\pi|z-y|^2/\beta}$ and tends to one. The canonical contact partition gives the archived reduction

$$N \leq sB + (1 - s) \int p_A(y), d\mu_{\text{contact}}(A, y),$$

where B is total contact mass and p_A is the forced-certificate projection diagonal. Thus the random remainder is controlled, while the contact-weighted predecessor term still needs geometry.

Two shortcuts were explicitly ruled out. First, two terminal clusters with gap $4r + d$ have zero one-grain contact mass at radius r , but two nearby insertions can bridge them. Second, at fixed β and zero bath a local connection requiring n points has leading contact mass $b_{n-1}s^{n-1}$, and its leverage-weighted mass has a positive leading coefficient of the same order. Their ratio therefore does not vanish like s as $s \downarrow 0$. This does not refute a bound restricted to $s \geq \beta/2$, or a uniform constant below one at small β .

9.10 Recovered half-Gaussian and twin-contact identities

Sections 5–6 of `research/current_law_contact_obstructions.md` calculate the portion of contact surgery that a half-space estimate actually controls. At a regular contact $|y - w| = 2r$, put $n = (w - y)/(2r)$ and $z = y + h$. Absence of a bypass for $|h| < 2r$ requires $h \cdot n \leq |h|^2/(4r)$. For $q_\beta(h) = \beta^{-1}e^{-\pi|h|^2/\beta}$ the full Gaussian mass of the allowed side obeys

$$p_\beta(r) \leq \frac{1}{2} + \frac{\sqrt{\beta}}{6\pi r} + e^{-\pi r^2/\beta}.$$

Indeed, on $|h| \leq r$, its positive normal coordinate is at most the square of the tangential coordinate divided by $3r$; the latter coordinate has variance $\beta/(2\pi)$.

This factor bounds only contacts which remain pivotal when z is deleted. In the contact measure $\delta(|y - w| - 2r)C(\eta + z; y, w) dy$, the complementary nonnegative part has $C(\eta; y, w) = 0$: the new point creates the contact's connectivity requirement. Denote its integrated mass by Q_β . The original contact mass can vanish while $Q_\beta > 0$, as in the gap example above. Swapping y, z moves its matching contact to a different artificial radius and does not cancel it at radius r . Equivalently, with the z -grain fixed at radius r and y -grain radius u , the difference $J(u) = F(\eta + B(y, u) + B(z, r)) - F(\eta + B(y, u))$ is an interval indicator when the background and z alone fail. The contact difference is the signed measure dJ/du .

There is also an exact symmetric perimeter identity. For fixed $0 < |h| < 2r$, define the successful dimer midpoint set

$$U_h(\eta) = \{c : F_r(\eta + \{c - h/2, c + h/2\}) = 1\}.$$

At a regular boundary point exactly one endpoint supplies the critical contact. Coarea, with both contact and midpoint translation speed one, gives

$$\int dc [A_{c-h/2}^r(\eta + c + h/2) + A_{c+h/2}^r(\eta + c - h/2)] = \text{Per}(U_h(\eta)).$$

Here A_y^r denotes the contact measure for increasing only the artificial grain at y ; identities may be read against radius test functions. Gaussian symmetry then yields

$$\int_{|h| < 2r} q_\beta(h) \int A_y^r(\eta + y + h) dy dh = \frac{1}{2} \int_{|h| < 2r} q_\beta(h) \text{Per}(U_h(\eta)) dh.$$

The comparison with $\text{Per}(U_0)$ remains unproved; the separate cofactor count term has additional weights. Thus neither identity gives a half-factor for the full leverage integral.

9.11 Deletion certificates and conditional count tails

The certificate method produced estimates normalized by the actual event probability. Suppose a point process η obeys reduced Campbell domination

$$\mathbb{E} \sum_{z \in \eta} g(z, \eta - z) \leq M \int \mathbb{E} g(z, \eta) dz$$

for nonnegative g . If an event E has a witness using at most k protected points in a cell Q , deletion of every other point preserves E , whence

$$(N(Q) - k)\mathbf{1}_E \leq \sum_{z \in \eta \cap Q} \mathbf{1}_E(\eta - z).$$

Applying Campbell gives

$$\mathbb{E}[N(Q)\mathbf{1}_E] \leq (M|Q| + k)\mathbb{P}(E). \quad (9.12)$$

A shortest path uses at most two vertices of a clique cell; if the artificial pivotal root lies in that clique, at most one removable witness vertex is needed there.

The exponential version is equally explicit. Multiplication by $e^{tN(Q)}$ and the same Campbell argument yield

$$F'(t) \leq (k + M|Q|e^t)F(t), \quad F(t) = \mathbb{E}[e^{tN(Q)}\mathbf{1}_E],$$

and hence

$$\mathbb{E}[e^{tN(Q)} \mid E] \leq \exp\{kt + M|Q|(e^t - 1)\}, \quad t \geq 0. \quad (9.13)$$

Factorial-moment domination supplies the needed exponential integrability, or one may first use finite generating-series truncations. For $\text{DPP}(sH_\beta) \cup \Pi_\nu$, $s < 1$, the applicable constant is $M = \nu + s/[\beta(1 - s)]$. These are actual event-weighted estimates, not independence claims.

They control counts but not a nearly singular coherent Gram inverse. At $s \uparrow 1$ the displayed Campbell constant also diverges. An abstract count-only attempt to absorb this divergence fails for a rank-one projection supported in a region Q : the event $\{N(Q) \geq 2\}$ has a two-point deletion certificate but probability zero, whereas one artificial insertion can make it occur whenever the single retained point is present. The residual leverage is then one. This is not a Ginibre pivotal counterexample; it disproves the proposed abstract count-only implication.

The canonical-certificate ordering is important. Choose candidates by an intrinsic ordering depending only on the candidate's own vertices and deterministic geometry. After forcing a chosen tuple A , absence of an earlier valid witness and absence of a bypass are decreasing conditions on the remainder. An ordering whose validity changes when unselected points are deleted would not justify the same partition.

For a normalized root row f of a residual projection, let $t = \int_{B^c} |f|^2$. The empty-region rank-one resolvent calculation gives, under an additional decreasing exterior event, the useful remainder bound

$$\frac{st}{1 - s + st}.$$

The sparse-Fock notes then aim to make t and the certificate projection small at small correlation scale. They use divided differences for near-colliding points before block Gram estimates. They do not assert well-conditioning of the original coherent Gram matrix or a deterministic localization theorem at $\beta = 1$.

9.12 A protected close-pair bound uniform at the projection endpoint

The recovered estimate in `pivotal_close_pair_estimate.md` concerns an internal consecutive DPP pair $z, z + h$ on an induced pivotal path, with $|h| \leq \delta < r$. Both outside neighbors must be old random centers, rather than the artificial root or a terminal. Chordlessness places them in the two disjoint crescents $B(z, 2r) \setminus B(z + h, 2r)$ and $B(z + h, 2r) \setminus B(z, 2r)$, each of area at most $4r|h|$.

For z in a fixed bounded set Q , let $K_{Q,\delta}$ count all eligible ordered quadruples. Choose nested fixed disks containing the quadruples, and let E_{ext} be measurable outside the larger disk. Its weighted factorial bound is preserved under deletion of the four points. The buffered exterior

Bergman estimate supplies constants b_0, b_1 on the smaller disk with diagonal at most b_0 and pair determinant at most $b_0 b_1^2 |h|^2$. For $\text{DPP}(sH_1) \cup \Pi_\nu$, summing the neighbor colors and integrating the pair separation gives

$$\mathbb{E}[\mathbf{1}_{E_{\text{ext}}} K_{Q,\delta}] \leq \frac{16\pi}{3} s^2 b_0 b_1^2 (sb_0 + \nu)^2 r^2 |Q| \delta^6 \mathbb{P}(E_{\text{ext}}). \quad (9.14)$$

The sixth power consists of two separation dimensions, two crescent widths and the quadratic determinantal zero. The earlier Campbell bound had constant $(16\pi^2/3)[s/(1-s)]^2[s/(1-s)+\nu]^2$ and diverged at $s = 1$. Equation (9.14) does not.

The note records submission for independent audit; the estimate is restored here as a recovered local calculation. A separate arm-preserving comparison $\mathbb{P}(E_{\text{ext}}) \leq C\mathbb{P}(\text{Piv}_x)$ is still needed. Root and terminal neighbors require additional strip arguments, and a shrinking close-pair cutoff cannot exclude the fixed-separation saturation family below.

9.13 Fock saturation, negative thin lenses, and positive ray examples

The exact bridge multiplier and its variational characterization are proved elsewhere in this report. Their significance here is that they separate two effects: conditioning on occupied arms removes coherent directions, while the necessarily empty bridge region increases the resolvent.

For two parents $A = \{-a, a\}$ with $R/2 < a < R$, let $B_a = B(-a, R) \cap B(a, R)$. Direct inversion of the two-by-two Gram matrix gives

$$P_A(0, 0) = \frac{2e^{-\pi a^2}}{1 + e^{-2\pi a^2}} = \text{sech}(\pi a^2) > 0.$$

As $a \uparrow R$, the lens area tends to zero. For the Lebesgue-integrated trace surplus $\int_{B_a} (M_{A,B_a,s}(x) - 1) dx$, the loss of coherent directions is bounded below by a positive constant times $|B_a|$, whereas the positive resolvent trace correction is $O(|B_a|^2)$. These are integrated estimates: the pointwise deficit at the origin does not vanish with the lens area. This proves the failure of a certificate-by-certificate nonnegative integrated surplus, including at $s = 1$ once $|B_a| < 1$.

The later saturation construction is stronger and is preserved as a recovered geometric calculation. Two induced paths traverse paired grid points with macronode spacing $D = R + \epsilon/2$ and fixed small within-pair displacement. If $D^2 < 2$, their eventual doubled-lattice subset has zero density $2/D^2 > 1$. Jensen's formula and the Fock evaluation bound $|f(z)| \leq \|f\| e^{\pi|z|^2/2}$ imply completeness of those coherent directions. The forced root deficits $q_N = 1 - P_{A_N}(0, 0)$ therefore tend to zero. Strict geometric inequalities retain two disconnected arms and an actual pivotal insertion, including circular terminals.

Their *entire* bridge region is one fixed thin lens B , independent of path length, with $|B| < 1$. The scalar estimate is elementary:

$$\|(Q_{A_N})_B\| \leq |B|, \quad \int_B |Q_{A_N}(z, 0)|^2 dz \leq q_N |B|,$$

so its multiplier satisfies

$$0 \leq M_{A_N,B,s}(0) \leq \frac{q_N}{1-s|B|} \leq \frac{q_N}{1-|B|} \longrightarrow 0 \quad (9.15)$$

uniformly for $0 \leq s \leq 1$. The full geometric verification is in the source notes. This construction shows why deterministic certificate positivity cannot be the missing estimate; it does not assess the probability of those certificates under actual pivotal conditioning.

The archive also contains a concrete *positive* family. Put

$$\pi a^2 = 11/5, \quad \pi t^2 = 2/5, \quad R = a + t, \quad \ell = 99R/100, \quad A_N = \{\pm(a + n\ell) : 0 \leq n \leq N\}.$$

Let $B_t = B(0, t)$, a disk contained in the two-parent bridge lens. The recovered rational bounds for a void in this disk give $M_{A_N, B_t, s}(0) > 1 + 3/1000$, uniformly in N and $24/25 \leq s \leq 1$. The proof retains the two-parent Schur complement before controlling all remaining ray vertices; it is not an estimate obtained by ignoring the arms. But no uniform positive fraction of pivotal configurations was shown to have that geometry, and the available analytic threshold bounds did not place its radius below the Poisson critical radius.

A second local construction moved two tight triples and increased their probability by a holomorphic circle average. This did not make them positive bridge targets. For its particular six-point family with $\pi R^2 = 9/2$, the recorded analytic calculation gives

$$P_A(0, 0) > 27/100, \quad |B| < 15/56, \quad M_{A, B, s}(0) < 1022/1025 < 1$$

uniformly in the phase and $s \leq 1$. Probability amplification and positive pivotal multiplier are separate requirements.

9.14 Count-preserving motions and exterior holomorphic fields

The conditional Ginibre density in a disk, given its full exterior and its rigid inside count, has a Vandermonde factor, a Gaussian factor, and the squared modulus of a nonvanishing holomorphic exterior factor. Consequently a fixed count-preserving motion must be compared under that exact density, not an independent Ginibre density with the exterior discarded.

For two parents moved from $\pm a$ to $\pm b$, the full Palm likelihood contains

$$\prod_w \left| \frac{w^2 - b^2}{w^2 - a^2} \right|^2$$

with the prescribed radial product and normalizing constant. The motion notes establish the exact inverse-moment range $0 < q < 1$ for the unrestricted likelihood. The obstruction at $q \geq 1$ is already local: a target Palm zero at b has density vanishing quadratically, whereas the source Palm density there is positive. Integrating the inverse likelihood therefore includes

$\int_{|w-b|<\delta} |w-b|^{-2q} dw$, which diverges exactly at $q \geq 1$. The same local obstruction persists under every nonzero thinning without an uncolored Poisson bath. Large-order exterior moment estimates under a genuine latent void cannot be substituted for these unrestricted moments.

Fixed-count support and averaged motion. The source `bridge_motion_continuation.md` gives a separate geometric obstruction. Put two exterior ports at $\pm(a + R - \delta)e_1$ and retain exactly two movable inside parents. Any replacement preserving their separate terminal attachments has parents u, v satisfying $|u - v| \geq 2a - 2\delta$. A proposed widening requiring $|u - v| \leq 2b$ is therefore impossible when $b < a - \delta$, even before probabilities are compared. The ports and paths admit open perturbations.

Under a genuine latent void of radius $D > a > b > 0$, the motion notes give an averaged quantitative repair. Put

$$\Delta = a^2 - b^2, \quad B_0 = \frac{2\Delta}{D^2 - a^2}, \quad V_0 = \frac{4\pi\Delta^2}{D^2 - a^2}, \quad \kappa = e^{2\pi\Delta} \frac{b^2}{a^2} \frac{1 - e^{-4\pi a^2}}{1 - e^{-4\pi b^2}}.$$

For the same residual event E under the reduced Palm laws μ_a, μ_b , with $E \subset \{X(B_D) = 0\}$ and $p = \mu_a(E) > 0$,

$$\frac{\mu_b(E)}{\mu_a(E)} \geq \kappa e^{-2B_0} p^{1/q} \exp[-\tfrac{1}{2} q e^{qB_0} V_0] \quad (q > 0). \quad (9.16)$$

The bounded far log-product has variance proxy V_0 and centered log-MGF at most $\frac{1}{2} t^2 e^{|t|B_0} V_0$; Holder gives this bound. Optimizing it, as checked in `ginibre_motion_independent_audit.md`, gives a relative factor $\exp[-(B_0 + o(1)) \log(1/p) / \log \log(1/p)]$ as $p \downarrow 0$. It still vanishes for rare events. This optimization concerns the void-restricted product and does not extend the unrestricted inverse-moment range $q < 1$.

More generally, for $L = d\mu_b/d\mu_a$ and independent thinning T_s , the exact pushed-forward likelihood is

$$\frac{d(T_s \mu_b)}{d(T_s \mu_a)}(\eta) = \mathbb{E}_{\mu_a}[L(X) \mid T_s X = \eta].$$

Adding a bath and forgetting colors is another Markov kernel with the same conditional-expectation rule. Finite full-law inverse moments transfer by conditional Jensen, but a thinned observed void does not imply the latent void needed in (9.16). Changing the parent-dependent definition of E also requires a geometric inclusion or a justified configuration map.

A holomorphic circle does not uniformly close a lone two-parent gap. If $z_1(t), z_2(t)$ are holomorphic in the unit disk and continuous on its boundary, the maximum modulus principle prevents $|z_1(t) - z_2(t)|$ from being strictly smaller at every boundary point than at $t = 0$. More generally a fixed witness path cannot have all its edge lengths uniformly improved on the whole parameter circle unless the central path was already improved.

Redundancy allows the active witness to change with phase. The archive constructs four points within η of c , moving with velocities $\rho(1, i, -1, -i)$. If, for example, $\rho = r/4$ and $\eta \leq r/100$, their radius- r occupied union contains the old occupied union at every boundary phase. Jensen's formula for the six Vandermonde factors gives the recovered density amplification

$$e^{-4\pi\rho^2} \left(\frac{\rho}{\sqrt{2\eta}} \right)^{12}.$$

The corresponding two-triple construction has factor

$$\frac{1}{20} e^{-6\pi\rho^2} \left(\frac{\sqrt{3}\rho}{2\eta} \right)^{12}.$$

The factor $1/20$ is a labeling allocation, not a free multiplicity gain. These are comparisons for *preexisting* redundant clusters. No uniform supply of such clusters under rare two-arm conditioning was proved.

Moving whole arms with tapered displacements preserves their slack edges, but integrating over the moving root lens introduces an additional obstruction. Its area, for parent separation d , is

$$A(d) = 2R^2 \arccos(d/(2R)) - \frac{d}{2} \sqrt{4R^2 - d^2}.$$

Although $\Delta A(d) > 0$ for thin lenses, one has $\Delta \log A(d) < 0$ for every $0 < d < 2R$. Indeed $A' = -\sqrt{4R^2 - d^2}$, $A'' = d/\sqrt{4R^2 - d^2}$, and $A(d) < (2R - d)\sqrt{4R^2 - d^2}$ imply $AA'' - (A')^2 < 0$; the radial term $A'/(dA)$ is also negative.

Under a holomorphic motion $Z + tv$, the exact density-weighted lens mass has the form

$$W(t) = e^{-\kappa|t|^2} A(|2a - 2\delta_0 t|) |f_O(t)|^2, \quad \kappa = \pi \sum_j |v_j|^2,$$

where f_O is nonvanishing holomorphic and depends on the exterior. Writing $u = \nabla \log A$ and $v_O = \nabla \log |f_O|^2$ at zero gives

$$\frac{\Delta W(0)}{W(0)} = 4\delta_0^2 \Delta \log A(2a) - 4\kappa + |u + v_O|^2.$$

An exterior field that cancels u makes this strictly negative. The later compact-factor support construction was needed to realize such fields by actual infinite-Ginibre exteriors; a finite polynomial ensemble example alone would not have supplied that conclusion. The fixed-Palm exterior-ratio result proved elsewhere in this report records the completed form of this obstruction. It still concerns arbitrary exteriors, not their weights in a pivotal average.

9.15 Density deficits, random zero sets, and the rare-event loss

A pivotal two-arm certificate can be chosen as two induced paths in *different* background components. Both paths cannot meet the same clique cell. Thus, for every such certificate A in the same realization,

$$|A \cap Q| \leq \min\{G(Q), 2\}. \quad (9.17)$$

This is cap two, not the cap four for an arbitrary union of two paths. Let $a = |Q|$ and $\Delta = a - \mathbb{E} \min\{G(Q), 2\} = \mathbb{E}(G(Q) - 2)_+ > 0$. Strict positivity follows by integrating the strictly positive three-point determinant over three separated target disks in Q . Negative association of disjoint cells and the bounded range $[0, 2]$ give, for a union U of n cells and suitable $\delta > 0$,

$$\mathbb{P} \left(\sum_{Q \subset U} \min\{G(Q), 2\} \geq (1 - \delta/2)|U| \right) \leq \exp\{-\delta^2 a |U|/8\}.$$

This bounds every certificate simultaneously, without a union bound over possible paths.

A recovered conditional density gap. Sections 5–6 of `research/induced_certificate_density_gap.md` strengthen the preceding unconditional estimate under a specific exterior conditioning. For concentric radii $M < R$, $M \leq R/10$, and $\pi R^2 \geq 5$, the weighted Bergman series gives

$$0 \leq B_R(z, z) - 1 \leq \varepsilon_R := 6e^{-c_0 \pi R^2}, \quad |z| \leq M, \quad c_0 = (1 - \log 2)/2.$$

To obtain the bound, split the monomial series at $\lfloor \pi R^2/2 \rfloor$. A Poisson Chernoff bound controls the low modes and the first term of the Poisson upper tail controls the high modes. Thus the constants are uniform on the smaller disk.

Let $U \subset B_M$ be a union of clique cells of area a , let $S_U = \sum_{Q \subset U} \min\{G(Q), 2\}$, and choose R large enough that $\varepsilon_R \leq \delta/4$, where the ordinary capped mean is at most $a(1 - \delta)$. Conditional domination by the fixed Bergman DPP, followed by its negative association, yields

$$\mathbb{P} \left(S_U \geq (1 - \delta/2)|U| \mid G|_{B_R^c} \right) \leq e^{-\delta^2 a |U|/32}. \quad (9.18)$$

Indeed monotone coupling with $G|_{B_M}$ increases each capped mean by at most $a\varepsilon_R$, because the cap is one-Lipschitz in the count. The bounded-variable exponential estimate then applies. For every event E measurable outside B_R , multiplying (9.18) by $\mathbf{1}_E$ retains the factor $\mathbb{P}(E)$, however small it is. The buffer is unobserved; arbitrary pivotal conditioning is not covered. Negative association is used for the dominating Bergman process, not assumed for every conditioned Ginibre law.

The advanced analytic continuation of this argument is preserved as recovered work. It builds an almost surely finite Lipschitz field of smoothing radii from these count tails, smears certificate atoms below density one, and solves a singular weighted $\bar{\partial}$ problem. The recorded conclusion is a simultaneous lower deficit

$$1 - P_A(x, x) \geq c_\delta h^{2+8\pi} e^{-CR(x)^2}, \quad h = \min\{\text{dist}(x, G)/2, 1\}, \quad (9.19)$$

with a Gaussian one-point tail for $R(x)$. A simpler product argument in the archive says that deleting any three actual Ginibre points leaves the exact zero set of a nonzero Fock function. It uses the Bufetov–Qiu Palm-product first moment and Tonelli; it does not assert that deletion of one or two points suffices. These claims concern analytic nonuniqueness and unconditional moments, not a near-critical pivotal comparison.

The probabilistic loss is exact even assuming (9.19). If $\mathbb{P}(R(x)^2 > u) \leq K e^{-cu}$ and $\mathbb{P}(E) = p > 0$, then

$$\mathbb{P}(R(x)^2 > u \mid E) \leq \min\{1, K e^{-cu}/p\}.$$

Integrating gives

$$\mathbb{E}[R(x)^2 \mid E] \leq c^{-1}(\log(K/p) + 1),$$

and for $0 < \eta < c$,

$$\mathbb{E}[e^{\eta R(x)^2} \mid E] \leq \frac{c}{c - \eta} (K/p)^{\eta/c}.$$

This loss is sharp from a tail bound alone: take an exponential random variable and condition on its upper tail. Consequently unconditional inverse moments of a deficit do not give a relative lower bound by a fixed constant times the rare pivotal probability.

The same issue appears in the variance-sensitive potential estimates. For a projection DPP statistic $S_f = \sum f(x)$, with oscillation M and variance V , the recovered exponential-tilt calculation gives

$$\log \mathbb{E} e^{t(S_f - \mathbb{E} S_f)} \leq \frac{V}{M^2} (e^{tM} - 1 - tM), \quad t \geq 0.$$

For $f_L(z) = \min\{L^{-4}, |z|^{-4}\}$, the Ginibre mean is $2\pi/L^2$ and the variance is at most L^{-8} . Hence

$$\mathbb{E}[S_{f_L} \mid E] \leq \frac{2\pi}{L^2} + L^{-4}(\sqrt{2e \log(1/p)} + 2 \log(1/p)).$$

Optimizing the Bennett expression improves the logarithmic profile but does not eliminate its dependence on p . The profile cannot be replaced by a window-independent constant without geometric information about E .

Thinning and a Poisson bath change the variance term. The recovered extension in `thinned_ginibre_potential_bennett.md` applies to $X = \text{DPP}(sH_1) \cup \Pi_\nu$, for a bounded nonnegative integrable f with $f \in L^2$. Put $M = \sup f$ and

$$V_* = s^2 \text{Var}_G\left(\sum f\right) + [s(1-s) + \nu] \int f^2.$$

The same Bennett expression holds with mean $(s + \nu) \int f$ and variance parameter V_* . A finite-rank proof embeds sP_N as the physical block of the projection

$$\begin{pmatrix} sP_N & \sqrt{s(1-s)} U_N \\ \sqrt{s(1-s)} U_N^* & (1-s)I_N \end{pmatrix},$$

where U_N is the basis isometry onto $\text{ran } P_N$. The statistic is zero on the auxiliary atoms. Trace-norm exhaustion and the independent Poisson MGF give the plane limit. For $f_L = \min(L^{-4}, |z|^{-4})$ one may use

$$V_* \leq s^2 L^{-8} + \frac{4\pi}{3} [s(1-s) + \nu] L^{-6}.$$

Consequently an event of probability $p > 0$ has conditional potential mean at most

$$(s + \nu) \frac{2\pi}{L^2} + \sqrt{2eV_* \log(1/p)} + 2L^{-4} \log(1/p).$$

For root-indexed events in a deterministic region of area A with integrated probability $I > 0$, entropy averaging gives the same bracket with $\log(A/I)$, multiplied by I . The L^{-6} term is required away from the full projection without a bath. This extension preserves the rare-event loss.

The corridor proposal aimed to supply that missing normalization. If an exterior-support event H contains the actual pivotal event E , conditional domination may bound a local bad event by

$$\mathbb{P}(E \cap \text{Bad}) \leq \tau_D \mathbb{P}(H).$$

The required next factor is $\mathbb{P}(H) \leq C_D \mathbb{P}(E)$. Emptying a large disk costs on the quartic scale, too much to absorb an area-exponential bad-density tail by that particular surgery. A fixed-width corridor has compressed Ginibre norm $\text{erf}(w\sqrt{\pi/2}) < 1$. The buffered conditional void theorem below makes the exponential-in-length cost precise. But creating a pivotal-producing filling with comparable conditional cost remained missing. Upper domination and hole bounds supply no insertion lower bound.

9.16 Recovered conditional corridor and annular clearing laws

The sources `research/rare_pivotal_weighting_and_corridors.md` and `rare_pivotal_corridor_spectral_audit.md` establish a conditional hole estimate under explicit buffer conditions. If $D \subset B_M$ lies in a width- w strip, write $\kappa_w = \text{erf}(w\sqrt{\pi/2}) < 1$. Choose $R \geq 10M$, $\pi R^2 \geq 5$, so that $\varepsilon_R \leq 1$ and $\varepsilon_R |D| \leq (1 - \kappa_w)/2$. The Bergman approximation then gives $\|B_R|_D\| \leq (1 + \kappa_w)/2$ and $\text{Tr}(B_R|_D) \leq 2|D|$. Using $\log(1 - u) \geq -u/(1 - \|B_R|_D\|)$ on its spectrum, buffered conditional domination yields

$$\mathbb{P}(N_K(D) = 0 \mid X|_{B_R^c}) \geq e^{-4|D|/(1-\kappa_w)} \quad (0 \leq K \leq H_1). \quad (9.20)$$

Fixed reduced-Palm remainders are included. A colored independent bath contributes $e^{-\nu|D|}$. A finite collection of fixed-width straight corridor pieces also has an exponential area cost under the buffered spectral conditions in the source. These statements give hole lower bounds, not occupied insertion lower bounds.

The count-preserving construction in `count_preserving_corridor_clearing.md` is a different positive result. Let U be a rectangle of length ℓ and width w , let U_b expand all four sides by b , and put $V = U_{2\delta}$. Move each unprotected point of U to the exit position of planar Brownian motion from U_b , with b uniform in $[\delta, 2\delta]$; points in $V \setminus U$ stay fixed. The resulting Markov kernel T preserves count and lands outside U . Its adjoint Lebesgue density is at most $1 + (w + 4\delta)/(2\delta)$, by the rectangle torsion bound.

Given the Ginibre exterior and inside count, the positional density is Gaussian times a squared holomorphic modulus. The exact gauge

$$e^{-\pi|z|^2} |F(z)|^2 = e^{-2\pi(\Im z)^2} |e^{-\pi z^2/2} F(z)|^2$$

and subharmonic stopping give a density cost per moved coordinate of at most $A = \exp[2\pi(w/2 + 2\delta)^2]$. Let $C = A[1 + (w + 4\delta)/(2\delta)]$. If almost every source configuration in E is sent into F_{tar} with probability one under the product move, then

$$\mathbb{P}(F_{\text{tar}} \mid \text{outside } V) \geq C^{-n} \mathbb{P}(E \mid \text{outside } V), \quad (9.21)$$

$$\mathbb{P}(F_{\text{tar}}) \geq \mathbb{E}[C^{-N(V)} \mathbf{1}_E]. \quad (9.22)$$

Here n is the conditioned count. The all-output geometric condition is essential: preserving an event under deletion does not ensure its preservation when the deleted points land in a surrounding layer. Protected fixed Palm points enter the holomorphic factor; subsequent canonical-certificate reintegration still needs multiplicity control.

The same construction applies to marked latent full Ginibre plus a Poisson bath, moving all unprotected latent points and carrying their retention marks. The count is the latent count plus bath count. If $p = \mathbb{P}(E)$ and $M = (1 + \nu)|V|$, its count MGF and Holder give

$$\mathbb{P}(F_{\text{tar}}) \geq p^{1+1/q} \exp[-M(C^q - 1)/q] \quad (q > 0).$$

For $|V| = O(t)$ and $\log(1/p) = O(t^2)$, a suitable q proportional to $\log t$ gives relative cost $\exp[-O(t^2/\log t)]$. There is a stronger normalization for an event H observed wholly outside B_t when V lies at depth at least $d > 0$ inside B_t : the conditional intensity is at most $b_d = (1 - e^{-\pi d^2})^{-1}$. If clearing every configuration in H has the target property, conditional Jensen gives

$$\mathbb{P}(F_{\text{tar}} \cap H) \geq e^{-(b_d + \nu)|V| \log C} \mathbb{P}(H).$$

No lower bound on $\mathbb{P}(H)$ is needed in this separated conditioning regime.

For an annulus of middle radius $a \geq W := w + 4\delta$ and width w , `count_preserving_annular_clearing.md` gives the same law with

$$C_{\text{ann}} = e^{3\pi W^2/4} (1 + 2W/\delta). \quad (9.23)$$

The harmonic gauge $2\pi a^2 \log(|z|/a)$ leaves a Gaussian potential of oscillation at most $3\pi W^2/4$, and annular torsion bounds the adjoint exit density by $2W/\delta$. These constants do not depend on a , while the landing region has area $2\pi aW$. Its two landing layers are disconnected. If $R < w < 2R$, all attachments and the protected arm of one terminal lie inside the ring, and those of the other lie outside, then every output keeps them separate. An inserted root within range of both protected arms is pivotal for every output. This is an eligible source geometry.

For a generic local bulk root both terminals may lie outside the annulus; clearing a full ring then disconnects the root from both. Ports, sufficient movable source vertices, and canonical multiplicity remain separate tasks. The independent checks in `count_preserving_clearing_audit.md` retain these restrictions. Thus the clearing laws resolve a density-comparison step without claiming a general pivotal-producing filling.

9.17 Omitted points, adaptive choices, and multi-pair optimization

The omitted-point route retained rather than discarded the event in a Palm norm identity. For distinct q, p, t , a normalized entire product $F_Y(q) = 1$ under the reduced (q, p, t) -Palm law gives a norm $N_q(Y)$ and the recovered exact formula

$$\mathbb{E}^{qpt}[N_q \mathbf{1}_E] = \int W_{qpt}(z) \mathbb{P}^{zpt}(E) dz, \quad W_{qpt}(z) = \frac{|q - p|^2 |q - t|^2 \det H(z, p, t)}{\det H(q, p, t) |z - p|^2 |z - t|^2}. \quad (9.24)$$

The same function E of the remainder appears on both sides; it is not recomputed after moving its geometric root. Determinantal zeros cancel the apparent poles and the far tail is quartic.

Vandermonde factors also cancel in the unconditioned norm mean as a fixed compact triple coalesces. These cancellations repair several earlier apparent singular constants.

Decreasingness alone still does not compare $\mathbb{P}^{zpt}(E)$ with $\mathbb{P}^{qpt}(E)$: their kernels differ by one positive and one negative rank-one direction. The notes also give large-disk voids for which the relative norm mean can grow with the disk's displacement. Thus retaining the event is necessary but not sufficient.

Steiner certificates and exact local capture. The deterministic result in `research/steiner_certificate_clique_excess.md` explains how local augmentation can retain every nearby pivotal alternative. In a finite graph, let an inclusion-minimal connected induced subgraph contain k prescribed vertices. Except for the trivial one-vertex case, deleting at most $k - 2$ exceptional vertices makes it triangle-free. To see this, every nonprescribed vertex is a cutvertex. If n_0 is the number of noncutvertices, c the number of cutvertices and b the number of blocks, the block-cut tree gives

$$\sum_{\text{blocks}} (|B| - 2) = n_0 + c - b - 1 \leq n_0 - 2 \leq k - 2.$$

Delete all but two vertices in each block of size at least three. This deletion is for the counting bound; it need not preserve connectivity or the prescribed vertices.

For two disconnected terminal components, prescribe every center in $B(q, L_0)$ on either side and connect each prescribed set to its virtual terminal by a minimal induced subgraph. Their union A has an exceptional set E with $|E| \leq N_\eta(B(q, L_0) \cap W)$, and $A \setminus E$ has cap two in every clique cell. Keeping the full A , and writing $R = 2r$, the exact capture identity is

$$B_A \cap B(q, L_0 - R) = \text{Piv}(\eta) \cap B(q, L_0 - R). \quad (9.25)$$

A pivotal insertion in the smaller disk has all its old neighbors in the prescribed local sets, with their paths retained; direct terminal attachments are retained as well. Conversely $A \subset \eta$ and the old background fails, so a bridge for A is an actual pivotal insertion. Canonical disintegration first fixes the local prescribed sets, then orders intrinsic candidate subgraphs. A global order whose validity changes when a new local point appears would not justify the needed decreasing remainder. The quantitative Fock treatment of the finite exceptional set is in `augmented_certificate_fock_calculation.md` and its audit.

Adaptive crowded-cell arguments were developed to obtain actual omitted marks. A cap-two certificate omits a pair from every cell containing at least four points. If the certificate is augmented by at most N exceptional local vertices, among its first $2N + 1$ crowded cells at least $N + 1$ still omit a pair. Averaging trial norms over those cells, then using Campbell and the exact Palm product, cancels a whole Gram determinant. The recovered result is a common inverse-first-moment bound polynomial in the local augmentation area, uniformly over the admissible augmented certificates. It is under the root Palm law *without* a rare pivotal conditioning. Restoring a forced mark without restoring it as a zero of the trial function would invalidate this argument.

For $k \geq 2$ actual omitted marks P , the exact relocation kernel is

$$W_{q,P}(z) = \frac{\det H(z, P)}{\det H(q, P)} \prod_{p \in P} \frac{|q - p|^2}{|z - p|^2}.$$

It has tail $|z|^{-2k}$. Campbell selection weights must be evaluated on the restored configuration; for a uniform ordered choice from a pool of size m the weight is $(m)_k^{-1}$. After reversing Campbell,

the pool and certificate are evaluated *after deleting the relocated point*, not under an unrelated ordinary configuration. This dependence is part of the transport kernel.

Several pair-deletion trials for one common omitted set span only

$$F_Y\{\text{polynomials of degree at most } k-2\},$$

a space of dimension $k-1$, not $\binom{k}{2}$. With outside-norm Gram matrix $G^B(Y)$, random optimizing coefficients are legitimate only if retained inside the transported expectation. They cannot be replaced by an optimizer of the mean Gram matrix. For example, if G is equally likely to be $\text{diag}(\epsilon, 1)$ or $\text{diag}(1, \epsilon)$ and $c_1 + c_2 = 1$, then

$$\mathbb{E} \min c^* G c = \frac{\epsilon}{1+\epsilon}, \quad \min c^* (\mathbb{E} G) c = \frac{1+\epsilon}{4}.$$

The difference is real, but its favorable size under pivotal weighting is not supplied by this example.

The decisive variational limitation is deterministic. Every admissible trial vanishes on the same retained certificate A . Its normalized full Fock norm is at least $1/(1 - P_A(q, q)) \geq 1$. If $|B| < 1$, its outside norm is at least

$$\frac{1 - |B|}{1 - P_A(q, q)}.$$

Thus no choice of many trial functions repairs the saturation examples in (9.15) pointwise.

The confluent benchmark in the later notes takes distinct marks to a cluster with k fixed, then lets $k \rightarrow \infty$, retaining fixed arm zeros and an explicitly stated local event. Its optimized limiting cost is precisely $1/M_{A,B}(q)$, the original outside-norm optimum. It can reach that multiplier, including its favorable examples, but cannot turn an unfavorable certificate into a favorable one. The orders of the two limits and the full spatial-integral domination are part of the recorded calculation. Neither prescribed clusters nor regular polygons can simply be assumed present with a useful selection weight; sufficiently dense polygons may form an occupied circuit and exclude bulk pivotality themselves.

9.18 Explicit confluent means and a finite-Palm outside-norm gain

The scalar calculations of `coalesced_multi_deletion_fock_mean.md` and `multipair_fock_gram_optimization.md` give more than the trial-space dimension. Fix a normalization root q and let k distinct omitted marks approach zero, first at fixed k . For $u = \pi|q|^2$, define

$$Q_k(u) = 1 - e^{-u} \sum_{j=0}^{k-1} \frac{u^j}{j!}, \quad a_k(u) = \frac{u^k}{k! Q_k(u)}, \quad w_k(z) = k \int_0^1 t^{k-1} e^{-\pi t|z|^2} dt.$$

Removable values are taken continuously. The limiting scalar expected density of the normalized common product is $a_k(u)w_k(z)$. This is a limit of exact distinct-Palm expectations, not an assertion of convergence of random products under a confluent law. Tonelli gives the complete finite polynomial moments

$$\int |z|^{2j} w_k(z) dA = \frac{k j!}{\pi^j (k-j-1)!}, \quad 0 \leq j \leq k-2;$$

higher moments diverge. Put $D_k(v) = \sum_{j=0}^{k-2} (k-1-j)v^j/j!$. The deterministic optimal polynomial with value one at q and degree at most $k-2$ is

$$P_k(z) = \frac{D_k(\pi z \bar{q})}{D_k(u)}, \quad \min_P a_k(u) \int |P|^2 w_k = \frac{a_k(u)k}{D_k(u)}.$$

These follow from the diagonal monomial Gram matrix. At $q = 0$ the continuous value of the minimum is $k/(k-1)$.

As $k \rightarrow \infty$, the optimized densities converge pointwise to $e^{-\pi|z-q|^2}$ and their total masses converge to one. Scheffe's lemma upgrades this to L^1 , so for every fixed Borel B ,

$$a_k(u) \int_{B^c} |P_k(z)|^2 w_k(z) dA \longrightarrow \int_{B^c} e^{-\pi|z-q|^2} dA. \quad (9.26)$$

For $B = B(q, b)$ this is $e^{-\pi b^2} < 1$. First select finite k giving a strict gain, then choose sufficiently close distinct marks; divided-kernel continuity and the integrable $|z|^{-4}$ tail preserve it on an open set of tuples. The omitted cluster can lie outside the desired bridge disk. This is an unconditional finite-Palm improvement. High-order Palm depletion empties every fixed local compact in the limit, so the required arms are not automatically present. Retaining forced arms and pivotal events leads back to the outside-norm optimum already described, with its possible unfavorable sign.

9.19 The complete outside-norm partition and its required coefficient

The source `pivotal_outside_norm_campbell.md`, checked in `neutral_janossy_outside_norm_audit.md`, makes a precise distinction between complete certificate coverage and an eligible pair in one selected cell. In this subsection set

$$I_F = \int \mathbb{E}_G D_q F dq, \quad J_F = \int \mathbb{E}^! q D_q F dq;$$

J_F here is the full reduced-Palm pivotal mass, not the leverage-weighted quantity of the small-beta section. For the intrinsic canonical certificate A , let E_A be its selection and root-pivotal event in the reduced A -Palm remainder. It is decreasing and contains its full bridge void. Conditional-void Palm order gives

$$Q_A(q, q) \mathbb{P}^{!(A, q)}(E_A) \geq M_{A, B(A), 1}(q) \mathbb{P}^{!A}(E_A).$$

Divide by the positive multiplier and sum the full Campbell partition. The determinant identity $\det H(A \cup \{q\}) = \det H(A) Q_A(q, q)$ and Tonelli yield

$$I_F \leq S_F := \int \mathbb{E}^! q \left[\frac{D_q F(X)}{M_{C_q(X), B(C_q(X)), 1}(q)} \right] dq. \quad (9.27)$$

No integrability of S_F is presumed before a transport bound. If one proved $S_F \leq c_0 I_F + c_1 J_F$, with $c_1 > 0$ and $c_0 + c_1 < 1$, then

$$J_F - I_F \geq \frac{1 - c_0 - c_1}{c_1} I_F.$$

This identifies a sufficient strict coefficient, not a proved estimate. Uniform continuation and endpoint control would still be needed for a critical-radius comparison.

The outside-norm transport removes redundant insertions in the same certificate bridge. Its remaining redundancy is exactly $z \in B_{\text{full}}(Y) \setminus B(C_q(Y))$, with q, z individually successful in the same failing background. Ordinary contributions still evaluate mark selection after deleting the relocated point. Equation (9.25) removes this residual in a local ball, but does not normalize the remaining rational potential to one. The fixed-cell eligibility condition of at least two omitted points is not decreasing and cannot be inserted into the order argument for (9.27). Adaptive complete selection must retain its unbounded spatial scale. Moreover choosing marks beyond an augmentation radius L_0 leaves the existing far bound with coefficient $D^4/(L_0 - R)^2 = O(L_0^2)$ when D is comparable to L_0 ; enlargement alone does not make it small.

9.20 Higher-mark angular gains and the annular obstruction

The detailed refinements of `multiple_omitted_marks_far_transport.md` and `k_omitted_marks_radial_bennett_audit.md` retain geometry beyond the tail exponent $2k$. For root zero and regular polygon marks $p_j = D\omega^j$, $\omega = e^{2\pi i/k}$, the rational factor is

$$R(0, z; P) = \frac{D^{2k}}{|z^k - D^k|^2}.$$

For $|z| \geq L > D$, it is at most $M_{\text{poly}}(L/|z|)^{2k}$, where $M_{\text{poly}} = (D/L)^{2k}/[1 - (D/L)^k]^2$. Extending the majorant inside by $M_{\text{poly}}(|z|/L)^{2k}$ gives mean $2\pi k L^2 M_{\text{poly}}/(k^2 - 1)$, variance at most $k M_{\text{poly}}^2$, and far integral $\pi L^2 M_{\text{poly}}/(k - 1)$. Marks within $\epsilon < L - D$ of their polygon slots incur only

$$C_{\text{slot}} = (1 + \epsilon/D)^{2k} (1 - \epsilon/(L - D))^{-2k}.$$

Thus these far coefficients can decay exponentially in k for $D = cT$, $c < 1$, and $L = T - R$. Actual admissible slot occupancy and its selection weights remain necessary.

Conversely, if all marks lie outside B_T , any associated rational function analytic inside B_T and normalized to one at zero has circle mean squared modulus at least one. Hence

$$\int_{T-R < |z| < T} |R_0(z)|^2 dA \geq \pi[T^2 - (T - R)^2].$$

Even retaining the Gram depletion does not remove the mass for sparse polygons. For marks on radius $T \geq 2R$ set

$$\alpha = e^{-\pi R^2/4}, \quad \sigma = 2 \sum_{m \geq 1} e^{-8\pi(T/k)^2 m^2}, \quad \epsilon_* = 2 \sum_{m \geq 1} e^{-2\pi(T/k)^2 (2m-1)^2}.$$

If $\sigma, \epsilon_* \leq (1 - \alpha)/8$, on $A = \{T - R < |z| < T - R/2\}$ the residual diagonal is at least $c_R = (1 - \alpha)/2$, the parent determinant is at least $(1 - \sigma)^k$, and exact angular averaging yields

$$\int_A \det H(z, P) R(0, z; P) dA \geq (1 - \sigma)^k c_R \frac{\pi T(T - R)}{k} \log 2. \quad (9.28)$$

For k of order $\log T$ this has order T^2/k . This is a kernel bound; the relocated pivotal expectation could still suppress that annulus and must not be discarded.

The correlated polygon calculation is also explicit. Set $a = \pi D^2$, $b = \pi r^2$, $E_\ell(a) = \sum_{n \geq 0} a^{\ell+n} / (\ell + nk)!$, and $c_{\ell n}(r) = (ab)^{(\ell+n)/2} / (\ell + nk)!$. For $r < D$ put $t = (r/D)^k$. Then

$$\begin{aligned} & \frac{1}{2\pi} \int_0^{2\pi} Q_P(re^{i\theta}, re^{i\theta}) R(0, re^{i\theta}; P) d\theta \\ &= \frac{1 - e^{-b} \sum_{\ell=0}^{k-1} \frac{\sum_{n,m \geq 0} c_{\ell n} c_{\ell m} t^{|n-m|}}{E_\ell(a)}}{1 - t^2}. \end{aligned} \quad (9.29)$$

For $r > D$, use $t = (D/r)^k$ and multiply the right side by t^2 . At $r = D$ its removable value is $\frac{1}{2k} \sum_\ell w_\ell \mathbb{E}|N_\ell - N'_\ell|$, where $w_\ell = e^{-a} E_\ell(a)$ and N_ℓ, N'_ℓ are independent Poisson(a) variables conditioned to be congruent to ℓ modulo k . Residue-class coherent expansions and Fourier averaging prove the formula; separate angular means cannot be multiplied. For the normalized BQ density divide by $Q_P(0, 0) = 1 - 1/E_0(a)$; for the Campbell-cancelled kernel multiply by $\det H(P)$ instead.

Finally, very dense restored polygons may exclude the source event: if $2D \sin(\pi/k) \leq R = 2r_0$ and $D \cos(\pi/k) > r_0$, their occupied circuit surrounds the entire root grain. When both terminal attachment sets lie outside this circuit, any two arms reaching the root already meet the circuit, so the root is not pivotal. The independent correlated-kernel audit preserves this distinction between prescribed Palm marks and actual restored marks.

9.21 What buffered Bergman domination does and does not supply

The conditional Bergman theorem proved elsewhere conditions outside a larger disk D while leaving the buffer between D and the target C unobserved. It bounds the conditional process in C above by the weighted Bergman DPP, including prescribed Palm zeros. Its unconditioned disk compression has eigenvalues $p_n(M)/p_n(R)$ for concentric radii $M < R$, and hence norm strictly below one. The limit to the full exterior is a conditional-expectation martingale limit; it is not the assertion that an inverse on the entire exterior remains a bounded operator.

Three limitations matter. Observing immediately outside C removes the free buffer and permits rigidity to force its count. Adding a decreasing condition in the buffer is additional conditioning not covered by the same theorem. Finally, an upper conditional intensity or a lower hole probability does not provide a lower probability of inserting a prescribed scaffold.

Prescribed clustered zeros make local conditional occupancy very small: the recorded bound is of the form $C(k+1)\rho^{2k}$ for a compact target with relative radius $\rho < 1$. But comparing through a common exterior law incurs a root normalization. For a k -fold jet at the disk center, the asymptotic factor recorded in the notes is

$$\frac{B_D^{\text{jet}}(q, q)}{Q_k(q, q)} \sim \frac{e^{\pi R^2} (k+1)!}{(\pi R^2)^{k+1} (1 - |q|^2/R^2)}.$$

Its factorial growth can overwhelm the local hole gain. Direct fixed-Palm exterior comparison was therefore investigated separately; the resulting unbounded likelihood ratios are a confirmed obstruction to uniform arbitrary-exterior normalization, not a proof that the actual pivotal average has the wrong sign.

The zero/one-neutral-point Janossy calculation illustrates the same discipline. With residual projection Q , a bounded retained region V , and $L = Q_V(I - Q_V)^{-1}$, the zero-point root factor is $L(q, q)$; with one neutral point u it is

$$L(q, q) - \frac{|L(q, u)|^2}{L(u, u)}.$$

The neutral set must exclude direct one-sided terminal contacts as well as the bridge. These factors can still vanish along saturated certificates. Few neutral points are not automatically a favorable pivotal configuration.

9.22 Reversible moves and genuine Goldman deletion weights

A finite-rank reversible move was an intermediate repair for the dependence problem. Given a configuration η and $y \in \eta$, let $Q = H - P_{\eta-y}$. Then

$$q_\eta^y(x) = \frac{|Q(x, y)|^2}{Q(y, y)}$$

is a normalized move density because Q is a projection. Schur factorization gives

$$\det H(\eta) q_\eta^y(x) = \det H(\eta - y) |Q(x, y)|^2,$$

which is symmetric in x, y . This establishes detailed balance of the move for the thinned finite-rank DPP. It does not establish stationarity at intermediate positions on the straight segment from y to x , which the Poisson movement argument used.

The incoming density is a nonorthogonal dual-vector frame, not the coherent projection diagonal. For n points coalescing near zero in the rank- n Fock space, that frame tends to $n|\phi_{n-1}(x)|^2$,

while their coherent span is the entire rank- n space. At the origin the ratio can vanish, and at distant points it can grow with n . The conditional squared jump distance tends to n/π . Thus a uniform conditional Gaussian jump tail or a configuration-uniform comparison of frame and leverage is false. Unweighted frame and leverage expectations agree, but a pivotal weight introduces an uncontrolled covariance. A process-level infinite-Ginibre limit of this finite-rank move was not constructed.

The final representation uses the *genuine* Goldman coupling, rather than inventing an insertion rule from a convenient trial norm. Disintegrating its deletion given G gives weights $w_q(G, z)$ on actual points with $\sum_{z \in G} w_q(G, z) = 1$. Its true conditional insertion density from a reduced Palm remainder X is

$$k_q(X, z) = w_q(X + z, z) \frac{d\mu^{!z}}{d\mu^{!q}}(X), \quad \int k_q(X, z) dz = 1, \quad \mathbb{E}^{!q} k_q(X, z) = e^{-\pi|z-q|^2}.$$

The Gaussian formula is a marginal, not independence. The exact signed redundancy-minus-cooperation identity, and its thinning extension, are proved in the principal-results section of this report.

Only an *annealed* spatial column identity follows automatically:

$$\mathbb{E}^{!z} \int w_q(X + z, z) dq = 1.$$

It does not imply column mass one for every realized point. The archive contains a finite projection-DPP counterexample to that automatic inference; it does not disprove the existence of a specially balanced Ginibre coupling. The rank-three failure of normalized Lagrange-square weights, proved elsewhere in this report, rules out another convenient candidate. Thus the final unresolved sign is attached to actual nonlinear deletion weights. Removing those weights would change the quantity being compared.

9.23 Local Palm insertion and excluded analytic weight representations

Sections 6–8 of `ginibre_palm_relocation_markov.md`, with `palm_relocation_local_independent_audit.md`, contain additional recovered results. For $D = B(q, R)$, condition ordinary Ginibre on its exterior ξ . Its inside law is a rank- n polynomial-ensemble projection P_ξ , with n fixed by the exterior; its conditional q -Palm law has rank $n - 1$. The exterior density of the reduced q -Palm law relative to the ordinary exterior is

$$t(\xi) = P_\xi(q, q), \quad 0 \leq t \leq (1 - e^{-\pi R^2})^{-1}, \quad \mathbb{E}t = 1.$$

Couple these two finite conditional projections by adding one point inside D . Starting from the actual reduced q -Palm law reconstructs the conditional ordinary inside law, but leaves its exterior tilted by t . The resulting full law $\tilde{\mu}$ therefore satisfies

$$d_{\text{TV}}(\tilde{\mu}, \mu) = \mathbb{E}(t - 1)_+ \leq e^{-\pi R^2}. \quad (9.30)$$

The inequality follows from $(t - 1)_+ \leq e^{-\pi R^2} t$. The source audit supplies a measurable selection of the finite couplings using their compact sections on configuration space. This is an absolute local approximation, not relative control of an arbitrarily rare pivotal event, and no limiting identification with a particular Goldman coupling is claimed.

A different exact obstruction excludes holomorphic-square weights. For an actual configuration G with at least two points, suppose entire functions $f_{G,z}$ vanish on $G \setminus \{z\}$ and obey

$$w_q(G, z) = e^{-\pi|q|^2} |f_{G,z}(q)|^2, \quad \sum_{z \in G} w_q(G, z) = 1$$

for almost every q . Submean bounds give local uniform convergence of the covariance series. Analytic polarization forces

$$e^{\pi q \bar{w}} = \sum_{z \in G} f_{G,z}(q) \overline{f_{G,z}(w)}.$$

At two distinct actual nodes the right side vanishes termwise and the left side is nonzero, a contradiction.

The nodal vanishing is itself forced for any genuine deletion family continuous in q . The reduced-Palm intensity gives

$$\mathbb{E}^! q N(B(q, \epsilon)) = \pi \epsilon^2 - (1 - e^{-\pi \epsilon^2}) = O(\epsilon^4).$$

Integrating over roots in an enlarged compact set, expressing the Palm law by deletion, dividing by $\pi \epsilon^2$, and applying Fatou to the nonnegative sum over actual pairs yields $w_x(G, z) = 0$ simultaneously for actual $x \neq z$. This avoids differentiating at a random point.

It also excludes a positive-operator representation $w_q(G, z) = \langle H(\cdot, q), A_{G,z} H(\cdot, q) \rangle$ with $A_{G,z} \geq 0$ and $\sum_z A_{G,z} = I$ weakly. The nodal zeros would imply $A_{G,z}^{1/2} H(\cdot, x) = 0$ for $x \neq z$; summing mixed inner products would make two distinct coherent vectors orthogonal. These impossibility statements concern the specified analytic and positive-operator classes. They do not contradict general measurable Goldman deletion weights. A root-dependent nonlinear normalization leaves the analytic-square class and still needs both marginal laws checked.

9.24 Finite-rank reveal dynamics and total insertion influence

The source `residual_projection_dynamics.md` gives finite-rank identities separate from the reversible move. Let H be a rank- N projection with finite spatial moments. Reveal the points of $\text{DPP}(H)$ in independent uniform mark order, writing X_s for those already revealed. Conditional on $X_s = \eta$, the unrevealed process has residual projection $Q_\eta = H - P_\eta$. Its future birth intensity is $Q_\eta(z, z)/(1 - s)$, and a birth at z changes the residual kernel by

$$Q_\eta^z(x, y) = Q_\eta(x, y) - \frac{Q_\eta(x, z) Q_\eta(z, y)}{Q_\eta(z, z)}.$$

Zero-denominator locations have zero birth intensity. With $F_s(x, y) = \mathbb{E}|Q_s(x, y)|^2$, the generator and projection identity give

$$\partial_s F_s(x, y) = -\frac{2F_s(x, y)}{1 - s} + \frac{1}{1 - s} \int \mathbb{E} \frac{|Q_s(x, z)|^2 |Q_s(z, y)|^2}{Q_s(z, z)} dz.$$

Define $M_p(s) = \iint |x - y|^p F_s(x, y) dx dy$. Subadditivity of $|\cdot|^p$ for $0 < p \leq 1$ bounds the integrated positive term by $2M_p(s)$; the squared triangle inequality bounds it by $4M_2(s)$. Consequently

$$M_p(s) \leq M_p(0) \quad (0 < p \leq 1), \quad M_2(s) \leq M_2(0)/(1 - s)^2. \quad (9.31)$$

For the rank- N Bargmann projection at scale beta, the monomial trace calculation gives $M_2(0) = 2N\beta/\pi$ and $M_1(0) \leq N\sqrt{2\beta/\pi}$.

The thinned finite-rank law has Papangelou intensity $c_s(x, \eta) = aQ_\eta(x, x)$, $a = s/(1 - s)$. Adding y changes it by $a|Q_\eta(x, y)|^2/Q_\eta(y, y)$, whose integral in x is exactly a whenever the denominator is positive. A common-birth-proposal and common-death coupling for the associated birth-death process therefore contracts expected Hamming disagreements at rate at least $1 - a$ for $s < 1/2$. This is a count estimate. Neither (9.31) nor the contraction permits multiplication by a rare pivotal indicator without additional generator terms. No infinite-Ginibre process limit is asserted by this recovered finite-rank calculation.

9.25 Endpoint passages and separate analytic threshold bounds

Several local calculations survived after their intended large-scale interpretation failed. The replacement for an unproved rectangular crossing step law is the full-plane seed-to-outer event with uncut grains. At fixed seed m , the events of reaching ∂B_L decrease as $L \rightarrow \infty$. Only finitely many components meet the seed, so their intersection is exactly the event that an unbounded component meets it. Letting $m \rightarrow \infty$ and using ergodicity gives the percolation zero–one law away from the critical radius. A deterministic slow diagonal and bounded capped thresholds then recover r_c in probability and mean. No assertion at criticality is needed. The required comparison must nevertheless hold uniformly along that diagonal; the diagonal does not create uniformity in a fixed-window Taylor remainder.

Similarly, finite local thinning converges to the full projection endpoint, and independent baths of intensity tending to zero disappear locally. These facts justify bounded-window endpoint limits when the comparison itself has controlled coefficients. They do not justify integrating $(1-s)^{-1}$ errors or exchanging the $s \uparrow 1$ and $L \rightarrow \infty$ limits without a bound. Number rigidity is one reason that an arbitrary conditional argument need not extend uniformly to $s = 1$.

The separate-bounds route was also checked. The published general sub-Poisson upper bound used in the later audit is $r_c(G) \leq \sqrt{2 \log 7}$. A fully elementary induced-path exclusion gives

$$r_c(\Pi_1) \geq \frac{1}{2\sqrt{\pi/3 + \sqrt{3}/2}}.$$

The intervals do not overlap in the useful direction. The history records sharper contour work, but it likewise does not establish an upper Ginibre radius below a proved Poisson lower radius.

The threshold audit excluded Monte Carlo confidence intervals as deterministic theorem inputs. It also found an unjustified substitution of an unconditional clustering coefficient for a history-selected overlap in one proposed published bound. For a parent at normalized distance h , the actual overlap fraction is

$$c(h) = \frac{2 \arccos(h/2) - (h/2)\sqrt{4-h^2}}{\pi},$$

and at $h = 1$ it differs by $1/3 - \sqrt{3}/(4\pi) > 0$ from the asserted averaged value. This rejects that supplied proof step, not necessarily the truth of the numerical inequality it claimed. No numerical experiment was performed for the present investigation or report; the earlier history’s numerical explorations remain historical material only.

9.26 Result-level map of the restored source discoveries

The following table maps every finding of the coverage audit to its mathematical treatment in this amended report. The associated source filenames and scope are given in those passages; the complete source index remains a separate archival inventory. The restored arguments retain the verification distinctions stated at the start of the report.

Audit	Restored result or refinement	Location
1	Quantitative stretching-limit rate and its extra Poisson input	7.8
2	Integrated critical-radius orientation, Riesz coefficient, and two-Palm response	7.9 , 7.11

Audit	Restored result or refinement	Location
3	Switching atoms, rotated cusps, and smooth negative curvature	7.10
4	Inclusion, displacement, overcrowding, and covering barriers	8.12
5	Strict positive-bath mixed-model threshold implication	8.10
6	Buffered corridor holes, Brownian-exit clearing, and annular repair	9.16
7	Entire fixed-kernel expansion and positive arms with neutral points retained	9.8
8	Residual reveal dynamics and total Papangelou influence	9.24
9	Protected internal close-pair estimate through the projection endpoint	9.12
10	Local Palm reconstruction and excluded analytic/positive-operator weights	9.23
11	Steiner clique excess and exact local capture	9.17 , (9.25)
12	Buffered conditional cap-density tail	9.15 , (9.18)
Q1	Complete outside-norm partition and sufficient strict coefficient	9.19
Q2	Confluent polynomial moments, optimizer, and finite-Palm outside-norm gain	9.18
Q3	Half-Gaussian contact split and dimer-perimeter identity	9.10
Q4	Balanced-mark tails and the correlated polygon/annular obstruction	9.20
Q5	Thinning-plus-bath Bennett variance and entropy bound	9.15
Q6	Fixed-count motion support, pushed-forward likelihood, and optimized rare-event cost	9.14 , (9.16)

9.27 Source map and precise remaining boundary of the investigation

The following map identifies the detailed working notes underlying this section. Their text is preserved in the source supplement, including historical confidence language; that language is subject to the status conventions at the beginning of this section. A note ending in **audit** records a collaborator’s check at the time, not a proof that all later uses of its conclusion were valid.

Earlier history. The attached research-history PDF, Sections 3–9, contains the initial void/Voronoi, structure-factor, cluster, contact, dynamics, static-energy, and square-replacement routes. The supplied ellipse note is preserved separately. The extracted text is a reading aid, not a substitute for the original mathematical typesetting.

Affine and fixed-window signs. `isotropic_affine_counterexample.md`; `research/anisotropic_variation.md`; `ellipse_critical_variation.md`; `ellipse_minmax_switching.md`; `research/fixed_kernel_poisson_second_variation.md`.

Cloning and contacts. `research/bottleneck_cloning.md`; `poisson_clone_decomposit`

ion.md; poisson_uniform_cloning_bound.md; mixed_clone_and_compensation.md; research/contact_certificate_disintegration.md; research/contact_scaffold_absorption.md; research/current_law_contact_obstructions.md.

Certificates and clearance. replacement_count_lemma.md; conditional_count_tails.md; canonical_void_clearance.md; current_law_clearance.md; sparse_fock_projection.md; finite_window_leverage_operator.md; the bulk, boundary and zero-bath separation notes and their audits.

Bridge signs and saturation. bridge_lens_resolvernt.md; full_ginibre_transport_and_certificate_obstruction.md; two_parent_wide_lens_resolvernt.md; straight_arm_wide_lens_resolvernt.md; ginibre_triple_resolvernt_certificate.md; pivotal_zero_one_neutral_janossy.md.

Motions and exterior support. ginibre_motion_inverse_moments.md; ginibre_full_circle_cluster_motion.md; ginibre_triangle_cluster_motion.md; tapered_arm_area_motion_obstruction.md; count_preserving_corridor_clearing.md; corridor_filling_source_obstructions.md; ginibre_infinite_compact_factor_support.md; fixed_palm_infinite_exterior_ratio_obstruction.md.

Random deficits and rare-event normalization. research/induced_certificate_density_gap.md; random_certificate_fock_nonuniqueness.md; ginibre_finite_deletion_fock_zero_sets.md; research/rare_pivotal_weighting_and_corridors.md; projection_dpp_variance_bennett.md; thinned_ginibre_potential_bennett.md.

Omitted points and Gram optimization. three_palm_weighted_fock_identity.md; adaptive_crowded_cell_palm_fock.md; augmented_crowded_cell_palm_fock.md; pivotal_omitted_pair_campbell.md; multipair_fock_cross_moments.md; multiple_omitted_marks_far_transport.md; forced_arm_confluent_scalar_limit.md; retained_arms_multipair_variational_limit.md; research/multipair_projection_and_marking_obstructions.md.

Conditional kernels and genuine coupling. exterior_palm_bergman_bounds.md; buffered_forced_marks_bergman_hole.md; reversible_projection_move.md; ginibre_palm_relocation_markov.md; thinned_goldman_relocation_identity.md; finite_ginibre_deletion_weights.md; research/goldman_column_mass_audit.md.

Limits, constants, and chronology. annulus_threshold_convergence.md; analytic_critical_radius_bounds_audit.md; kong_yeh_clustering_audit.md; checkpoint_front_version2.md; continuation_front.md. The latter two record successive states of the investigation and must not be read as simultaneous certifications of every evolving claim.

The final mathematical boundary is specific. The full Ginibre endpoint has exact current-law interpolation and genuine deletion identities. There are local bridge configurations of both signs, arbitrary-exterior likelihood obstructions, and unconditional analytic estimates that lose their constants under rare pivotal conditioning. What was not obtained is a signed, quantitatively positive comparison of the actual weighted pivotal contributions, uniform over a sufficient interpolation and growing windows, with controlled endpoints. Consequently no strictly positive radius gap for G versus Π_1 follows from the investigation as it stands.

10 The remaining comparison and where strictness would enter

The original target is a strict inequality between two numbers at equal intensity. It is weaker than monotonicity of every crossing probability, every finite-cluster event, every affine deformation, or an entire interpolation. Counterexamples to those stronger assertions eliminate particular proofs, not the target itself.

10.1 A finite-window route that would produce a radius gap

The good-edge construction in the ellipse section supplies a universal number $b_0 > 0$: a stationary negatively associated process percolates if, for one square scale $L > 2r$, both orientations of its local good-edge event have failure probability less than b_0 . This is an infinite-volume criterion proved by a dual-circuit sum, not an informal inference from a high crossing probability.

Consequently, the following precise missing assertion is sufficient: there is a finite scale $L > 2r_P$ such that

$$\max_{o \in \{\text{horizontal}, \text{vertical}\}} \mathbb{P}_G(\text{the scale-}L \text{ edge of orientation } o \text{ is bad at } r_P) < b_0. \quad (\text{S})$$

If (S) were proved with a strict margin, it would persist at some $r_P - \delta > 0$. To justify this last step, fix one slightly enlarged bounded observation window containing every relevant center for radii near r_P . The point count there is almost surely finite. Conditional on the count, its Janossy law is absolutely continuous. The finitely many disk-intersection and side-contact incidences are stable in radius except on tangency sets of Lebesgue measure zero. Equivalently one can describe intersection of the finitely many clipped disks by finitely many semialgebraic inequalities; the nontrivial equality cases have zero probability. Thus each fixed block-event probability is continuous in radius, by bounded convergence. The same fixed L and a small positive δ retain the criterion, giving

$$r_c(G) \leq r_P - \delta < r_P.$$

Here the strict margin in (S), followed by finite-window continuity, is the exact source of a positive radius gap. No proof of (S) is supplied by this investigation.

10.2 What the two recovered endpoint results do and do not say

The verified limit

$$r_c(S_t^{-1}G) \longrightarrow r_P$$

does not order the value at $t = 0$. This is a logical issue, not a lost estimate: convergence alone allows approach from either side. The large-area comparison holds for each fixed nonzero eccentricity beyond a shape-dependent area threshold. It does not identify that threshold with a neighborhood of a critical area, uniformly in eccentricity. The joint hole regime makes the nonuniformity of the two limits explicit.

If the archived small- β geometric estimate is fully verified, its squeeze proves

$$r_c(G_\beta) = r_P(1 - \beta/4) + o(\beta)$$

and hence a strict comparison for every sufficiently small positive β . The effective Poisson intensity exceeds one because $L_\beta = 1 + \beta/2 + O(\beta^2)$, while its geometric loss is $o(\beta)$. That is a valid, explicit strictness mechanism at the Poisson endpoint. It gives no continuation to $\beta = 1$ without additional control. Even a right derivative at zero does not determine a function's value at one. No monotonicity or uniform derivative estimate on the whole interval has been established here.

10.3 The event-weighted obstruction at full Ginibre

The exact signed identity reduces the local interpolation derivative to redundancy minus cooperation under a coupled reduced Palm law. Conditional Fock and Bergman estimates retain substantial information, but the required quantity weights that information by actual pivotality. An unconditional inverse-moment bound, or a positive multiplier for a particular deterministic certificate, does not control that weighted quantity. The report contains explicit mechanisms preventing the replacement of the weighted estimate by a uniform pointwise or arbitrary-exterior assertion.

The unresolved step is therefore not an omitted formal integration or a Taylor expansion. It is a quantitative estimate retaining the pivotal geometry and its probability, strong enough either to prove (S) or to produce a positive intensity/radius gain uniformly over the growing connection windows of a valid interpolation. No such full-Ginibre estimate has been proved or disproved here.

11 Published inputs and audit record

The following are external mathematical inputs, not results first proved in this investigation. Normalizations and hypotheses are stated where they enter the proofs.

1. A. Goldman, Ginibre and reduced-Palm coupling.

[Original paper](#), [arXiv:math/0610243](#). Theorem 1 supplies the one-point coupling, in intensity- $1/\pi$ normalization. Scaling gives the kernels used here. The added point's location is not assumed independent of the Palm remainder.

2. R. Lyons, determinantal negative dependence and order. [Determinantal Probability: Basic Properties and Conjectures](#), [arXiv:1406.2707v1](#).

Theorem 3.7 supplies negative association; Theorem 3.8 supplies stochastic domination from operator order for locally trace-class positive contractions on the same Radon-measure space. For a possibly irregular observation set, its compressed kernels can be extended by zero to the plane. Area-preserving changes of variables retain the hypotheses.

3. D. Ahlberg, V. Tassion, A. Teixeira, planar Poisson crossings. [Sharpness of the phase transition for continuum percolation in \$\mathbb{R}^2\$](#) , [arXiv:1605.05926v1](#).

Theorem 1.1(i),(iii) supplies off-critical rectangle-crossing limits under the radius-distribution second-moment hypothesis. Deterministic disks satisfy that hypothesis. Their intensity formulation converts to a radius formulation by spatial dilation, which also scales the rectangle.

4. K. Adhikari, N. K. Reddy, Ginibre hole asymptotics. [Hole probabilities for finite and infinite Ginibre ensembles](#), [arXiv:1604.08363v3](#).

Theorem 6 and Example 17 supply the ellipse coefficient $(ab)^3/[2(a^2 + b^2)]$ in their normalization. Ellipses satisfy the required exterior-ball regularity. The report performs the intensity and area changes explicitly.

5. A. Bufetov, Y. Qiu, multiplicative functionals and Palm laws. [arXiv:1608.03736v2](#).

The cited Proposition 1.1 and Corollary 1.6 concern equal-order Palm equivalence and the radial conditional-density formula. The proof here controls the radial analytic tail and handles fixed tuples explicitly; it does not infer a uniform likelihood bound from equivalence.

6. S. Ghosh, Y. Peres, Ginibre rigidity and tolerance. [Rigidity and Tolerance in Point Processes: Gaussian Zeros and Ginibre Eigenvalues](#), [arXiv:1211.2381](#).

For Ginibre, the outside configuration determines the inside count, and the conditional inside law is equivalent

to Lebesgue measure on that fixed-count configuration space. The argument does not impose the additional sum rigidity of the distinct Gaussian-zero model.

7. **Strassen’s closed-order coupling theorem.** The local-to-global coupling extension uses the theorem for probability measures on a Polish space with a closed partial order; see [Lindvall’s account](#) and its [published correction](#). The critical-radius lower bound can also be obtained by the direct local event exhaustion given in the proof, without this global coupling step.
8. **I. Manolescu, L. V. Santoro, near-critical Poisson crossings.** [Widths of crossings in Poisson Boolean percolation](#), [arXiv:2211.11661v1](#). Theorem 2.4 and equation (2.16) are the additional inputs to the recovered rate in Section 7.8. The paper states the near-critical result for fixed-radius disks and sketches its proof. Only those inputs are used; radius conversion here uses $\lambda = r^2$.

11.1 What was checked in preparing this report

The six core statements were reread, with separate checks of the coupling normalizations, conditional operator bounds, fixed-tuple exterior argument, and finite-rank deletion counterexample. The ellipse critical-radius proof received a separate reconstruction of its Gaussian estimates, support colorings, planar gluing, source hypotheses, and order of limits. The small- β archive was recovered and its dependency chain documented; the entire geometric estimate was not newly certified by this review. The additional branch survey distinguishes short reproduced calculations from advanced historical derivations.

For the amendment, each coverage finding was matched to its original derivation and audit where available; formulas, hypotheses, normalization, and scope were restored in the relevant section. The additional near-critical source statement was checked directly. This coverage and consistency review does not promote all recovered arguments to the six core statements’ verification status.

These checks justify the statuses stated in this report. They do not turn an unresolved geometric assertion into a theorem, and they do not establish the original strict inequality. Compilation and typesetting checks of the document are separate from these mathematical checks.

A Complete source index

The companion archive contains the standalone report source and PDF, the two original research attachments, the consolidated historical checkpoint, and the individual notes indexed below. The originals are preserved byte for byte. Their titles and old audit labels are historical metadata, not fresh status assignments.

There are 130 individual mathematical Markdown notes. The file `manifest.json` records source paths, sizes, and SHA-256 digests. This is a preservation check, not a mathematical validation.

Source path	Original title
<code>active_fock_density_calculation.md</code>	Active Fock calculation after the capped-count theorem
<code>adaptive_crowded_cell_independent_audit.md</code>	Independent audit of the adaptive crowded-cell Fock estimate
<code>adaptive_crowded_cell_palm_fock.md</code>	Adaptive crowded cells and a uniform Palm first inverse Fock moment

Source path	Original title
<code>analytic_critical_radius_bounds_audit.md</code>	Audit of explicit analytic critical-radius bounds
<code>annulus_threshold_convergence.md</code>	A seed-to-boundary threshold that converges to the Boolean critical radius
<code>audit_sources_findings.md</code>	Independent audit: primary inputs and finite-window counterexamples
<code>augmented_certificate_fock_audit.md</code>	Audited quantitative Fock bound for augmented certificates
<code>augmented_certificate_fock_calculation.md</code>	Augmented certificates: finite exceptions and local bridge capture
<code>augmented_crowded_cell_palm_fock.md</code>	First inverse moments for augmented certificates under one-point Palm
<code>bridge_lens_resolvent.md</code>	Bridge-lens resolvent: exact full-correlation calculation
<code>bridge_lens_resolvent_audit.md</code>	Independent audit: bridge-lens resolvent
<code>bridge_motion_continuation.md</code>	Fixed-count bridge motion: exact obstructions and averaged identities
<code>buffered_bergman_domination_audit.md</code>	Independent audit of buffered Bergman domination
<code>buffered_forced_bergman_independent_audit.md</code>	Independent audit of buffered holes with forced Palm marks
<code>buffered_forced_marks_bergman_hole.md</code>	Buffered conditional holes retain prescribed Palm zeros
<code>canonical_certificate_audit.md</code>	Canonical exterior certificates and the weighted Palm decomposition
<code>canonical_void_clearance.md</code>	Canonical certificates, empty-ball conditioning, and residual leverage
<code>checkpoint_front_version2.md</code>	Ginibre versus Poisson: continuation checkpoint
<code>coalesced_multi_deletion_fock_mean.md</code>	Exact multi-deletion mean in a confluent Palm family
<code>conditional_count_tails.md</code>	Exponential count bounds under deletion certificates
<code>conditional_interior_audit.md</code>	Finite-rank conditional interior law and its planar probability limit
<code>continuation_front.md</code>	Ginibre versus Poisson: active research checkpoint
<code>cooperative_pair_boundary.md</code>	Boundary cooperative insertions: an explicit one-arm Poisson surgery
<code>cooperative_pair_boundary_audit.md</code>	Independent audit of the boundary cooperative-insertion estimate
<code>cooperative_pair_bulk.md</code>	A uniform bulk estimate for cooperative nearby Poisson insertions
<code>cooperative_pair_dpp_bath.md</code>	Cooperative insertion estimates for a thinned projection DPP with a positive Poisson bath
<code>corridor_filling_source_obstructions.md</code>	Source-count and exterior-weight obstructions to corridor filling
<code>count_preserving_annular_clearing.md</code>	Count-preserving annular clearing and its exact geometric scope
<code>count_preserving_clearing_audit.md</code>	Independent audit of Brownian-exit corridor clearing
<code>count_preserving_corridor_clearing.md</code>	A count-preserving conditional Ginibre corridor-clearing comparison
<code>current_law_clearance.md</code>	Averaged exterior clearance for the current interpolation law
<code>current_law_clearance_proposal.md</code>	Proposed clearance estimate for current-law surgery — unproved

Source path	Original title
direct_pivotal_comparison_route.md	Direct insertion-pivotal route to the small- β pure endpoint
direct_pivotal_void_audit.md	Independent audit: direct pivotal small-ball split
direct_pivotal_void_split.md	Direct pivotal leverage bound by a small empty ball
ellipse_bridge.md	Audited ellipse results and a critical-radius limit
ellipse_critical_variation.md	Exact ellipse critical variation and the signed obstruction
ellipse_critical_variation_audit.md	Independent audit: ellipse variation and quantitative Poisson limit
ellipse_minmax_switching.md	Stretch derivatives of a finite bottleneck radius
exterior_palm_bergman_bounds.md	Exterior Palm likelihood bounds and local Bergman domination
exterior_palm_bergman_independent_audit.md	Independent audit: exterior Palm and Bergman bounds
finite_ginibre_deletion_weights.md	Explicit finite Ginibre deletion weights and a rank-three test
finite_window_leverage_operator.md	A positive operator version of local Palm leverage
fixed_kernel_poisson_second_variation_audit.md	Independent audit of fixed-kernel second variation at Poisson
fixed_palm_exterior_transfer_independent_audit.md	Independent audit of the fixed-Palm infinite-exterior transfer
fixed_palm_exterior_transfer_pending.md	Fixed-Palm exterior transfer: superseded continuation record
fixed_palm_infinite_exterior_ratio_obstruction.md	Fixed-Palm upgrade of the actual infinite-exterior ratio obstruction
fixed_region_zero_bath_audit.md	Independent audit of the fixed-region zero-bath scaffold theorem
forced_arm_confluent_scalar_limit.md	High-order deletion mean with fixed forced arms retained
full_ginibre_obstructions_audit.md	Independent audits of two full-Ginibre obstructions
full_ginibre_transport_and_certificate_obstruction.md	Full Ginibre transport and a saturated pivotal certificate
ginibre_compact_factor_support_audit.md	Audit of infinite-Ginibre compact-factor realization
ginibre_finite_deletion_fock_zero_sets.md	Finite deletion and Fock zero sets for an actual Ginibre configuration
ginibre_finite_deletion_fock_zero_sets_audit.md	Independent audit of the three-deletion Fock zero-set argument
ginibre_full_circle_cluster_motion.md	A full-circle Ginibre motion with a quantitative density gain
ginibre_full_circle_motion_independent_audit.md	Independent audit: full-circle redundant-cluster motion
ginibre_infinite_compact_factor_support.md	Actual infinite-Ginibre realization of a compact exterior factor
ginibre_motion_audit.md	Ginibre two-parent motion: conditional ratios and the averaging gap
ginibre_motion_independent_audit.md	Independent audit of Ginibre parent motion
ginibre_motion_inverse_moments.md	Exact inverse-moment range for moving two Ginibre Palm parents
ginibre_palm_relocation_markov.md	Genuine Ginibre Palm relocation density and its normalization
ginibre_quad_jensen_audit.md	Independent audit: four-particle angular surgery

Source path	Original title
ginibre_triangle_cluster_motion.md	Three-point version of the full-circle cluster repair
ginibre_triple_resolvent_certificate.md	Exact uniform resolvent certificate for two triple clusters
induced_certificate_density_gap_audit.md	Independent audit: induced-certificate density gap
isotropic_affine_counterexample.md	Isotropy and ergodicity do not make disks minimize critical radius
k_omitted_marks_radial_bennett_audit.md	k omitted marks: sharp radial tail and Bennett audit
kong_yeh_clustering_audit.md	Independent audit of the Kong–Yeh clustering lower-bound argument
mixed_clone_and_compensation.md	Current-law clone bound and differential compensation with a positive bath
mixed_cooperative_audit.md	Independent audit of the positive-bath cooperative estimate
multipair_confluent_independent_audit.md	Independent audit of multipair Gram identities and confluent scalar means
multipair_fock_cross_moments.md	Polarized multi-pair Fock moments with exact Palm weights
multipair_fock_gram_optimization.md	Multi-deletion trial spans and exact Gram optimization
multiple_omitted_marks_far_transport.md	Multiple omitted Ginibre marks: exact transport and far-potential bounds
neutral_janossy_outside_norm_audit.md	Independent audit: neutral Janossy states and the complete outside-norm partition
palm_continuation_checkpoint.md	Palm branch continuation checkpoint
palm_current_law.md	Current-law form of the Palm interpolation loss
palm_independent_audit.md	Independent audit of current-law Palm and deletion estimates
palm_relocation_local_independent_audit.md	Independent audit of local Palm relocation and measurable coupling
pivotal_close_pair_estimate.md	A local protected close-pair estimate with its event weight retained
pivotal_omitted_pair_campbell.md	Two omitted points: exact pivotal Campbell transport
pivotal_outside_norm_campbell.md	Bridge-void Fock norm and exact alternative-certificate residual
pivotal_pair_bennett_independent_audit.md	Independent audit: omitted-pair transport and projection Bennett bound
pivotal_zero_one_neutral_janossy.md	Exact zero/one neutral-point law after forcing augmented arms
poisson_clone_decomposition.md	Uniform Poisson movement bound and the remaining cloning term
poisson_cloning_bv_audit.md	Independent audit: the Poisson cloning estimate at a fixed radius
poisson_uniform_cloning_bound.md	Uniform Poisson cloning upper bounds
polygon_correlated_kernel_independent_audit.md	Audit of the correlated polygon kernel and source geometry
positive_bath_pde_comparison.md	From the positive-bath differential inequality to a strict mixed-model threshold gap
projection_dpp_variance_bennett.md	A variance-sensitive exponential bound for projection DPP statistics
projection_dpp_variance_bennett_audit.md	Independent audit of the projection-DPP Bennett estimate

Source path	Original title
pure_small_beta_theorem_audit.md	Independent complete audit: strict small-beta threshold improvement
random_certificate_fock_nonuniqueness.md	Fock nonuniqueness from a logarithmic-scale density gap
random_certificate_fock_nonuniqueness_audit.md	Independent audit of the random-certificate Fock construction
rare_pivotal_corridor_spectral_audit.md	Independent audit of strip and polygonal-corridor clearing estimates
replacement_count_lemma.md	A deletion bound for crossing pivotals and the fine-mesh replacement limit
research/anisotropic_variation.md	Anisotropic Ginibre expansion and opposite crossing signs
research/bottleneck_cloning.md	Bottleneck thresholds: exact identities, counterexample, and Poisson cloning coefficient
research/contact_certificate_disintegration.md	Canonical certificate disintegration of artificial-grain contact
research/contact_scaffold_absorption.md	Contact-scaffold reduction of the zero-bath leverage term
research/current_law_contact_obstruction_s.md	Artificial-grain contact: geometric facts and two ruled-out stronger bounds
research/fixed_kernel_poisson_second_variation.md	Fixed-kernel expansion at Poisson and the combined two-parent geometry
research/ginibre_coupling_bridge_obstructions.md	Exact obstructions to a thinning or bounded-displacement bridge
research/ginibre_exterior_analytic_product_audit.md	The distant Ginibre analytic product tends to one
research/goldman_column_mass_audit.md	Audit of genuine Palm insertion and the separate column-mass question
research/induced_certificate_density_gap.md	A strict density gap for all induced two-arm certificates
research/multipair_projection_and_marking_obstructions.md	Adversarial checks for optimized multi-pair deletion trials
research/rare_pivotal_weighting_and_corridors.md	Rare pivotal weighting and a conditional corridor estimate
research/small_beta_critical_radius_asymptotic.md	Exact Poisson domination and the first-order small-beta critical radius
research/steiner_certificate_clique_excess.md	Minimal Steiner certificates: clique excess and capture of nearby relocations
residual_projection_dynamics.md	Finite-rank residual projection dynamics
retained_arms_multipair_variational_limit.md	Retained finite arms in the high-order multi-deletion limit
reversible_projection_move.md	A reversible one-point move for a thinned finite-rank projection DPP
small_beta_asymptotic_audit.md	Independent audit: exact domination and small-beta critical asymptotic
sparse_fock_projection.md	Gaussian localization of a sparse coherent-state projection
sparse_fock_projection_audit.md	Independent audit of sparse coherent-state localization
straight_arm_wide_lens_resolvent.md	Positive wide-lens surplus with arbitrarily long straight forced arms
tapered_arm_area_motion_obstruction.md	Tapered arm motion: the integrated lens-area obstruction
thinned_ginibre_potential_bennett.md	Bennett control for a thinned Ginibre process with a Poisson bath

Source path	Original title
<code>thinned_goldman_relocation_identity.md</code>	Thinned Goldman weights and the exact interpolation coefficient
<code>thinned_relocation_independent_audit.md</code>	Independent audit of the thinned Goldman interpolation identity
<code>three_palm_weighted_fock_identity.md</code>	Exact event-weighted three-Palm Fock identity and its limitation
<code>two_parent_wide_lens_resolvent.md</code>	An explicit positive two-parent wide-lens multiplier
<code>weak_hybrid_threshold_audit.md</code>	Independent audit of the strict weak-hybrid threshold comparison
<code>wide_lens_independent_audit.md</code>	Independent audit of the positive wide-lens calculation
<code>zero_bath_boundary_audit.md</code>	Independent audit: fixed-region boundary separation
<code>zero_bath_boundary_separation.md</code>	Fixed-region boundary separation with no Poisson bath
<code>zero_bath_bulk_audit.md</code>	Independent audit: fixed-region zero-bath bulk separation
<code>zero_bath_bulk_separation.md</code>	Two-arm separation by deterministic holes and explicit disk geometry
<code>zero_bath_certificate_route.md</code>	Canonical certificates and a proposed zero-bath clearance bound

Original attachments and consolidated record

- `attachments/ginibre_poisson_attempts_summary.pdf`: the supplied research history, dated 5 September 2026.
- `attachments/ginibre_ellipses_large_area.tex`: the supplied ellipse note, unchanged.
- `history/ginibre_poisson_continuation_2026-09-15.md`: the consolidated historical checkpoint, including its full proof-note appendices.
- `history/ginibre_proved_results.md`: the earlier selective account, retained to make the subsequent restoration of the ellipse theorem traceable.

The source archive intentionally retains assertions that the present report does not adopt as established. Consult the status discussion and the small- β section before using those historical claims.